

Digital Electronics

Detailed Solutions • 1991–2024

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Digital Electronics (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EE	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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SOLUTIONS

1 Logic Gates and Boolean Algebra

Topics

1. Introduction to Digital Systems
2. Basic Gates and Concept of Multivibrators Using NOT Gate
3. Universal and Arithmetic Gates
4. Implementing Boolean Expressions Using Universal Logic Gates
5. Timing Diagrams
6. Boolean Algebra, Venn Diagrams, and the Concept of Dual

Unit Snapshot*

Stream: EE	**Questions:** 14	**MCQ:** 14	**MSQ:** 0	**NAT:** 0
1M: 10	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2024)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 2M Ans: B ●●○

Given logic circuit is shown below,

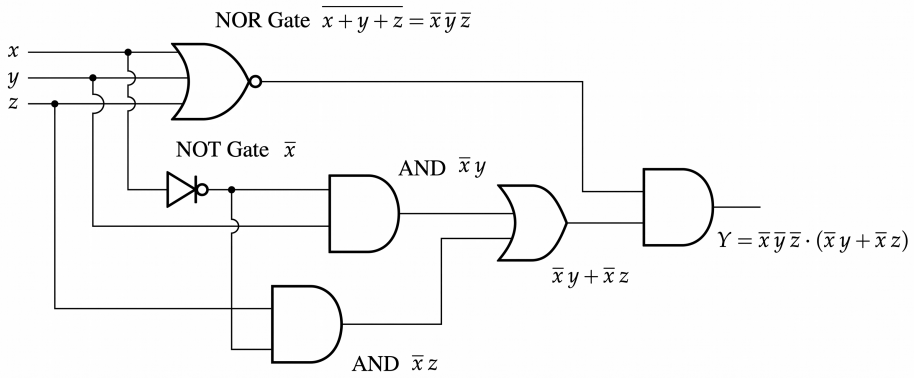


Fig. Solution Q1.26.1

From the circuit, the output can be expressed as:

$$\text{OUT} = \bar{x}\,\bar{y}\,\bar{z}\cdot(\bar{x}\,y+\bar{x}\,z)$$

$$=\bar{x}\,\bar{y}\,\bar{z}\cdot\bar{x}(y+z)$$

$$=\bar{x}\,\bar{y}\,\bar{z}(y+z)$$

$$=\bar{x}(\bar{y}\,\bar{z}\,y+\bar{y}\,\bar{z}\,z)$$

$$=\bar{x}(0+0)=0$$

Hence, the correct option is (B).

(B) 0

Q1.24.1 MCQ 2024 1M Ans: A ●●○

Given function F is in SOP form:

$$F(P,Q,R,S)=\bar{P}\bar{Q}+\bar{P}QS+P\bar{Q}\bar{R}\bar{S}+P\bar{Q}R\bar{S}$$

Method 1

$$F(P,Q,R,S)=\bar{P}\bar{Q}+\bar{P}QS+P\bar{Q}\bar{R}\bar{S}+P\bar{Q}R\bar{S}$$
$$=\bar{P}\bar{Q}+\bar{P}QS+P\bar{Q}\bar{S}\left(\bar{R}+R\right)$$

OR Law:

(1) $1+A=1$ (2) $A+\bar{A}=1$

(3) $A+A=A$ (4) $A+0=A$

$$F(P,Q,R,S)=\bar{P}\bar{Q}+\bar{P}QS+P\bar{Q}\bar{S}$$
$$=\bar{P}\bar{Q}\left(S+\bar{S}\right)+\bar{P}QS+P\bar{Q}\bar{S}$$
$$=\bar{P}\bar{Q}S+\bar{P}\bar{Q}\bar{S}+\bar{P}QS+P\bar{Q}\bar{S}$$

$$= \bar{P}S\left(Q + \bar{Q}\right) + \bar{Q}\bar{S}\left(P + \bar{P}\right)$$
$$= \bar{P}S + \bar{Q}\bar{S}$$

Hence, the correct option is (A).

Method 2

$$F(P, Q, R, S) = \bar{P}\bar{Q} + \bar{P}QS + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}RS$$
$$= \bar{P}\bar{Q}\left(R + \bar{R}\right)\left(S + \bar{S}\right) + \bar{P}QS\left(R + \bar{R}\right) + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}RS$$
$$F(P, Q, R, S) = \bar{P}\bar{Q}\bar{R}\bar{S} + \bar{P}\bar{Q}\bar{R}S + \bar{P}\bar{Q}R\bar{S} + \bar{P}\bar{Q}RS + \bar{P}QSR + \bar{P}QS\bar{R} + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}RS$$

k-map of function f in SOP form is,

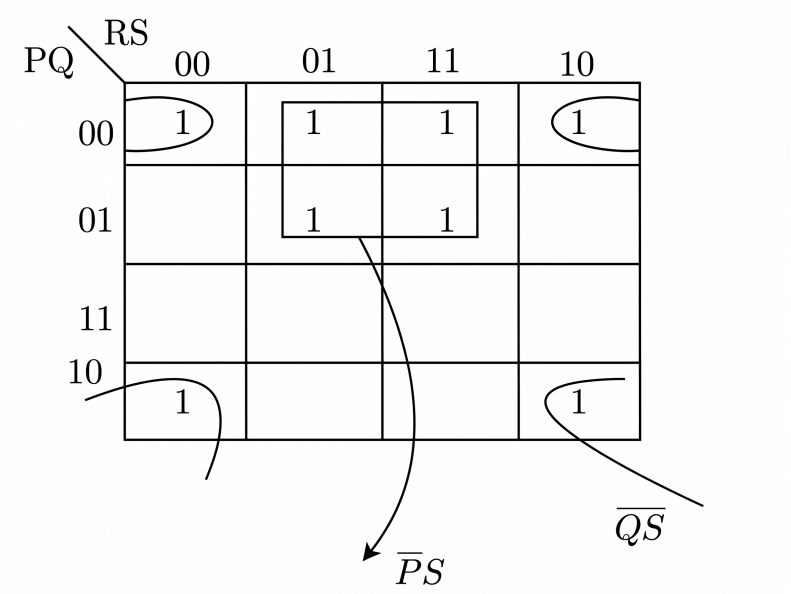


Fig. Solution Q1.24.1

$$F(P, Q, R, S) = \bar{P}S + \bar{Q}\bar{S}$$

Hence, the correct option is (A).

(A) $\bar{P}S + \bar{Q}\bar{S}$

Q1.19.1 MCQ | 2019 | 2M | Ans: A | ● ○ ○

Given circuit is shown below,

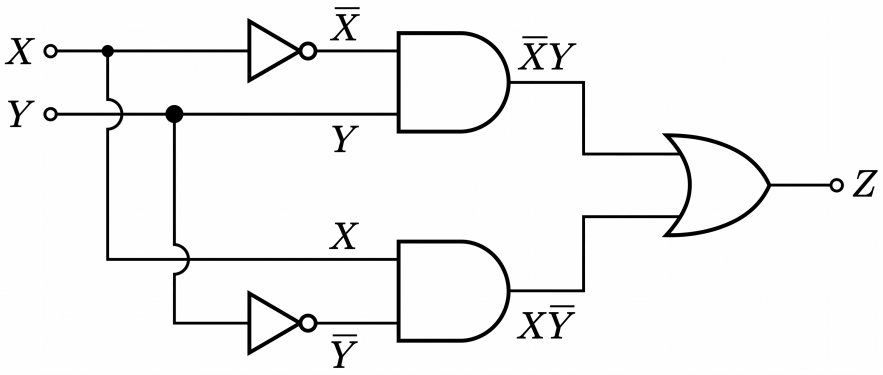


Fig. Solution Q1.19.1

From the circuit diagram,

$$Z = \bar{X}Y + X\bar{Y} = X \oplus Y$$

This is the expression for an **XOR gate**.
Hence, the correct option is (A).

(A) XOR gate

Q1.18.1 MCQ 2018 1M Ans: D

Given logic circuit is shown below,

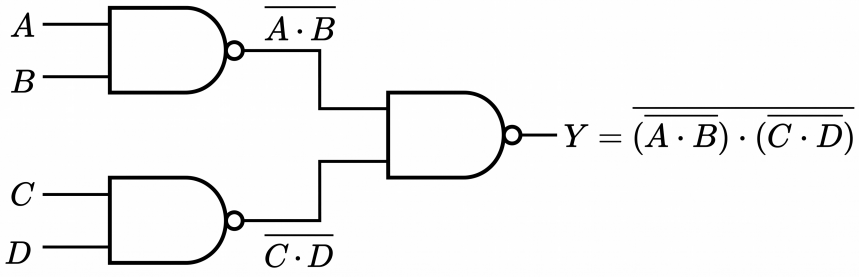


Fig. Solution Q1.18.1

From the circuit, the two NAND gates produce:

$$G_1 = \overline{AB}, \qquad G_2 = \overline{CD}$$

The final NAND gate gives:

$$Y = \overline{G_1 \cdot G_2} = \overline{\overline{AB} \cdot \overline{CD}}$$

By De-Morgan's law ($\overline{A \cdot B} = \bar{A} + \bar{B}$):

$$Y = \overline{\overline{AB}} + \overline{\overline{CD}} = AB + CD$$

Hence, the correct option is (D).

(D) $Y = AB + CD$

Q1.17.1 MCQ | 2017 | 1M | Ans: C ● ● ○

Given logic circuit is shown below,

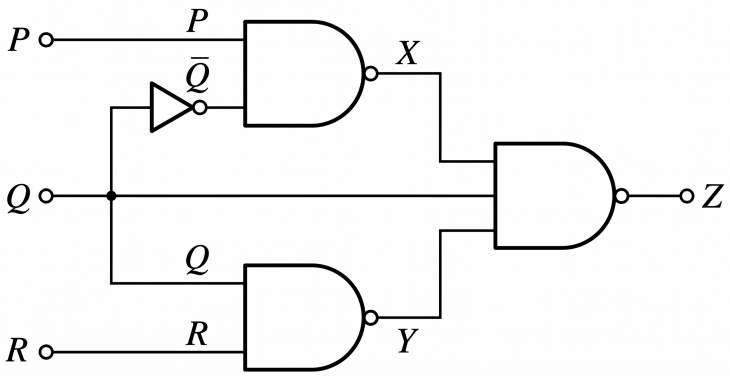


Fig. Solution Q1.17.1

From the above circuit diagram,

$$X = \overline{P\bar{Q}}$$

$$Y = \overline{QR}$$

$$Z = \overline{X \cdot Y \cdot Q} = \bar{X} + \bar{Y} + \bar{Q}$$

$$Z = \overline{\overline{P\bar{Q}}} + \overline{\overline{QR}} + \bar{Q} = P\bar{Q} + QR + \bar{Q}$$

$$Z = (1 + P)\bar{Q} + QR \qquad [\because 1 + P = 1]$$

$$Z = \bar{Q} + QR$$

By distributive law,

$$A + BC = (A + B)(A + C)$$

$$Z = (\bar{Q} + Q)(\bar{Q} + R) \qquad [\because \bar{Q} + Q = 1]$$

$$Z = \bar{Q} + R$$

Hence, the correct option is (C).

(C) $\bar{Q} + R$

Q1.13.1 MCQ | 2013 | 1M | Ans: C ● ○ ○

Let the two staircase switches be represented by inputs A and B.
The bulb should change its state whenever any one of the switches is toggled, irrespective of the state of the other switch.
The bulb is ON when the switches are in opposite positions and OFF when the switches are in the same position.
The truth table is

A	B	Bulb
0	0	0
0	1	1
1	0	1
1	1	0

The above truth table corresponds to the Exclusive-OR (XOR) gate.
Therefore,

$$\text{Bulb} = A \oplus B$$

(C) An XOR gate

Key Points

- An XOR gate produces HIGH output when its inputs are different.
- The Boolean expression of XOR is
$$A \oplus B = \overline{A}B + A\overline{B}.$$
- In staircase wiring, toggling either switch changes the state of the bulb, which is the characteristic behavior of an XOR gate.

Mistakes to be avoided

- Do not confuse staircase switching with an OR gate. In an OR gate, the output remains HIGH when both inputs are HIGH, whereas the bulb is OFF when both switches are in the same position.
- Remember that XOR gives HIGH only when the inputs are unequal.

Q1.11.1 MCQ | 2011 | 1M | Ans: A | ● ○ ○

Given logic circuit is shown below,

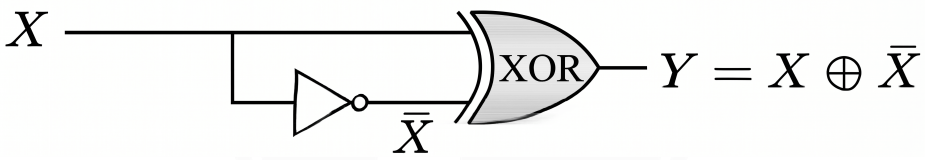


Fig. Solution Q1.11.1

From the circuit, using XOR with its complement:

$$Y = X \oplus \bar{X}$$
$$Y = X \cdot \overline{\bar{X}} + \bar{X} \cdot \bar{X} = X \cdot X + \bar{X} \cdot \bar{X} = X + \bar{X}$$
$$\boxed{Y = 1}$$

Hence, the correct option is (A).

(A) 1

Q1.09.1 MCQ | 2009 | 1M | Ans: C | ● ○ ○

Using either NOR or NAND gate, all possible logic gates can be implemented. Hence, NOR and NAND are designated as universal logic gates.

Key Points

- **NAND gate:** Can implement NOT, AND, OR, NOR, XNOR, XOR.
- **NOR gate:** Can implement NOT, OR, AND, NAND, XOR, XNOR.
- XOR and XNOR are *special purpose* gates, not universal.

Hence, the correct option is (C).

(C) NOR and NAND Gates

Q1.07.1 MCQ | 2007 | 1M | Ans: B | ● ○ ○

Given logic circuit is shown below,

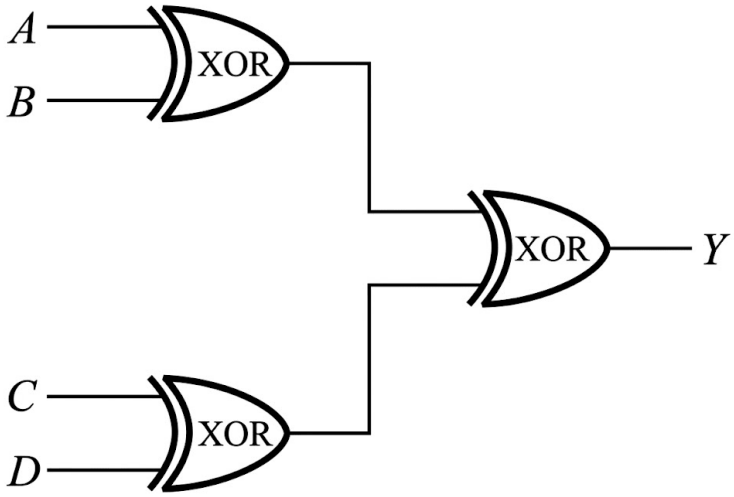


Fig. Solution Q1.07.1

From the cascaded XOR gates,

$$Y = A \oplus B \oplus C \oplus D$$

Hence, $Y = 1$ when the number of 1's among A, B, C, D is odd, and $Y = 0$ when it is even.
The required sum is

$$S = A + B + C + D + Y$$

- If A, B, C, D contain an odd number of 1's, then $Y = 1$.

$$S = (\text{odd}) + 1 = (\text{even})$$

- If A, B, C, D contain an even number of 1's, then $Y = 0$.

$$S = (\text{even}) + 0 = (\text{even})$$

Also, for $A = B = C = D = 0$,

$$S = 0$$

Therefore, S is always either zero or even.

(B) S is always either zero or even.

Key Points

- XOR output is 1 when the number of input 1's is odd.
- Cascaded XOR gates act as a parity generator.
- Adding the parity bit makes the total number of 1's even.

Q1.02.1 MCQ 2002 2M Ans: B ● ● ○

Given logic circuit is shown below,

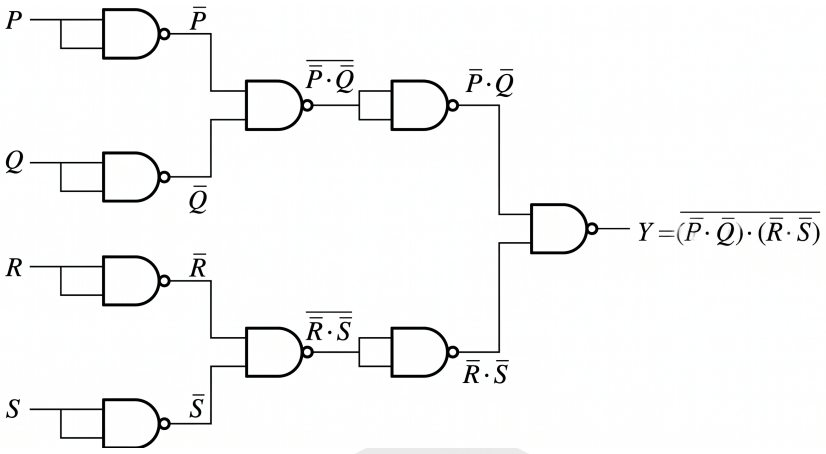


Fig. Solution Q1.02.1

$$Y = \overline{(\bar{P}\cdot\bar{Q})\cdot(\bar{R}\cdot\bar{S})}$$

From De-Morgan's Law,

(i) $\overline{A\cdot B} = \bar{A} + \bar{B}$

(ii) $\overline{A + B} = \bar{A}\cdot\bar{B}$

$$Y = \overline{(\bar{P}\cdot\bar{Q})} + \overline{(\bar{R}\cdot\bar{S})}$$

$$Y = P + Q + R + S$$

Hence, the correct option is (B).

(B) $P + Q + R + S$

Q1.01.1 MCQ 2001 1M Ans: D ● ○ ○

Given: Output of a logic gate is "1", when all its inputs are at logic "0".

There are only three gates which produce output 1 when all inputs are 0: NOR, NAND, and EX-NOR. Among the given options, only NOR and EX-NOR appear together.

NOR gate:

$$Y_0 = \overline{X + Y}, \qquad Y_0 = 1 \text{ when } X = Y = 0$$

EX-NOR gate:

$$Y_0 = XY + \bar{X}\bar{Y}, \qquad Y_0 = 1 \text{ when } X = Y = 0$$

EX-OR gate gives $Y_0 = 0$ when $X = Y = 0$, so it does *not* qualify.

Hence, the correct option is (D).

(D) A NOR or an EX-NOR gate

Q1.99.1 MCQ 1999 2M Ans: B ● ● ○

Given:

$$f = \overline{(x \cdot \bar{y}) + (\bar{x} \cdot y)}$$

Method 1

Recognise that $(x \cdot \bar{y}) + (\bar{x} \cdot y) = x \oplus y$, so:

$$f = \overline{x \oplus y} = x \odot y = xy + \bar{x}\bar{y}$$

Now check option (B):

$$\overline{(\bar{x} + \bar{y})(x + y)} = \overline{\bar{x} + \bar{y}} + \overline{(x + y)} = xy + \bar{x}\bar{y}$$

This matches f. Hence, the correct option is (B).

Method 2

By De-Morgan's law applied directly:

$$f = \overline{(x \cdot \bar{y}) + (\bar{x} \cdot y)} = \overline{(x \cdot \bar{y})} \cdot \overline{(\bar{x} \cdot y)}$$
$$= (\bar{x} + y)(x + \bar{y})$$
$$= \bar{x}x + \bar{x}\bar{y} + xy + y\bar{y} = \bar{x}\bar{y} + xy$$
$$= \overline{(\bar{x} + \bar{y})(x + y)}$$

Hence, the correct option is (B).

$$\boxed{\text{(B) } f = \overline{(\bar{x} + \bar{y})(x + y)}}$$

Q1.96.1 MCQ 1996 1M Ans: A ● ● ○

Given logic circuit is shown below,

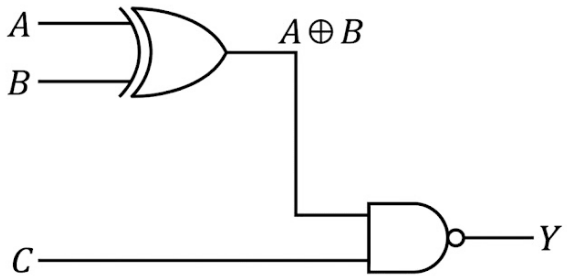


Fig. Solution Q1.96.1

$$Y = \overline{(A \oplus B) \cdot C}$$

From De-Morgan's law,

$$Y = \overline{(A \oplus B) \cdot C}$$
$$Y = \overline{(A \oplus B)} + \bar{C}$$
$$Y = (A \odot B) + \bar{C} \qquad \left[\because \overline{W \oplus X} = W \odot X\right]$$
$$Y = AB + \bar{A}\bar{B} + \bar{C}$$

Hence the correct option is (A).

$$\boxed{\text{(A) } Y = \bar{A}\bar{B} + AB + \bar{C}}$$

Q1.91.1 MCQ | 1991 | 1M | Ans: A

Given logic circuit is shown below,

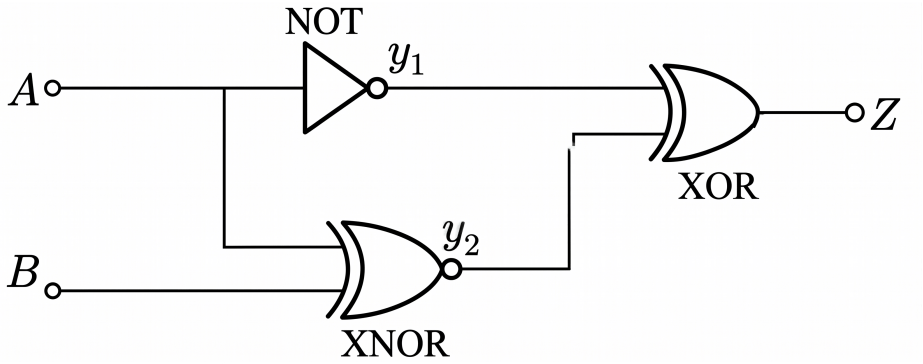


Fig. Solution Q1.91.1

We can make truth table for this two input combinational circuit,

A	B	$y_1 = \bar{A}$	$y_2 = A \odot B$	$z = y_1 \oplus y_2$
0	0	1	1	0
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1

Above truth table can be reduced as,

A	B	$z = y_1 \oplus y_2$
0	0	0
0	1	1
1	0	0
1	1	1

Hence, the correct option is (A).

(A)

SOLUTIONS

11

Representation of Boolean Expressions and K-Maps

Topics

1. Minterms and Maxterms
2. SOP and POS Representation
3. Karnaugh Maps (K-Maps)
4. Concept of Implicants

Unit Snapshot*

Stream: EE	**Questions:** 17	**MCQ:** 17	**MSQ:** 0	**NAT:** 0
1M: 7	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2015)–2M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.25.1 MCQ | 2025 | 2M | Ans: A ● ● ○

Given : Boolean function

$$f = (\bar{u} + \bar{v} + \bar{w} + \bar{x}) \cdot (\bar{u} + \bar{v} + \bar{w} + x) \cdot (\bar{u} + v + \bar{w} + \bar{x}) \cdot (\bar{u} + v + \bar{w} + x)$$

f can also write in the form of maxterms as,

$$f = \underset{1111}{(\bar{u} + \bar{v} + \bar{w} + \bar{x})} \cdot \underset{1110}{(\bar{u} + \bar{v} + \bar{w} + x)} \cdot \underset{1011}{(\bar{u} + v + \bar{w} + \bar{x})} \cdot \underset{1010}{(\bar{u} + v + \bar{w} + x)}$$

$$f = \pi(1111, 1110, 1011, 1010)$$

$$x = \pi(1010, 1011, 1110, 1111)$$

$$x = \pi(10, 11, 14, 15)$$

k-map for f is shown below,

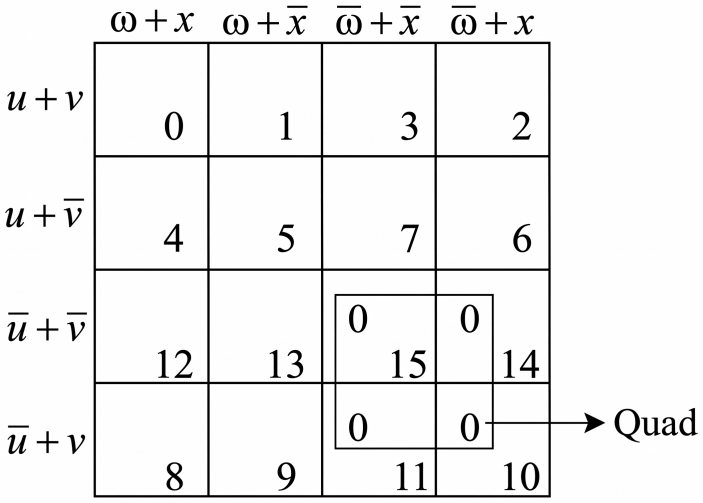


Fig. Solution Q2.25.1

So, minimised POS form is,

$$f = \bar{u} + \bar{w}$$

(A) The minimised POS expression is $f = \bar{u} + \bar{w}$.

Q2.19.1 MCQ | 2019 | 2M | Ans: B ● ○ ○

Given K-map is shown below,

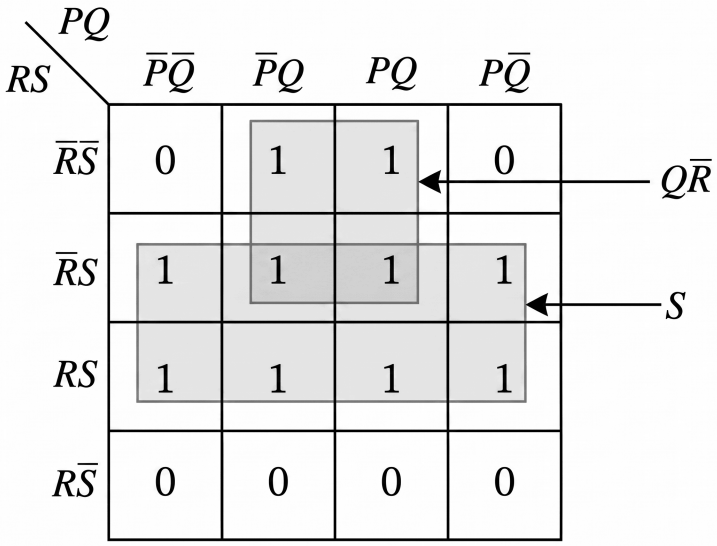


Fig. Solution Q2.19.1

From the K-map, grouping the 1s:

- A horizontal group gives $Q\bar{R}$
- A vertical group gives S

$$y = Q\bar{R} + S$$

(B) The minimised SOP expression is $y = Q\bar{R} + S$.

Q2.18.1 MCQ 2018 2M Ans: B

Given:

$$F = m_0 + m_2 + m_3 + m_5 + m_7, \quad \text{don't care at } m_1$$

$$F(A, B, C) = \Sigma m(0, 2, 3, 5, 7) + \Sigma\phi(1)$$

K-map for $F(A, B, C)$:

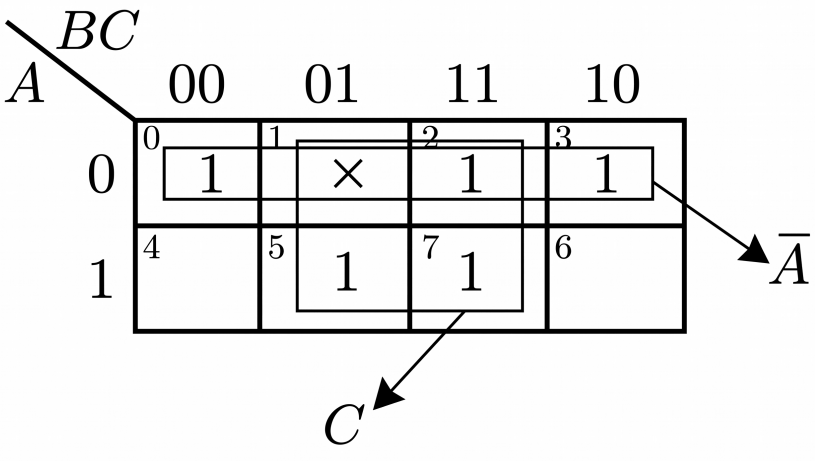


Fig. Solution Q2.18.1

Minterm table for reference:

Min-term	SOP / Binary			Decimal
m_0	$\bar{A}$	$\bar{B}$	$\bar{C}$	0
m_1	$\bar{A}$	$\bar{B}$	C	1 (don't care)
m_2	$\bar{A}$	B	$\bar{C}$	2
m_3	$\bar{A}$	B	C	3
m_5	A	$\bar{B}$	C	5
m_7	A	B	C	7

Grouping 1s (including don't care m_1):

- m_0, m_1, m_2, m_3 (quad) $\rightarrow \bar{A}$
- m_3, m_5, m_7 and wrapping $\rightarrow C$

$$F = \bar{A} + C$$

(B) The minimised SOP expression is $F = \bar{A} + C$.

Q2.17.1 MCQ | 2017 | 1M | Ans: A

Given Boolean function is

$$F(A, B, C) = AB + A\bar{C} + BC$$

Using K-map,

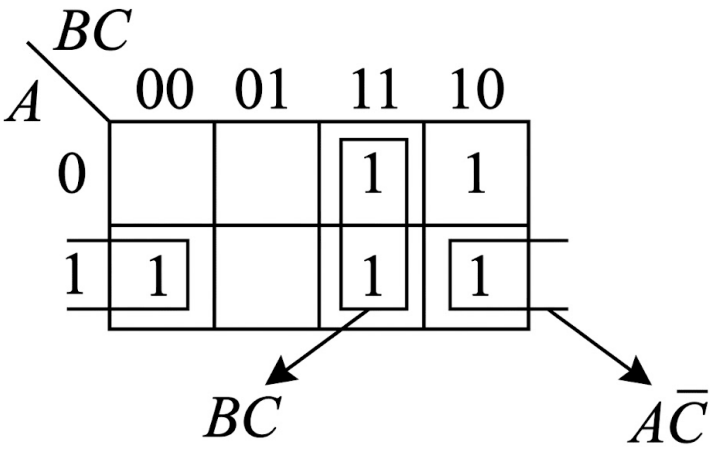


Fig. Solution Q2.17.1

Hence, the minimized SOP expression is

$$F(A, B, C) = BC + A\bar{C}$$

(A) $BC + A\bar{C}$

Q2.17.2 MCQ | 2017 | 2M | Ans: D ● ● ○

Given K-map is shown below,

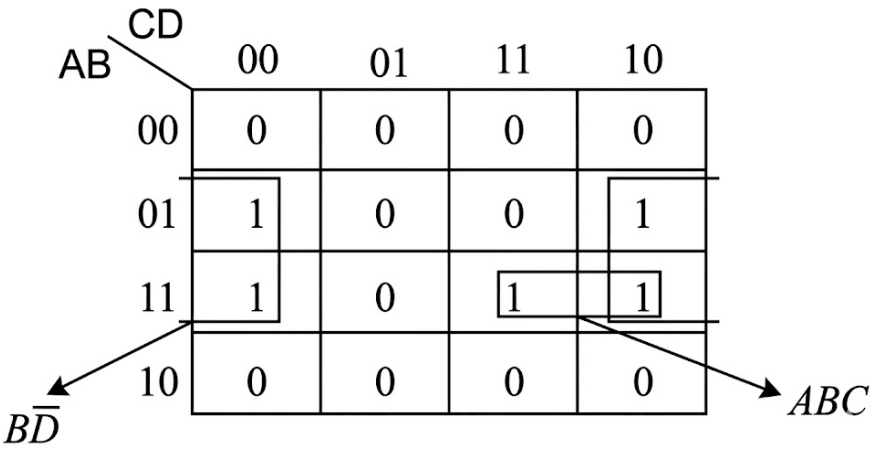


Fig. Solution Q2.17.2

Thus, the SOP expression of function F is,

$$F = B\bar{D} + ABC$$

Hence, the correct option is (D).

(D) $B\bar{D} + ABC$

Q2.16.1 MCQ | 2016 | 1M | Ans: B ● ● ○

Given K-map is shown below,

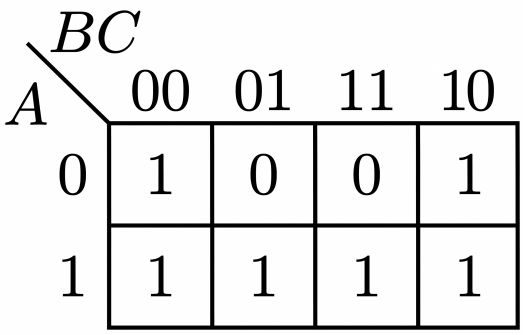


Fig. Solution Q2.16.1

The K-map in POS form (grouping 0s):

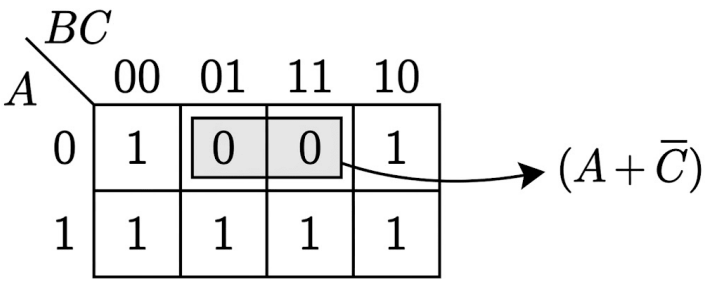


Fig. Solution Q2.16.1

The 0s form a group giving factor $(A + \bar{C})$.
The POS expression of the function is:

$$Y = A + \bar{C}$$

(B) The minimised expression is $Y = A + \bar{C}$.

Q2.16.2 MCQ 2016 2M Ans: D

Method 1

From De-Morgan's law,

(i) $\overline{A \cdot B} = \bar{A} + \bar{B}$

(ii) $\overline{A + B} = \bar{A} \cdot \bar{B}$

Thus,

$$F = \overline{(a + \bar{b} + c + \bar{d}) + (b + \bar{c})}$$

$$F = \overline{(a + \bar{b} + c + \bar{d})} \cdot \overline{(b + \bar{c})}$$

$$F = (\bar{a} \cdot b \cdot \bar{c} \cdot d)(\bar{b} \cdot c) \qquad [\because\ X \cdot \bar{X} = 0]$$

$$F = 0$$

Hence, the correct option is (D).

Method 2

Given,

$$F = \overline{(a + \bar{b} + c + \bar{d}) + (b + \bar{c})}$$

$$F = \overline{a + \bar{b} + b + c + \bar{c} + \bar{d}}$$

$$F = \overline{a + \bar{d} + 1 + 1} \qquad [\because\ x + \bar{x} = 1]$$

$$F = \overline{a + \bar{d} + 1} \qquad [\because\ 1 + x = 1]$$

$$F = \overline{1}$$

$$F = 0$$

Hence, the correct option is (D).

(D) 0

Q2.15.1 MCQ | 2015 | 1M | Ans: C

Given : The POS function,

$$f(A, B, C, D) = \Pi M(0, 1, 3, 4, 5, 7, 9, 11, 12, 13, 14, 15)$$

The minterms of the given function are the states not present in the maxterm list:

$$f(A, B, C, D) = \Sigma m(2, 6, 8, 10)$$

Plotting these minterms on a 4-variable K-map to find the minimized Sum-of-Products (SOP) form:

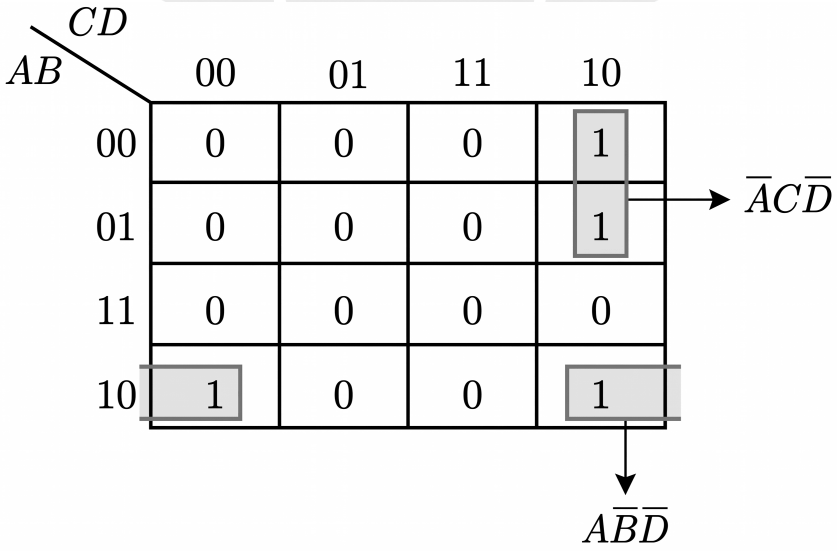


Fig. Solution Q2.15.1

From the K-map, we group the adjacent cells:

- Group 1 (m2, m6): $\bar{A}C\bar{D}$
- Group 2 (m8, m10): $A\bar{B}\bar{D}$

Hence, the minimized SOP expression is

$$f(A, B, C, D) = \bar{A}C\bar{D} + A\bar{B}\bar{D}.$$

Option (C) is

$$\bar{A}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D}.$$

Using Boolean algebra,

$$A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} = A\bar{B}\bar{D}(C + \bar{C}) = A\bar{B}\bar{D},$$

which simplifies Option (C) to

$$\bar{A}C\bar{D} + A\bar{B}\bar{D}.$$

(C) is equivalent to the minimised expression $f(A, B, C, D) = \bar{A}C\bar{D} + A\bar{B}\bar{D}$.

Taking a dual does not, in general, preserve the truth table. For example, at $A = B = C = D = 0$, the given maxterm list requires $f = 0$, whereas option (A) evaluates to $(0 + 1 + 0)(1 + 0 + 0) = 1$. Thus (A) is not equivalent to the given function.

Q2.15.2 MCQ | 2015 | 1M | Ans: A | ● ● ○

Method 1

Given:

$$F = ABC + \bar{A}\bar{B}C + A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C}$$

$$F = \Sigma m(111, 001, 101, 011, 000)$$

$$F = \Sigma m(7, 1, 5, 3, 0)$$

$$F = \Sigma m(0, 1, 3, 5, 7) \qquad \text{(SOP form)}$$

$$F = \Pi M(2, 4, 6) \qquad \text{(POS form)}$$

$$F = \Pi M(010, 100, 110)$$

$$F = (A + \bar{B} + C)(\bar{A} + B + C)(\bar{A} + \bar{B} + C)$$

Hence, the correct option is (A).

Method 2

Given function F is in SOP form.

$$F = ABC + \bar{A}\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C} + A\bar{B}C$$

Take out remaining terms which shows complement of F.

$$\bar{F} = \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}$$

Take again complement of $\bar{F}$ we will get POS form of F.

$$\bar{\bar{F}} = \overline{\bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}}$$

$$\bar{\bar{F}} = F = (A + \bar{B} + C)(\bar{A} + B + C)(\bar{A} + \bar{B} + C)$$

Hence, the correct option is (A).

(A) $F = (A + \bar{B} + C)(\bar{A} + B + C)(\bar{A} + \bar{B} + C)$

Q2.14.1 MCQ | 2014 | 2M | Ans: A

Method 1

Given : The SOP (Sum of Product) form of a Boolean function,

$$f(A, B, C, D) = \Sigma m(0, 1, 3, 7, 11)$$

(where A is MSB, D is LSB).

As the given options are in **product of sum form**, therefore convert it into **canonical product of sum form**.

$$f(A, B, C, D) = \Pi M(2, 4, 5, 6, 8, 9, 10, 12, 13, 14, 15)$$

Thus, K-map of function f in POS form is,

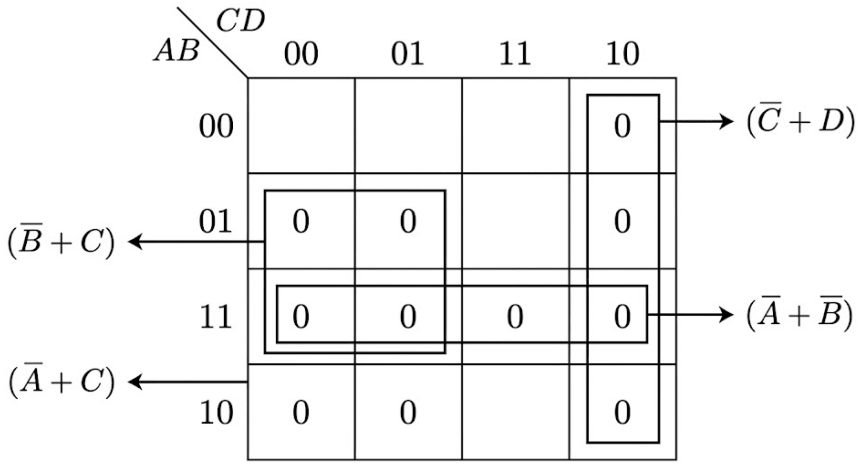


Fig. Solution Q2.14.1

$$f = (\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{B})(\bar{C} + D)$$

Method 2

The SOP form of Boolean function is given as,

$$f(ABCD) = \Sigma m(0, 1, 3, 7, 11) \qquad \ldots (i)$$

Options are given in POS form we have to convert given SOP into POS form but in this method, no need to convert given function in POS form, we can solve this using SOP only.

Take complement term of equation (i),

$$\overline{f(ABCD)} = \Sigma m(2, 4, 5, 6, 8, 9, 10, 12, 13, 14, 15)$$

K-map simplification for above function in SOP,

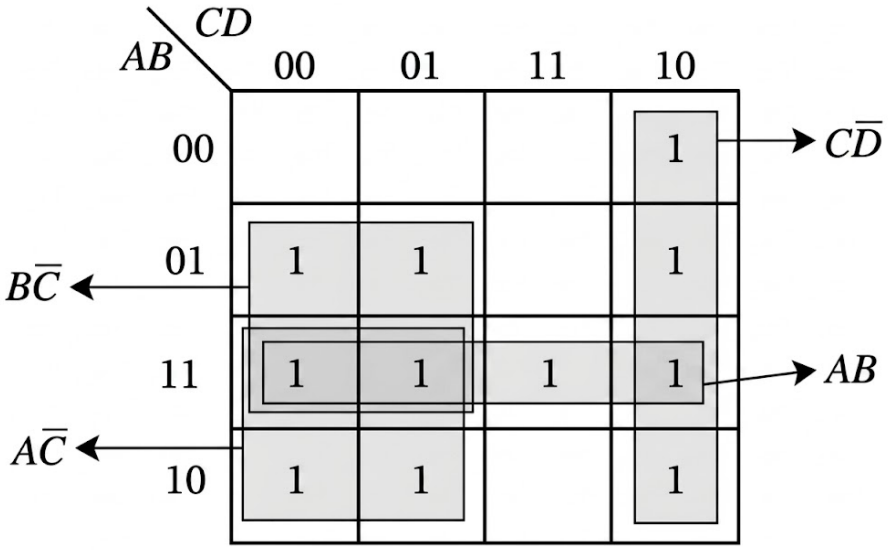


Fig. Solution Q2.14.1

$$\overline{f(ABCD)} = B\bar{C} + A\bar{C} + AB + C\bar{D}$$

Take complement on both sides (see, key point below this question)

$$\overline{\overline{f(ABCD)}} = \overline{B\bar{C} + A\bar{C} + AB + C\bar{D}}$$

$$f(ABCD) = (\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{B})(\bar{C} + D)$$

Tip : This method-2 is present for those aspirants who are not comfortable with POS form.

(A) The minimised POS expression is $f = (\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{B})(\bar{C} + D)$.

Q2.14.2 MCQ 2014 2M Ans: C ●●○

Given K-map is shown below,

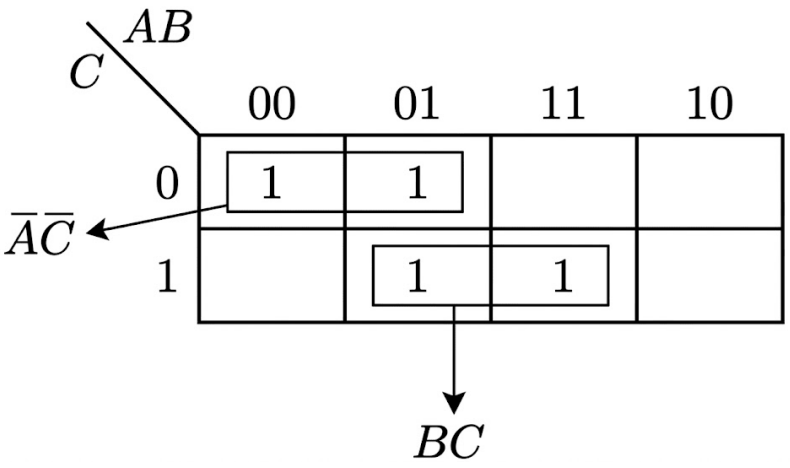


Fig. Solution Q2.14.2

From above K-map,

$$F = BC + \bar{A}\bar{C}$$

The above function can be realized as,

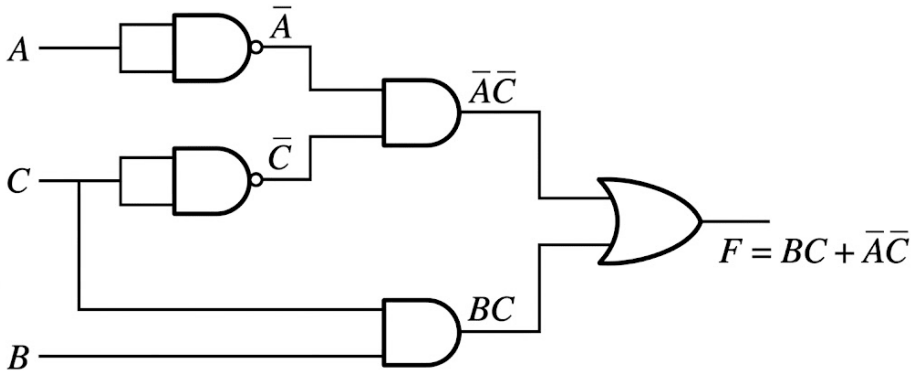


Fig. Solution Q2.14.2

Hence, the correct option is (C).

(C) Realization of $F = BC + \bar{A}\bar{C}$

Q2.12.1 MCQ 2012 1M Ans: A

Given:

$$f(X, Y, Z) = \Sigma m(2, 3, 4, 5)$$

$$f(X, Y, Z) = \bar{X}Y\bar{Z} + \bar{X}YZ + X\bar{Y}\bar{Z} + X\bar{Y}Z$$

K-map for function $f(X, Y, Z)$:

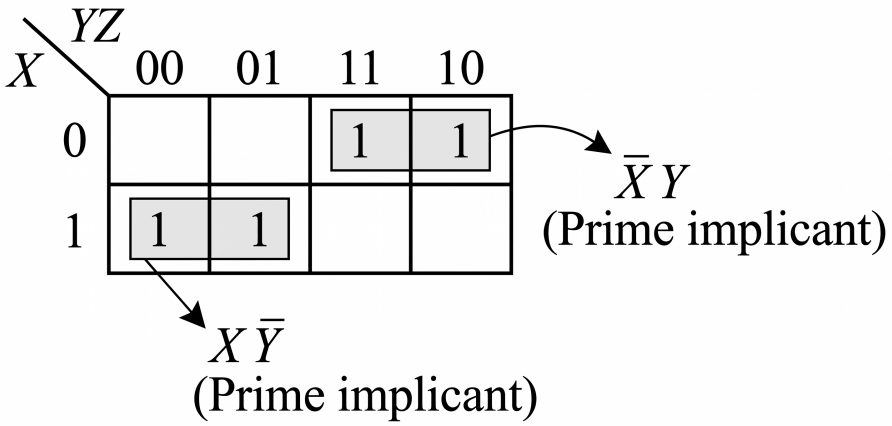


Fig. Solution Q2.12.1

Identifying prime implicants (maximum possible groupings):

- m_2, m_3 (pair) $\rightarrow \bar{X}Y$ (Prime Implicant)
- m_4, m_5 (pair) $\rightarrow X\bar{Y}$ (Prime Implicant)

(A) The prime implicants of $f(X, Y, Z)$ are $\bar{X}Y$ and $X\bar{Y}$.

Q2.10.1 MCQ 2010 1M Ans: B ● ○ ○

Given K-map is shown below,

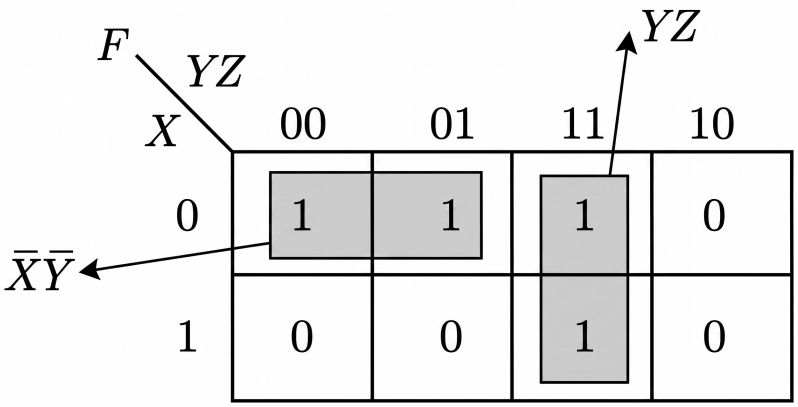


Fig. Solution Q2.10.1

Grouping the 1s in the K-map:

- Horizontal group ($YZ = 01$ and $YZ = 11$, $X = 0$) $\rightarrow \bar{X}\bar{Y}$
- Vertical group ($YZ = 11$, both rows) $\rightarrow YZ$

$$F = \bar{X}\bar{Y} + YZ$$

Key Points

- To find the minimised expression in POS form, 0s are grouped.
- To find the minimised expression in SOP form, 1s are grouped.

(B) The minimised SOP expression is $F = \bar{X}\bar{Y} + YZ$.

Q2.10.2 MCQ 2010 1M Ans: D ● ● ○

Given: $F = \bar{X}\bar{Y} + YZ$

To realise F, an OR gate is needed to sum the two terms $\bar{X}\bar{Y}$ and YZ.

Product term YZ can be generated by an AND gate.

Product term $\bar{X}\bar{Y} = \overline{X + Y}$ can be generated by a NOR gate (NOR gate is equivalent to a bubbled AND gate):

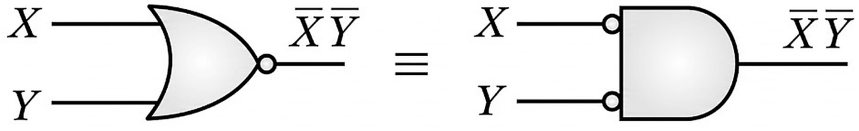


Fig. Solution Q2.10.2

Hence, F can be realized as shown below:

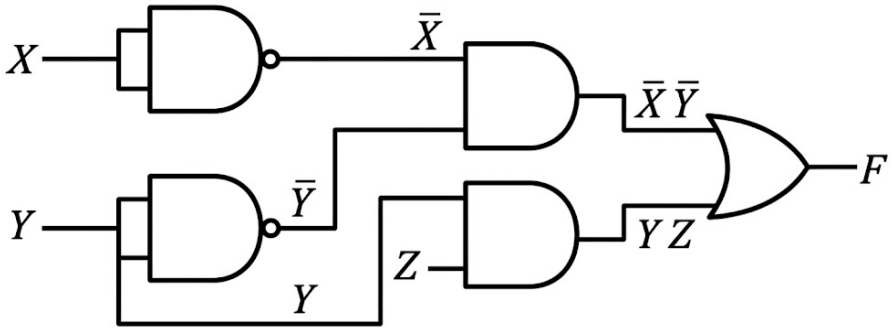


Fig. Solution Q2.10.2

(D) $F = \bar{X}\bar{Y} + YZ$ can be realised using one NOR gate, one AND gate, and one OR gate.

Q2.04.1 MCQ 2004 2M Ans: A ● ● ○

Given:

$$Y = (\bar{A}BC + D)(\bar{A}D + B\bar{C})$$

$$Y = \bar{A}\bar{A}BCD + \bar{A}DD + \bar{A}BC\bar{B}\bar{C} + B\bar{C}D$$

$$[\because\ X \cdot \bar{X} = 0 \text{ and } X \cdot X = X]$$

$$Y = \bar{A}BCD + \bar{A}D + B\bar{C}D$$

$$Y = \bar{A}D(BC + 1) + B\bar{C}D$$

$$[\because\ 1 + BC = 1]$$

$$Y = \bar{A}D + B\bar{C}D$$

Hence, the correct option is (A).

(A) $\bar{A}D + B\bar{C}D$

Key Points

- Two important laws of Boolean algebra:
  1. 1+ Any literal = 1
  2. 0+ Any literal = Any literal

Q2.03.1 MCQ | 2003 | 2M | Ans: B | ● ● ○

Given:

$$f = \bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z + XY\bar{Z} + X\bar{Y}Z + XYZ$$

K-map of function f in SOP form is,

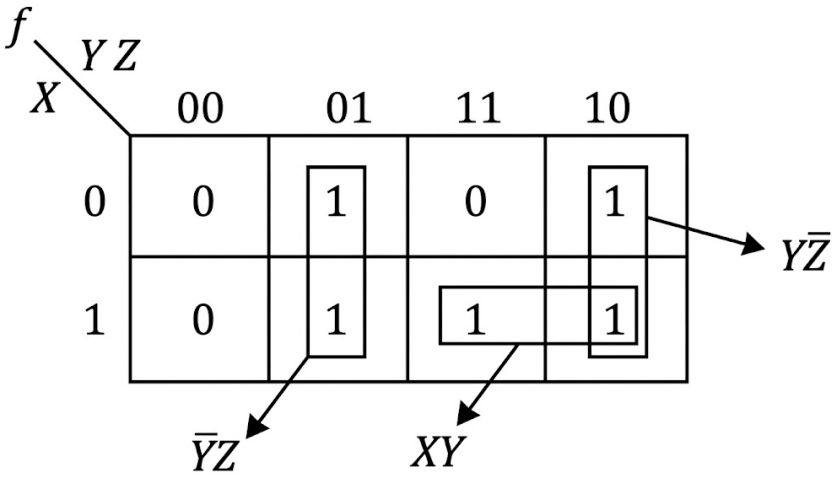


Fig. Solution Q2.03.1

The simplified form is

$$f = XY + \bar{Y}Z + Y\bar{Z}$$

Hence, the correct option is (B).

(B) $XY + \bar{Y}Z + Y\bar{Z}$

Q2.00.1 MCQ | 2000 | 2M | Ans: A | ● ● ○

Given K-map is shown below,

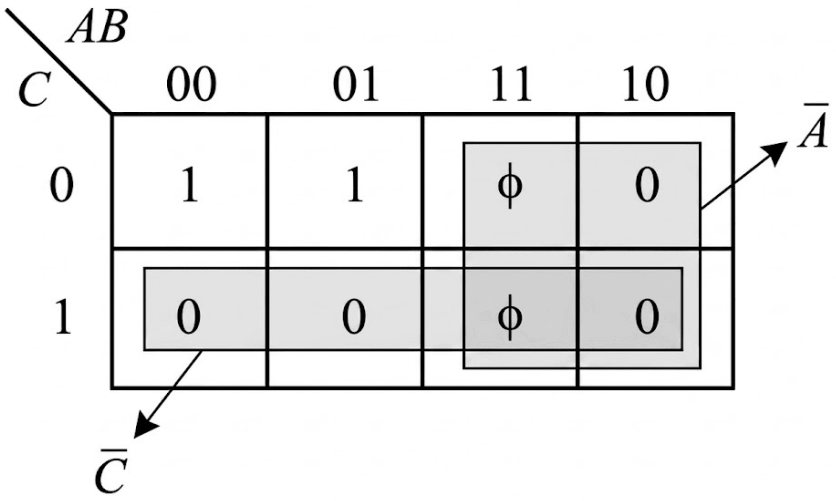


Fig. Solution Q2.00.1

Since 0's are grouped, so it is a case of POS form.
Thus, $f = \bar{A} \cdot \bar{C}$

(A) The minimised POS expression is $f = \bar{A} \cdot \bar{C}$.

PrepFusion

SOLUTIONS

III

Number Systems

Topics

1. Introduction to Number Systems
2. Arithmetic Operations on Numbers
3. Binary Codes and Their Operations
4. Signed Number Representations
5. Complement Arithmetic
6. Floating-Point Representation

Unit Snapshot*

Stream: EE	**Questions:** 2	**MCQ:** 2	**MSQ:** 0	**NAT:** 0
1M: 2	**2M:** 0	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 0M(2014)–0M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q3.14.1 MCQ 2014 1M Ans: D ● ○ ○

Checking each option against valid BCD states (0–9 only):

Options	8	4	2	1	Decimal	BCD
Option (A)	1	0	0	0	8	Valid
Option (B)	1	0	0	1	9	Valid
Option (C)	0	0	1	1	3	Valid
Option (D)	1	1	0	0	12	Invalid

BCD counter counts from 0 to 9 only; therefore from option (D), i.e. 1100 whose decimal equivalent is 12, is an invalid state.

Key Points

- BCD is an abbreviation for Binary Coded Decimal.
- BCD is a numeric code/weighted code in which each digit of a decimal number is represented by a separate group of bits.
- Examples of BCD code: 8-4-2-1 BCD code, 2-4-2-1 BCD code, 4-2-2-1 BCD code, 7-4-2-1 BCD code, etc.
- 8-4-2-1 code is a type of BCD code and is the most commonly used.

(D) 1100 (decimal 12) is an invalid BCD state.

Q3.07.1 MCQ 2007 1M Ans: B ● ○ ○

Given: HEX number $(AB.CD)_{16}$

Integer part → Octal:

To convert the integer part of the binary number into octal, form groups of three bits starting from the LSB:

$$\underbrace{010}_{2}\ \underbrace{101}_{5}\ \underbrace{011}_{3} \quad \Rightarrow \quad \text{Integer part (octal)} = 253$$

(A leading 0 is added to complete the leftmost group of 3.)

Fractional part → Octal:

To convert the fractional part into octal, form groups of three bits starting from the MSB:

$$\underbrace{110}_{6}\ \underbrace{011}_{3}\ \underbrace{010}_{2} \quad \Rightarrow \quad \text{Fractional part (octal)} = .632$$

(A trailing 0 is added to complete the rightmost group of 3.)

Thus, the octal equivalent is:

$$(253.632)_8$$

(B) The octal equivalent of $(AB.CD)_{16}$ is $(253.632)_8$.

SOLUTIONS

IV

Combinational Circuits

Topics

1. Introduction to Combinational Circuits
2. Code Converters and 7-Segment Display
3. Parity Generator and Checker
4. Magnitude Comparators
5. Multiplexers (MUX) and Demultiplexers (DEMUX)
6. Implementing Logical Expressions Using MUX
7. Decoders, Encoders and Priority Encoders
8. Half Adder and Full Adder
9. Delay-Based Problems in Adders
10. Half Subtractor and Full Subtractor
11. Binary Parallel Adder with Delay-Based Problems
12. Look-Ahead Carry Adder
13. Binary Parallel Subtractor
14. Adder and Subtractor Circuits of Different Binary Codes
15. Binary Multiplier

Unit Snapshot*

Stream: EE	**Questions:** 13	**MCQ:** 12	**MSQ:** 0	**NAT:** 0
1M: 6	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 1M(2024)–2M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q4.24.1 MCQ | 2024 | 1M | Ans: B ● ○ ○

Given : $F(X,Y) = X\bar{Y} + \bar{X}$

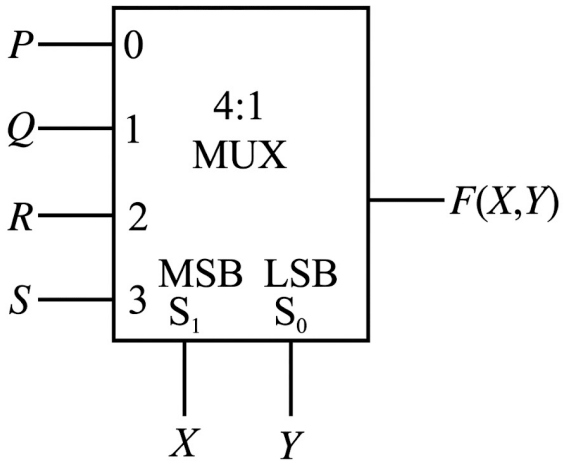


Fig. Solution Q4.24.1

Truth table is shown below,

$$F = X\bar{Y} + \bar{X} \qquad \ldots (i)$$

The output of 4×1 Mux is given by,

$$F = \bar{S}_1\bar{S}_0I_0 + \bar{S}_1S_0I_1 + S_1\bar{S}_0I_2 + S_1S_0I_3$$

$$F = \bar{X}\bar{Y}I_0 + \bar{X}YI_1 + X\bar{Y}I_2 + XYI_3$$

$$F = \bar{X}\bar{Y}P + \bar{X}YQ + X\bar{Y}R + XYS \qquad \ldots (ii)$$

Comparing equation (i) and (ii),

$$P = 1,\ Q = 1,\ R = 1,\ S = 0$$

Key Points

- For a $2^n \times 1$ MUX with n select lines, any function of n variables can be implemented by appropriately driving the data inputs with 0 or 1.
- Expand the Boolean function into minterms, then match each minterm to the corresponding MUX input.

Hence, the correct option is (B).

(B) 1110.

Q4.16.1 MCQ | 2016 | 1M | Ans: D ● ○ ○

Given: A circuit using a 2×1 MUX as shown.

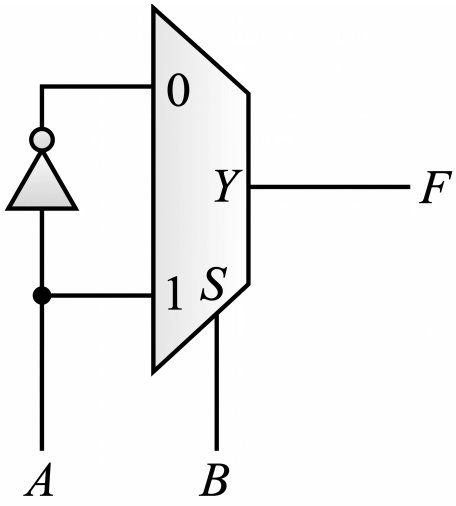


Fig. Solution Q4.16.1

From the circuit, the 2×1 MUX has:

$$S = A, \quad I_0 = \bar{A}B = AB \text{ (through AND with } B), \quad I_1 = \bar{A}B$$

The output of a 2×1 MUX is given by:

$$F = \bar{S} \cdot I_0 + S \cdot I_1$$

From the figure, with $S = B$, $I_0 = \bar{A}$, $I_1 = A$:

$$F = \bar{B} \cdot \bar{A} + B \cdot A = \overline{A \oplus B} = A \odot B$$

The output of the above circuit:

$$Y = AB + \bar{A}\bar{B}$$

This is the XNOR function, which is the complement of XOR:

$$\boxed{F = \overline{A \oplus B}}$$

Key Points

- XNOR gate output is 1 only if both inputs are equal; hence it is also called an **equality detector**.
- $A \odot B = AB + \bar{A}\bar{B} = \overline{A \oplus B}$

Hence, the correct option is (D).

$$\boxed{\text{(D) } \overline{A + B}}$$

Q4.15.1 MCQ | 2015 | 1M | Ans: B | ● ○ ○

Given 4×1 MUX is shown below,

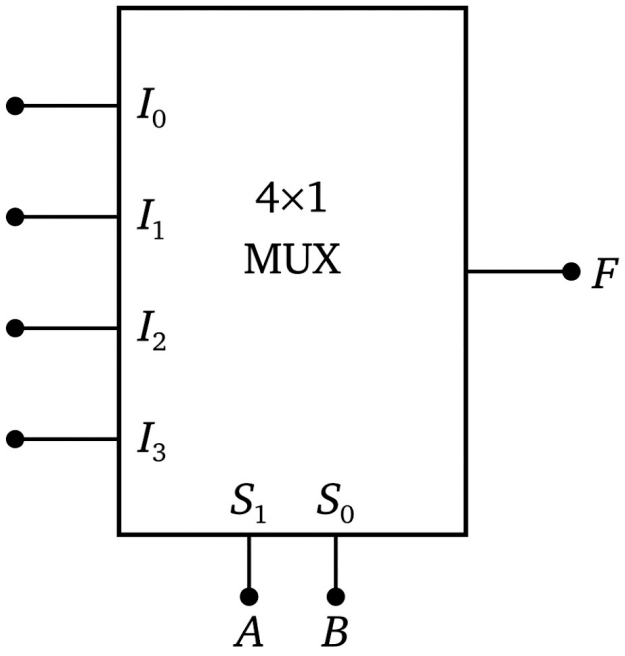


Fig. Solution Q4.15.1

$$F = A \oplus B$$

$$F = \bar{A}B + A\bar{B} \qquad \ldots (i)$$

The output of 4×1 MUX is given by,

$$F = \bar{S}_1\bar{S}_0I_0 + \bar{S}_1S_0I_1 + S_1\bar{S}_0I_2 + S_1S_0I_3$$

$$F = \bar{A}\bar{B}I_0 + \bar{A}BI_1 + A\bar{B}I_2 + ABI_3 \qquad \ldots (ii)$$

Comparing equation (i) and (ii),

$$I_0 = 0,\ I_1 = 1,\ I_2 = 1,\ I_3 = 0$$

Hence, the correct option is (B).

(B) 0110.

Q4.15.2 MCQ 2015 2M Ans: B ●●○

Given 8×1 MUX is shown below,

$$f(A, B, C, D) = \Pi M(1, 5, 12, 15)$$
$$= \Sigma m(0, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14)$$

To implement this, SOP expression using 8×1 MUX is,

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\overline{D}$	⓪	②	④	⑥	⑧	⑩	12	⑭
D	1	③	5	⑦	⑨	⑪	⑬	15
	$\overline{D}$	1	$\overline{D}$	1	1	1	D	$\overline{D}$

Fig. Solution Q4.15.2

Thus, the required connection is shown below,

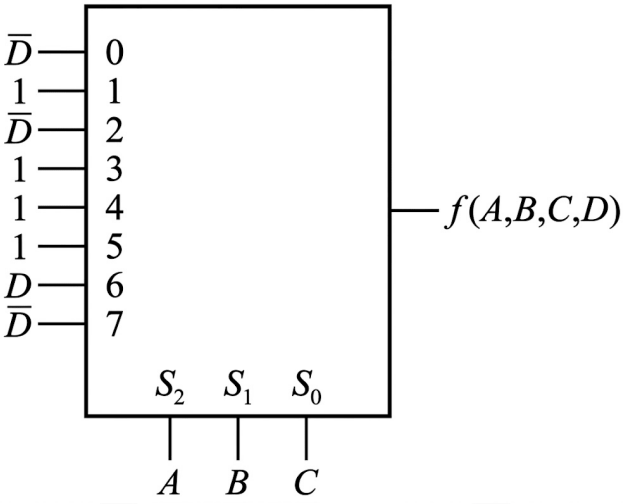


Fig. Solution Q4.15.2

Hence, the correct option is (B).

(B) $\bar{D}$, 1, $\bar{D}$, 1, 1, 1, D, $\bar{D}$.

Q4.14.1 MCQ | 2014 | 2M | Ans: A ● ● ○

Given circuit is shown below,

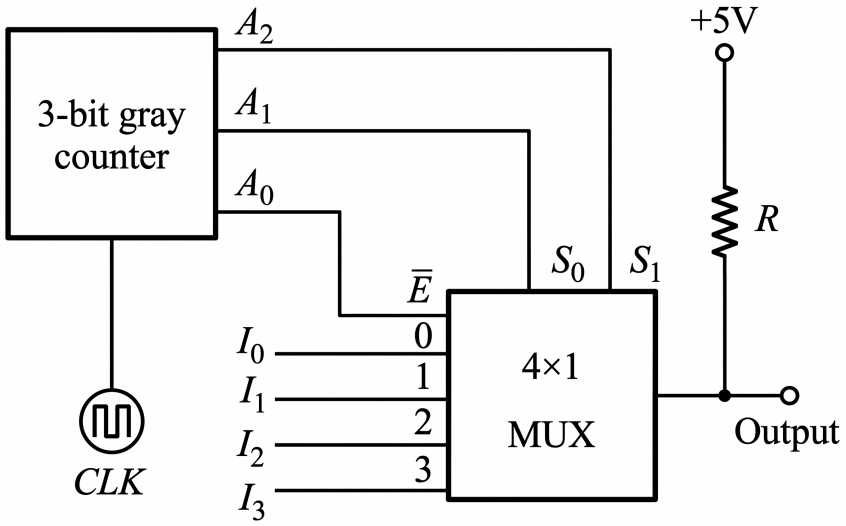


Fig. Solution Q4.14.1

The 4×1 MUX in the given circuit is active low enable, since $\bar{E}$ is present.

(i) For $A_0 = 0$, the 4×1 MUX will be enabled.

(ii) For $A_0 = 1$, the 4×1 MUX will be disabled.

The initial state of the counter (i.e. 3 bit gray counter) is 000.
Truth table for gray code counter is shown below,

A_2	A_1	A_0
0	0	0
0	0	1
0	1	1
0	1	0
1	1	0
1	1	1
1	0	1
1	0	0

When $E = 1(\bar{E} = 0)$, MUX is not enable. Therefore, output of MUX is NOT connected to any input of MUX (i.e. I_0, I_1, I_2 and I_3). Hence, final Output of circuit is connected to +5 V i.e. logic high.

A_2	A_1	A_0	$\bar{E} = A_0$	$S_1 = A_2$	$S_0 = A_1$	F
0	0	0	0	0	0	I_0
0	0	1	1	0	0	1
0	1	1	1	0	1	1
0	1	0	0	0	1	I_1
1	1	0	0	1	1	I_3
1	1	1	1	1	1	1
1	0	1	1	1	0	1
1	0	0	0	1	0	I_2

Key Points

- Gray code changes only 1 bit per step, reducing glitches in MUX-based circuits.
- When the enable pin $\bar{E} = 1$ (active low disabled), the MUX output is determined by the pull-up/pull-down, not the selected input.

Hence, the correct option is (A).

(A) I_0, 1, 1, I_1, I_3, 1, 1, I_2.

Q4.12.1 MCQ 2012 1M Ans: B ● ○ ○

Given: A 2-bit comparator with output $Y = 1$ when 2-bit input $A > B$.
Let $A = A_1A_0$ and $B = B_1B_0$. Total combinations = $2^2 \times 2^2 = 16$.
The truth table for $A > B$:

A_1	A_0	B_1	B_0	$Y = A > B$
0	0	0	1	0
0	1	0	0	1
1	0	0	0	1
1	0	0	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1

Counting all pairs where $A > B$: the number of such combinations equals:

$$\frac{2^n(2^n - 1)}{2} = \frac{4 \times 3}{2} = 6$$

6

Key Points

- For an n-bit comparator, total combinations = 2^{2n}.
- Combinations where $A = B$: 2^n.
- Combinations where $A > B$ (or $A < B$): $\frac{2^{2n} - 2^n}{2} = \frac{2^n(2^n - 1)}{2}$.

Hence, the correct option is (B).

(B) 6.

Q4.08.1 MCQ 2008 2M Ans: A ● ● ○

The **bubbled (negative) OR gate** in the given circuit is equivalent to **NAND gate**.
The equivalent circuit will be shown below,

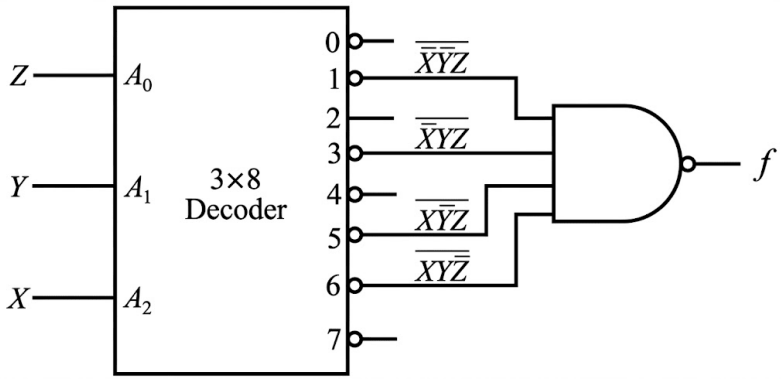


Fig. Solution Q4.08.1

$$f = \overline{\overline{\bar{X}\bar{Y}Z} \cdot \overline{\bar{X}YZ} \cdot \overline{X\bar{Y}Z} \cdot \overline{XY\bar{Z}}}$$

$$f = \bar{X}\bar{Y}Z + \bar{X}YZ + X\bar{Y}Z + XY\bar{Z}$$

The min-terms expression is given by,

$$f(X, Y, Z) = \Sigma m(1, 3, 5, 6)$$

Therefore, the max-terms expression will be,

$$f(X, Y, Z) = \Pi M(0, 2, 4, 7)$$

K-map of function f in POS form is,

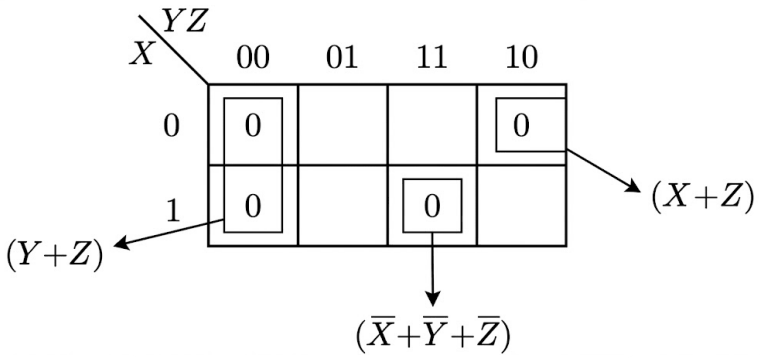


Fig. Solution Q4.08.1

From K-map, function $f(X, Y, Z)$ is given by,

$$f(X, Y, Z) = (X + Z)(\bar{X} + \bar{Y} + \bar{Z})(Y + Z)$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } (X + Z) \cdot (\bar{X} + \bar{Y} + \bar{Z}) \cdot (Y + Z).}$$

Q4.06.1 MCQ | 2006 | 2M | Ans: A ● ○ ○

The output of 4×1 MUX is given by,

$$F(A, B, C) = \bar{S}_1\bar{S}_0I_0 + \bar{S}_1S_0I_1 + S_1\bar{S}_0I_2 + S_1S_0I_3 \qquad \ldots (i)$$

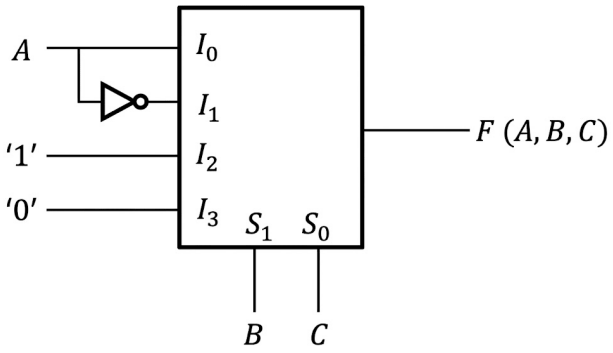


Fig. Solution Q4.06.1

Selection lines are : $S_1 = B,\ S_0 = C$
Inputs are : $I_0 = A,\ I_1 = \bar{A},\ I_2 = 1,\ I_3 = 0$
From equation (i),

$$F(A, B, C) = \bar{B}\bar{C}A + \bar{B}C\bar{A} + B\bar{C} \cdot 1 + BC \cdot 0$$

$$F(A, B, C) = A\bar{B}\bar{C} + \bar{A}\bar{B}C + (A + \bar{A})B\bar{C}$$

$$\left[\because A + \bar{A} = 1\right]$$

$$F(A, B, C) = \underset{(100)}{A\bar{B}\bar{C}} + \underset{(001)}{\bar{A}\bar{B}C} + \underset{(110)}{AB\bar{C}} + \underset{(010)}{\bar{A}B\bar{C}} \qquad \ldots (ii)$$

From the above SOP expression, min-terms are given below,

$$F(A, B, C) = \Sigma m(1, 2, 4, 6)$$

Key Points

- A 4×1 MUX with 2 select lines can implement any 3-variable Boolean function by using the third variable (or its complement) as data inputs.
- Build the truth table by evaluating the selected input for each combination of select line values.

Hence, the correct option is (A).

(A) $F(A, B, C) = \sum m(1, 2, 4, 6).$

Q4.04.1 MCQ | 2004 | 2M | Ans: D ● ○ ○

Given: A digital comparator circuit using XOR gates for 4-bit numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$. Output $Y = 0$ is required.

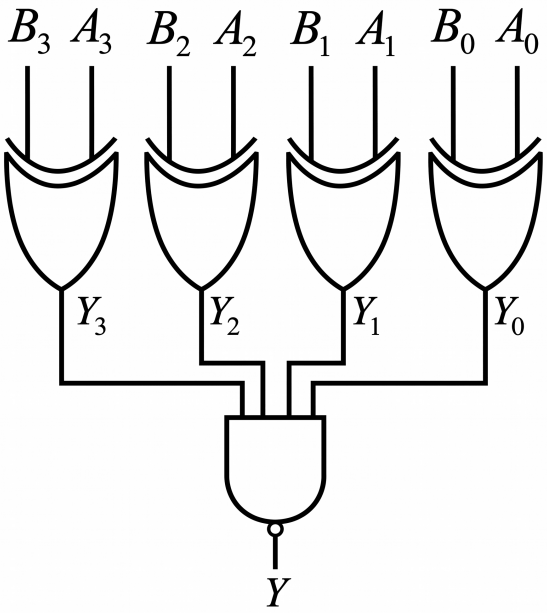


Fig. Solution Q4.04.1

From the circuit, each XOR gate compares corresponding bits:

$$Y_i = A_i \oplus B_i$$

The final output is the NAND of all XOR outputs:

$$Y = \overline{Y_3 \cdot Y_2 \cdot Y_1 \cdot Y_0}$$

For $Y = 0$, the NAND output must be 0, which requires all inputs to the NAND to be 1:

$$Y_3 = Y_2 = Y_1 = Y_0 = 1 \implies A_i \neq B_i \text{ for all } i$$

Checking option (D): $A = 1010$, $B = 0101$:

$$A_3 \oplus B_3 = 1 \oplus 0 = 1, \quad A_2 \oplus B_2 = 0 \oplus 1 = 1, \quad A_1 \oplus B_1 = 1 \oplus 0 = 1, \quad A_0 \oplus B_0 = 0 \oplus 1 = 1$$

$$Y = \overline{1 \cdot 1 \cdot 1 \cdot 1} = \overline{1} = 0 \quad \checkmark$$

Options (A), (B), (C) have $A = B$, so all XOR outputs are 0, giving $Y = \overline{0} = 1 \neq 0$.

$$\boxed{A = 1010,\ B = 0101}$$

Key Points

- An XOR gate acts as a 1-bit inequality detector: output is 1 when inputs differ.
- A NAND gate following the XOR chain produces 0 only when all bits differ between A and B.

Hence, the correct option is (D).

$$\boxed{\text{(D) } 1010,\ 0101.}$$

Q4.03.1 MCQ | 2003 | 1M | Ans: C

Given: A 4×1 MUX implementing the sum S of a 1-bit full adder with inputs P, Q and carry-in C_{in}. Select lines: $S_1 = P$, $S_0 = Q$.

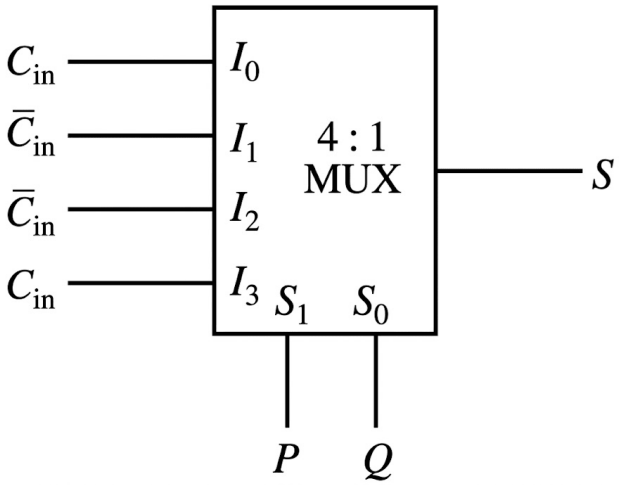


Fig. Solution Q4.03.1

The truth table for a 1-bit full adder sum:

P	Q	C_{in}	S
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$S(P, Q, C_{in}) = \sum m(1, 2, 4, 7)$$

The MUX output with select $S_1 = P$, $S_0 = Q$:

$$S = \bar{P}\bar{Q} \cdot I_0 + \bar{P}Q \cdot I_1 + P\bar{Q} \cdot I_2 + PQ \cdot I_3$$

Grouping by PQ:

$$\begin{aligned} PQ = 00: &\quad S = C_{in} \implies I_0 = C_{in} \\ PQ = 01: &\quad S = \bar{C}_{in} \implies I_1 = \bar{C}_{in} \\ PQ = 10: &\quad S = \bar{C}_{in} \implies I_2 = \bar{C}_{in} \\ PQ = 11: &\quad S = C_{in} \implies I_3 = C_{in} \end{aligned}$$

$$\boxed{I_0 = I_3 = C_{in}; \quad I_1 = I_2 = \bar{C}_{in}}$$

Key Points

- Full adder sum: $S = P \oplus Q \oplus C_{in}$.
- A 4×1 MUX with 2 select lines can implement any 3-variable function; use the third variable as data input.

- Full adder can also be built from two half adders and one OR gate.

Hence, the correct option is (C).

(C) $I_0 = I_3 = C_{in};\ I_1 = I_2 = \bar{C}_{in}.$

Q4.01.1 MCQ 2001 1M Ans: B ●●○

Given: A 4×1 MUX with select lines S_1, S_0 and data inputs as shown.

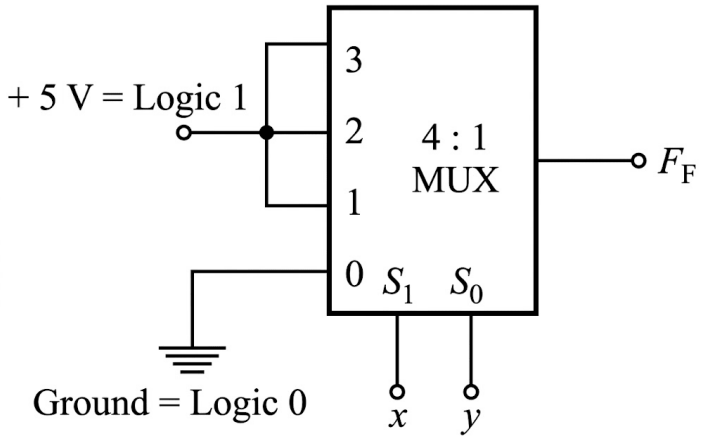


Fig. Solution Q4.01.1

Truth table is shown below,

x	y	Selected Input	F
0	0	I_0	0
0	1	I_1	1
1	0	I_2	1
1	1	I_3	1

The above truth table is equivalent to OR gate.
Therefore, $F = x + y$

Key Points

- Always expand MUX output using the standard formula and substitute the data input values from the circuit.
- Boolean simplification using K-map or algebraic methods can verify the final expression.

Hence, the correct option is (B).

(B) $x + y$.

Q4.99.1 Descriptive | 1999 | 2M | Ans: YES | ● ● ○

Given,

$$F = AC + ABD + ACD$$

Using absorption law,

$$AC + ACD = AC$$

Therefore,

$$F = AC + ABD$$

Since the select lines of the 8 : 1 MUX are D, C, A with A as LSB,

$$\text{Select Inputs} = (DCA)$$

Construct the truth table with respect to the select variables D, C, A.

D	C	A	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	B
1	1	0	0
1	1	1	1

Hence, the required inputs at terminals 0 to 7 are

MUX Input	Data Applied
I_0	0
I_1	0
I_2	0
I_3	1
I_4	0
I_5	B
I_6	0
I_7	1

Realization using a 4 : 1 **MUX**

Since the function contains four variables (A, B, C, D), one variable can be used as the data input while the remaining three variables determine the output.

After selecting two variables as select lines, the remaining variables can be connected through the data inputs.

Therefore, the function can also be realized using a 4 : 1 multiplexer.

YES

Key Points

- For MUX realization, first simplify the Boolean expression.
- Fix the select variables and express the output in terms of the remaining variables.
- For an 8 : 1 MUX, three variables are used as select lines and the remaining variable appears in the data inputs.
- Any 4-variable Boolean function can be realized using a 4 : 1 MUX by appropriately choosing select lines and data

inputs.

Q4.98.1 MCQ | 1998 | 2M | Ans: A

Given :

(A) $F = \bar{X} + \bar{Y}$

By De-Morgan law,

$$F = \overline{XY} \text{ i.e. NAND}$$

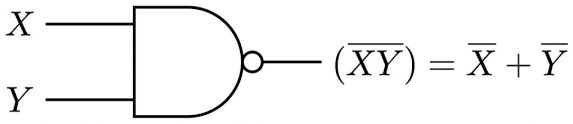


Fig. Solution Q4.98.1

$$\text{I} \rightarrow \text{Q}$$

(B) $F = XY$ i.e. carry of half adder.

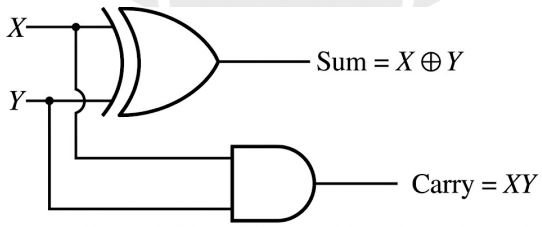


Fig. Solution Q4.98.1

$$\text{II} \rightarrow \text{R}$$

(C) $F = \bar{X}\bar{Y}$

By De-Morgan law,

$$F = \overline{X + Y} \text{ i.e. NOR}$$

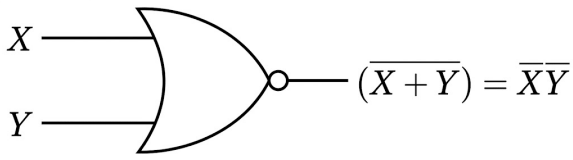


Fig. Solution Q4.98.1

$$\text{III} \rightarrow \text{S}$$

Hence, the correct option is (A).

(A) I-Q, II-R, III-S.

SOLUTIONS

V

Sequential Circuits

Topics

1. Latches — SR, JK, D and T Types, Their Drawbacks and Gated Latches
2. Flip-Flops — Introduction, Types, Conversions, Race Around Condition and Master-Slave Configuration
3. Asynchronous Counters — Binary, Non-Binary (Up/Down) and Output Decoding
4. Synchronous Counters — Propagation Delay, Setup/Hold Time, Critical Path Delay, Clear and Preset Input
5. Shift Registers and Applications — Ring Counter, Johnson Counter and Cascading
6. FSM and Sequence Detectors — Design, Generation, Detection and Overlapping/Non-Overlapping Types
7. Static Timing Analysis — Setup/Hold Time, Clock Constraints, Critical Path and Timing Violations

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 34	**MCQ:** 26	**MSQ:** 0	**NAT:** 7
1M: 16	**2M:** 15	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2024)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q5.24.1 MCQ | 2024 | 1M | Ans: C

Given: Present value of $Z = 1$. The circuit contains a D flip-flop with combinational logic $S = X \oplus Y \oplus Z$, where $X = 1$ and $Y = 0$.

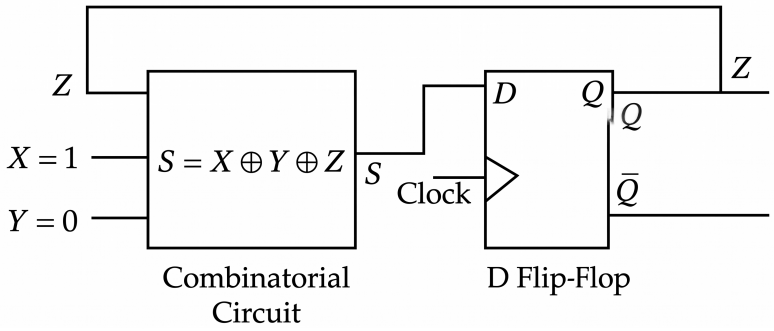


Fig. Solution Q5.24.1

1. State Before the Clock Edge:

Initially, the present state is $Z = 1$. The combinational logic computes the value of S (which feeds directly into the D input):

$$S = X \oplus Y \oplus Z$$

$$S = 1 \oplus 0 \oplus 1 = 0$$

Therefore, right before the clock edge arrives, $D = S = 0$.

2. State Immediately After the Clock Edge:

When the clock edge is applied, the D flip-flop latches the input value D to the output Q (which is Z):

$$Z_{new} = D = 0$$

3. Instantaneous Combinational Update:

Since we neglect the delay in the combinational circuit, the change in Z propagates immediately to update the value of S:

$$S_{new} = X \oplus Y \oplus Z_{new}$$

$$S_{new} = 1 \oplus 0 \oplus 0 = 1$$

Thus, after the application of the clock, the stable values are $S = 1$ and $Z = 0$.

Key Points

- In a synchronous circuit, the flip-flop samples its input precisely at the active clock edge based on the *present* state.
- Once the clock edge transitions the flip-flop to its *next* state, any zero-delay combinational logic connected to the output updates instantaneously.

Hence, the correct option is (C).

$$\boxed{\text{(C) } S = 1, Z = 0}$$

Q5.23.1 NAT | 2023 | 2M | Ans: 8 | ● ● ○

Given: Sequential circuit containing two D flip-flops labeled Q_2 (left) and Q_1 (right), a NOR gate configured as an inverter, and an XOR gate.

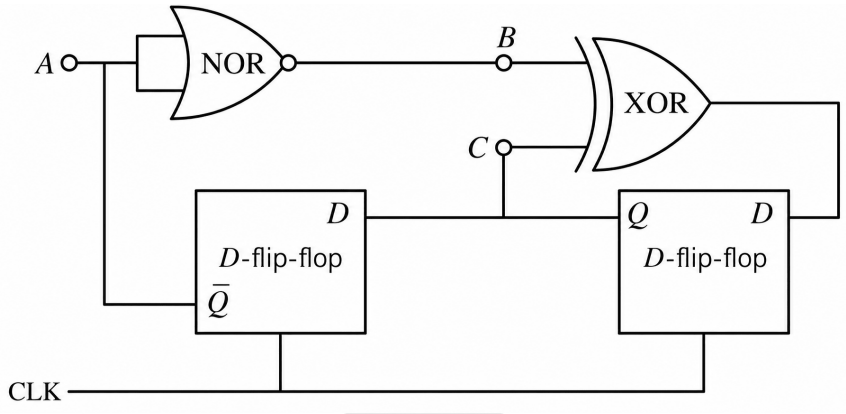


Fig. Solution Q5.23.1

1. Expressing Node Logic Equations:

From the circuit topology, the combinational nodes are expressed in terms of the flip-flop states as follows:

- Node A is connected to the inverted output of the left flip-flop: $A = \overline{Q}_2$.
- Node B is the output of the NOR inverter fed by A: $B = \overline{A} = Q_2$.
- Node C is the direct output of the left flip-flop: $C = Q_2$.
- Node D is the output of the XOR gate: $D = B \oplus C = Q_2 \oplus Q_1$.

2. Setting Up Flip-Flop Input Equations:

Looking closely at the routing to the D inputs of each flip-flop:

- The left flip-flop input is driven by C: $D_2 = Q_1$.
- The right flip-flop input is driven by D: $D_1 = D = Q_2 \oplus Q_1$.

3. Verifying the Stability (Fixed-Point) Condition:

For a state to not change with the application of a clock pulse, the next-state equations must satisfy $Q_{2,\text{next}} = Q_2$ and $Q_{1,\text{next}} = Q_1$:

$$Q_{2,\text{next}} = D_2 = Q_1 \implies Q_2 = Q_1$$

$$Q_{1,\text{next}} = D_1 = Q_2 \oplus Q_1 \implies Q_1 = Q_2 \oplus Q_1$$

Substituting $Q_2 = Q_1$ into the second condition:

$$Q_1 = Q_1 \oplus Q_1 \implies Q_1 = 0$$

Since $Q_2 = Q_1$, we find that the stable internal state is:

$$Q_2 Q_1 = 00$$

4. Finding the Decimal Equivalent of $[ABCD]$:

Substitute the stable state values ($Q_2 = 0, Q_1 = 0$) back into the sequence variables:

- $A = \overline{Q}_2 = \overline{0} = 1$
- $B = Q_2 = 0$
- $C = Q_1 = 0$
- $D = B \oplus C = 0 \oplus 0 = 0$

This yields the binary sequence:

$$[ABCD] = 1000_2$$

Converting this binary value to decimal:

$$1000_2 = 1 \times 2^3 = 8$$

Key Points

- A sequential system is in a stable state (fixed point) when the values present at the D inputs match the current outputs (Q) before the clock edge arrives.
- Correctly mapping the layout connections avoids false contradictions when setting up next-state balance equations.

$$\boxed{8}$$

Q5.22.1 NAT | 2022 | 1M | Ans: 10 | ● ○ ○

Given: MOD-2 and MOD-5 up-counters combined together.
When two counters are **cascaded** (combined), the overall modulus is the **product** of the individual moduli:

$$\text{MOD}_{\text{total}} = \text{MOD}_1 \times \text{MOD}_2 = 2 \times 5 = \boxed{10}$$

Key Points

- Cascading two counters of MOD-M and MOD-N gives an overall MOD-$(M \times N)$ counter.
- The clock output of the first counter drives the clock input of the second counter.

$$\boxed{10}$$

Q5.22.2 NAT | 2022 | 1M | Ans: 12.5 | ● ○ ○

Given:

1. 4-stage ripple counter
2. Number of bits $n = 4$
3. Propagation delay of each flip-flop $t_{pd} = 20$ nsec
4. Maximum clock frequency of n-stage ripple counter,

$$f_{CLK} = \frac{1}{n \cdot t_{pd}} = \frac{1}{4 \times 20 \times 10^{-9}} = 12.5 \text{ MHz}$$

$$\boxed{f_{\max} = 12.5 \text{ MHz}}$$

Key Points

- In a ripple counter, each flip-flop clocks the next, so delays **add** in series.
- Maximum clock frequency = $\frac{1}{n \cdot t_{pd}}$ for an n-stage ripple counter.
- In a synchronous counter, all flip-flops are clocked simultaneously, so the delay is only t_{pd} (not $n \cdot t_{pd}$).

Hence, the answer is **12.5** MHz.

Q5.21.1 NAT 2021 2M Ans: 2.05 ●●○

Given: 16-bit counter $Q_{15}\ Q_{14}\ Q_{13}\ \ldots\ Q_1\ Q_0$.
There are 2^{16} combinations as:

$$\begin{cases} 0\,0\,0\ \ldots\ 0\,0 \\ 1\,1\,1\ \ldots\ 1\,1 \end{cases}$$

If we fix $Q_{15}\ Q_{14}$ bits then (Q_{13} to Q_0) will take 2^{14} combinations.

$$Y = Q_{15} + Q_{14}$$

Q_{15}	Q_{14}	$Q_{13} \ldots Q_1 Q_0$	Y
0	0	2^{14} combinations	$Y = 0$ (25%)
0	1	2^{14} combinations	$Y = 1$
1	0	2^{14} combinations	$Y = 1$
1	1	2^{14} combinations	$Y = 1$

So $Y = 0$ for 2^{14} times [25%] and $Y = 1$ for 3×2^{14} times [75%].

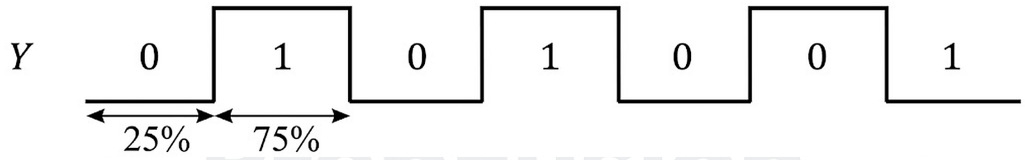


Fig. Solution Q5.21.1

Given that $Y = 1$ for 24 ms.
So $Y = 0$ for $\frac{24 \times 25}{75} = 8$ ms.
So total time period = 8 + 24 = 32 ms.
Time period of clock: $T_{clk} = \frac{32 \text{ ms}}{2^{16}}$

$$f_{clk} = \frac{2^{16}}{32 \times 10^{-3}}$$

$$\boxed{f_{clk} = 2.048 \text{ MHz} \approx 2.05 \text{ MHz}}$$

Key Points

- For a 16-bit counter, $Y = Q_{15} + Q_{14} = 0$ only for the lower $\frac{1}{4}$ of the count range (when both MSBs are 0).
- Y is high for $\frac{3}{4}$ of the total period: $\frac{3}{4} \times 2^{16} = 49152$ clock cycles.

$$\boxed{2.05}$$

Q5.21.2 MCQ | 2021 | 2M | Ans: C

Given Sequence: 000 → 001 → 011 → 010 → 110 → 111 → 101 → 100 → 000

Present State			Next State			Flip-Flop Input		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	1	0	1	1
0	1	1	0	1	0	0	1	0
0	1	0	1	1	0	1	1	0
1	1	0	1	1	1	1	1	1
1	1	1	1	0	1	1	0	1
1	0	1	1	0	0	1	0	0
1	0	0	0	0	0	0	0	0

K-map for D_1 ($D_1 = Q_1^+$):

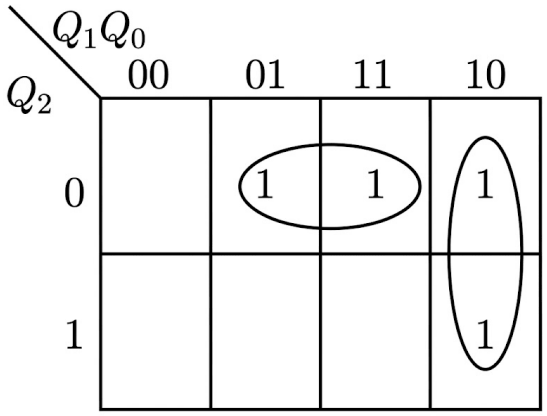


Fig. Solution Q5.21.2

From the K-map, grouping minterms m_1, m_3 ($\bar{Q}_2Q_0$) and m_2, m_6 ($Q_1\bar{Q}_0$):

$$D_1 = \bar{Q}_2Q_0 + Q_1\bar{Q}_0$$

$$\boxed{D_1 = \bar{Q}_2Q_0 + Q_1\bar{Q}_0}$$

Key Points

- For a D flip-flop, the excitation equation is simple: $D_i = Q_i^+$.
- Always ensure the state sequences are mapped sequentially from the given problem context. In this 3-bit Gray code sequence, the transition step after `010` is `110`.
- Grouping 1s across adjacent cells in the K-map minimizes the boolean logic expression directly into Sum-of-Products (SOP) form.

Hence, the correct answer is (C).

$$\boxed{\text{(C) } \bar{Q}_2Q_0 + Q_1\bar{Q}_0}$$

Q5.20.1 MCQ 2020 1M Ans: C ● ○ ○

Given: Input data is processed to detect the overlapping sequence 101 as shown below.

1	1	0	1	0	0	1	1	0	1	0	1	1	0

When sequence will be 101, then output will be 1 otherwise 0, so starting from first three bits, output bits for respective combination are as 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0.

Output sequence:

$$0,\ 1,\ 0,\ 0,\ 0,\ 0,\ 0,\ 1,\ 0,\ 1,\ 0,\ 0$$

Key Points

- In an **overlapping** detector, after detecting the target sequence, the tail bits are reused as potential start bits of the next occurrence.
- The output is asserted high at the clock cycle when the *last* bit of the sequence is received.

Hence, the correct option is (C).

(C) 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0

Q5.18.1 MCQ 2018 2M Ans: B ● ● ○

Given: Logical circuit is shown below.

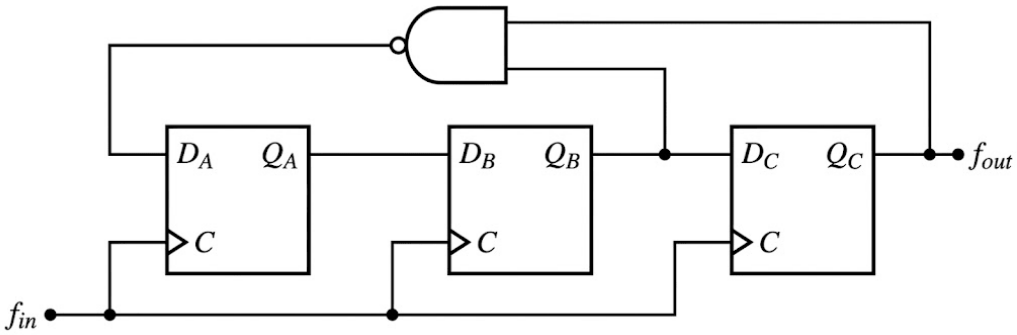


Fig. Solution Q5.18.1

From the above sequential circuit:

Present state			FF inputs			Next state		
Q_A	Q_B	Q_C	$D_A=\overline{Q_B \cdot Q_C}$	$D_B=Q_A$	$D_C=Q_B$	Q_A^+	Q_B^+	Q_C^+
0	0	0	1	0	0	1	0	0
1	0	0	1	1	0	1	1	0
1	1	0	1	1	1	1	1	1
1	1	1	0	1	1	0	1	1
0	1	1	0	0	1	0	0	1
0	0	1	1	0	0	1	0	0

To ensure complete rigor and eliminate any possibility of disjoint cycles, let us check the remaining unlisted possible states of the three flip-flops:

- Starting from state 010 (decimal 2):

$$010 \rightarrow 101 \rightarrow 110 \rightarrow 111 \rightarrow 011 \rightarrow 001 \rightarrow 100 \rightarrow \ldots$$

- Starting from state 101 (decimal 5):

$$101 \rightarrow 110 \rightarrow 111 \rightarrow 011 \rightarrow 001 \rightarrow 100 \rightarrow \ldots$$

This shows that both 010 and 101 are transient states that eventually merge into the same repeating sequence:

$$100 \rightarrow 110 \rightarrow 111 \rightarrow 011 \rightarrow 001 \rightarrow 100$$

Thus, there is only one stable, main cycle containing five states ($4 \rightarrow 6 \rightarrow 7 \rightarrow 3 \rightarrow 1 \rightarrow 4$), while the remaining states (0, 2, and 5) are transient.

State diagram:

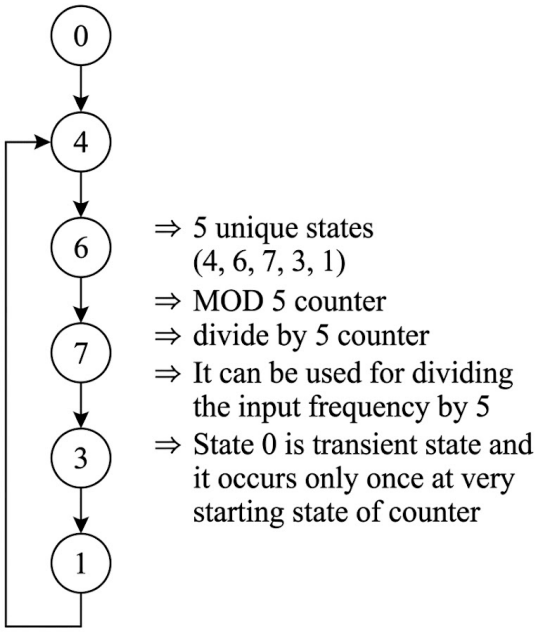


Fig. Solution Q5.18.1

PrepFusion

Consequently, the circuit behaves definitively as a MOD-5 counter and functions as a divide-by-5 frequency divider.

Key Points

- The modulus of a counter equals the number of distinct states in its main repeating cycle.
- If the counter has N states in its stable cycle, the output frequency = f_{clk}/N.
- Verification of all 2^n possible states confirms the absence of isolated disjoint loops.

Hence, the correct option is (B).

(B) It can be used for dividing the input frequency by 5.

Q5.17.1 NAT 2017 2M Ans: 6 ● ● ●

Given synchronous sequential circuit features two distinct shift register configurations driven by a common clock:

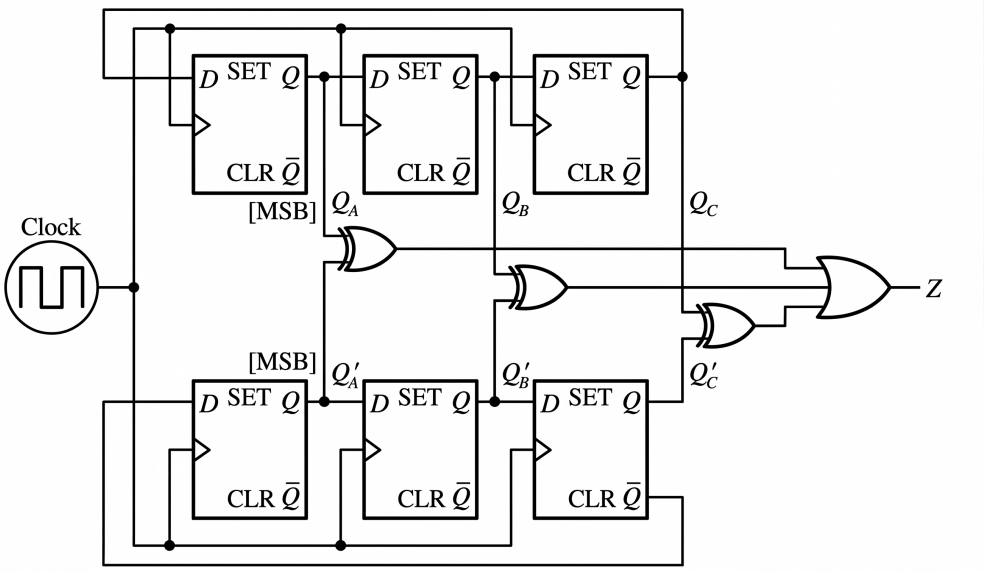


Fig. Solution Q5.17.1

1. Analyzing Register Configurations & Next-State Equations:

- **Top Register (Q_A, Q_B, Q_C):** The serial input to the first stage (D_A) is fed back directly from the true output of the last stage (Q_C). This forms a **Ring Counter** (period = 3).

$$Q_{A,n+1} = Q_{C,n}, \quad Q_{B,n+1} = Q_{A,n}, \quad Q_{C,n+1} = Q_{B,n}$$

- **Bottom Register (Q'_A, Q'_B, Q'_C):** The serial input to the first stage (D'_A) is fed back from the complementary output of the last stage ($\bar{Q}'_C$). This forms a **Johnson Counter / Twisted Ring Counter** (period = $2 \times 3 = 6$).

$$Q'_{A,n+1} = \bar{Q}'_{C,n}, \quad Q'_{B,n+1} = Q'_{A,n}, \quad Q'_{C,n+1} = Q'_{B,n}$$

2. Output Logic Expression: The combinational logic consists of three 2-input XOR gates feeding into a 3-input OR gate:

$$Z = (Q_A \oplus Q'_A) + (Q_B \oplus Q'_B) + (Q_C \oplus Q'_C)$$

For the output Z to be 0, the output of every individual XOR gate must be 0. This implies that the states of both counters must match bit-for-bit:

$$Z = 0 \iff (Q_A = Q'_A) \text{ AND } (Q_B = Q'_B) \text{ AND } (Q_C = Q'_C)$$

3. State Transition Table Tracking: Starting with the given initial conditions $Q_AQ_BQ_C = 100$ and $Q'_AQ'_BQ'_C = 100$:

Clock Cycle	Q_A	Q_B	Q_C	Q'_A	Q'_B	Q'_C	$Q_A \oplus Q'_A$	$Q_B \oplus Q'_B$	$Q_C \oplus Q'_C$	Output (Z)
Initially (0)	1	0	0	1	0	0	0	0	0	**0**
1	0	1	0	1	1	0	1	0	0	1
2	0	0	1	1	1	1	1	1	0	1
3	1	0	0	0	1	1	1	1	1	1
4	0	1	0	0	0	1	0	1	1	1
5	0	0	1	0	0	0	0	0	1	1
6	1	0	0	1	0	0	0	0	0	**0**

As shown in the state sequence trace, the states match identically again after precisely 6 clock cycles.

Key Points

- A 3-bit standard ring counter repeats its sequence every 3 clock cycles.
- A 3-bit Johnson counter repeats its sequence every $2N = 6$ clock cycles.
- The least common multiple (LCM) of their individual periods ($LCM(3, 6) = 6$) determines the minimum interval at which their initial aligned state configuration repeats.

Hence, the minimum number of clock cycles after which the output Z would again become zero is **6**.

6

Q5.16.1 MCQ 2016 2M Ans: C

Given: Given synchronous counter is shown below.

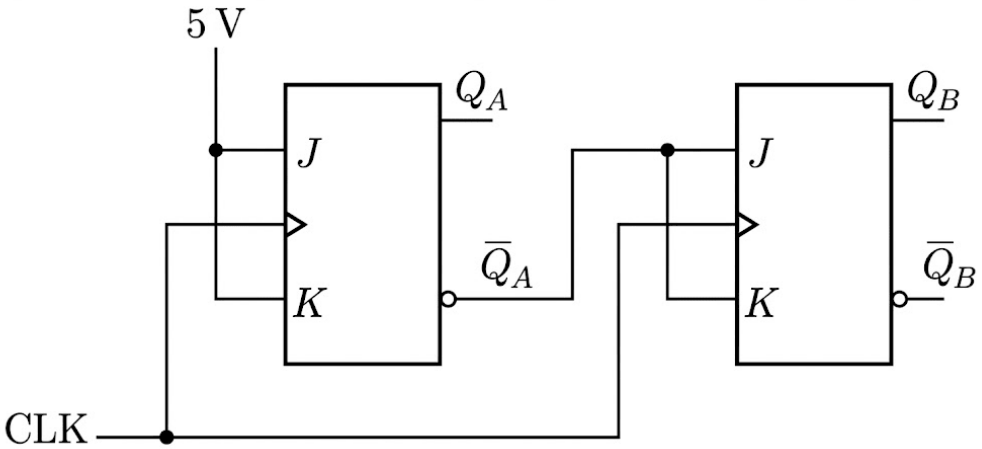


Fig. Solution Q5.16.1

From the connection schematic diagram:

- For Flip-Flop A: $J_A = K_A = 1$ (connected to +5 V, which represents Logic 1).
- For Flip-Flop B: $J_B = K_B = \bar{Q}_A$.

State Transitions Tracking: Given the initial/present state is $Q_AQ_B = 00$:

1. **Flip-Flop A inputs:** Since $J_A = K_A = 1$, Flip-Flop A operates in **Toggle mode**.

$$Q_A^+ = \bar{Q}_A = \bar{0} = 1$$

2. **Flip-Flop B inputs:** The inputs are driven synchronously by the present state value of $\bar{Q}_A$:

$$J_B = K_B = \bar{Q}_A = \bar{0} = 1$$

Since $J_B = K_B = 1$, Flip-Flop B also operates in **Toggle mode** during this clock cycle:

$$Q_B^+ = \bar{Q}_B = \bar{0} = 1$$

Summarizing this transition behavior in the state table:

Present State		Flip-Flop Inputs				Next State	
Q_A	Q_B	J_A	K_A	J_B	K_B	Q_A^+	Q_B^+
0	0	1	1	1	1	**1**	**1**

Therefore, the next state of the system is $Q_AQ_B = 11$.

Key Points

- In a synchronous circuit, all state-determining input equations are computed using the **present state** values before the clock edge takes effect.
- When both J and K control inputs are tied to logic high (1), the output toggles on the active clock transition ($Q_{n+1} = \bar{Q}_n$).

Hence, the correct option is (C).

(C) 11

Q5.15.1 MCQ | 2015 | 2M | Ans: B | ● ○ ○

Given: Initial state of Q_1Q_0 is 00.

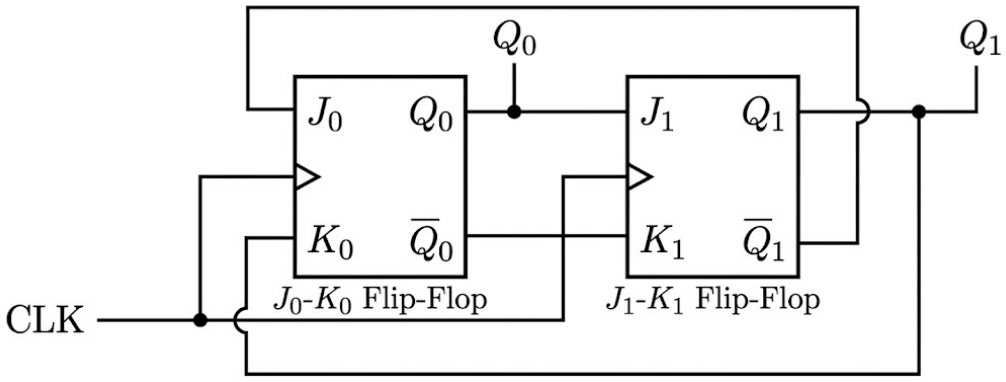


Fig. Solution Q5.15.1

From the above circuit:

$$J_1 = Q_0, \quad K_1 = \bar{Q}_0, \quad J_0 = \bar{Q}_1, \quad K_0 = Q_1$$

State table of the above circuit is shown below:

Present state		Flip flop Input				Next state	
Q_1	Q_0	$J_1=Q_0$	$K_1=\bar{Q}_0$	$J_0=\bar{Q}_1$	$K_0=Q_1$	Q_1^+	Q_0^+
0	0	0	1	1	0	0	1
0	1	1	0	1	0	1	1
1	1	1	0	0	1	1	0
1	0	0	1	0	1	0	0
0	0						← Repeated

Thus, state diagram of given circuit is:

$$00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00$$

The circuit has 4 distinct states (mod 4).
After every 4 clock pulses, the circuit goes back to the initial state $Q_1Q_0 = 00$.
Hence, after 332 (i.e. divisible by 4) clock pulse, $Q_1Q_0 = 00$ and after 333rd clock pulse $Q_1Q_0 = 01$.

Key Points

- Find the period of the counter sequence, then use modular arithmetic: state after n pulses = state after (n mod period) pulses.
- 333 mod 4 = 1, so the answer is the state after 1 clock pulse.

Hence, the correct option is (B).

(B) 01

Q5.15.2 NAT 2015 2M Ans: 6 ●●○

Given: Logic circuit is shown below,

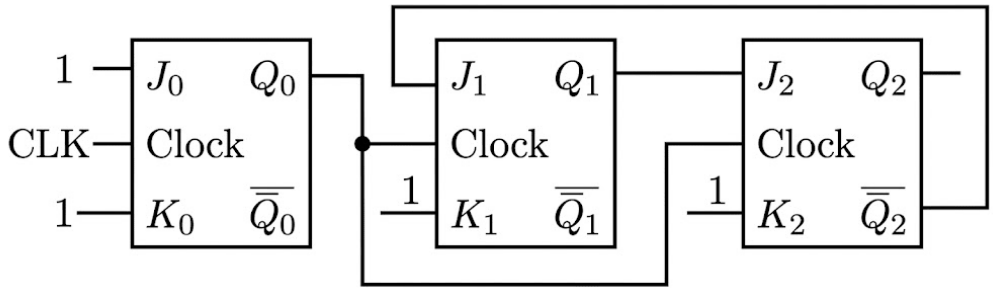


Fig. Solution Q5.15.2

As per the question, all the $J-K$ flip-flops are negative-edge triggered. To analyze this asynchronous circuit properly, we must track the specific triggering conditions for each stage:

- **FF0 Clock:** Driven directly by the main clock (CLK). Since $J_0 = K_0 = 1$, Q_0 toggles on every single falling edge (↓) of CLK.
- **FF1 & FF2 Clocks:** Both clock pins are tied directly to the output Q_0. Therefore, FF1 and FF2 can update **only** when Q_0 undergoes a falling edge transition ($1 \rightarrow 0$). On a rising edge of Q_0 ($0 \rightarrow 1$), they are not triggered and strictly hold their previous states regardless of input values.

Evaluating the circuit edge-by-edge starting from the initial state $Q_2Q_1Q_0 = 000$, we construct the true asynchronous state table, explicitly tracing the synchronous steering lines $J_1 = \bar{Q}_2$, $K_1 = 1$, $J_2 = Q_1$, and $K_2 = 1$:

Clock Pulse	Transition of Q_0	J_1	K_1	J_2	K_2	FF1 Action	FF2 Action	Next State ($Q_2Q_1Q_0$)
Initially	—	—	—	—	—	—	—	000
1st (↓ CLK)	0 → 1 (Rising Edge)	1	1	0	1	Hold State	Hold State	001
2nd (↓ CLK)	1 → 0 (**Falling Edge**)	1	1	0	1	Toggle	Reset	010
3rd (↓ CLK)	0 → 1 (Rising Edge)	1	1	1	1	Hold State	Hold State	011
4th (↓ CLK)	1 → 0 (**Falling Edge**)	1	1	1	1	Toggle	Toggle	100
5th (↓ CLK)	0 → 1 (Rising Edge)	0	1	0	1	Hold State	Hold State	101
6th (↓ CLK)	1 → 0 (**Falling Edge**)	0	1	0	1	Reset	Reset	000

From the state table, we can map out the corresponding state diagram:

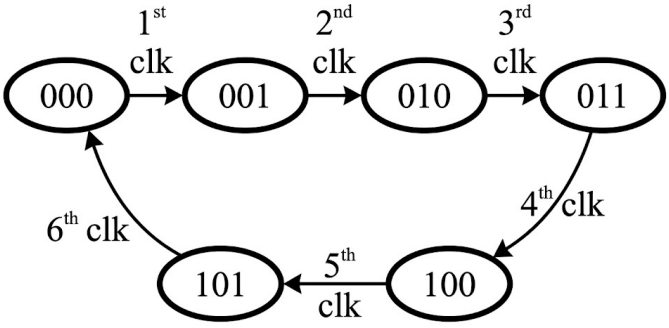


Fig. Solution Q5.15.2

Thus, the state vector $Q_2Q_1Q_0$ returns to its initial configuration after exactly 6 complete clock cycles of the main clock, confirming that it functions as a mod-6 counter sequence.

Detailed Behavioral Analysis:

(i) **First active trigger of secondary stages (CLK 2):** When Q_0 drops from $1 \rightarrow 0$, both FF1 and FF2 are clocked. Evaluating inputs based on the previous state (001):

- For FF1: $J_1 = \bar{Q}_2 = \bar{0} = 1, K_1 = 1 \implies$ **Toggle** $(0 \rightarrow 1)$.
- For FF2: $J_2 = Q_1 = 0, K_2 = 1 \implies$ **Reset** (Stays 0).

(ii) **Second active trigger of secondary stages (CLK 4):** When Q_0 drops from $1 \rightarrow 0$ again. Evaluating inputs based on the previous state (011):

- For FF1: $J_1 = \bar{Q}_2 = \bar{0} = 1, K_1 = 1 \implies$ **Toggle** $(1 \rightarrow 0)$.
- For FF2: $J_2 = Q_1 = 1, K_2 = 1 \implies$ **Toggle** $(0 \rightarrow 1)$.

(iii) **Third active trigger of secondary stages (CLK 6):** When Q_0 drops from $1 \rightarrow 0$ for the final time. Evaluating inputs based on the previous state (101):

- For FF1: $J_1 = \bar{Q}_2 = \bar{1} = 0, K_1 = 1 \implies$ **Reset** (Stays 0).
- For FF2: $J_2 = Q_1 = 0, K_2 = 1 \implies$ **Reset** $(1 \rightarrow 0)$.

$$\boxed{6}$$

Key Points

- Asynchronous clock cascades require careful evaluation of the specific falling edges (↓) generated by preceding state transitions.
- Next-state conditions must always evaluate the current logic level of the control steering lines (J, K) right before the active clock transition edge hits.

Hence, the answer is **6**.

Q5.14.1 MCQ | 2014 | 1M | Ans: D ● ● ○

Given: State diagram is shown below.

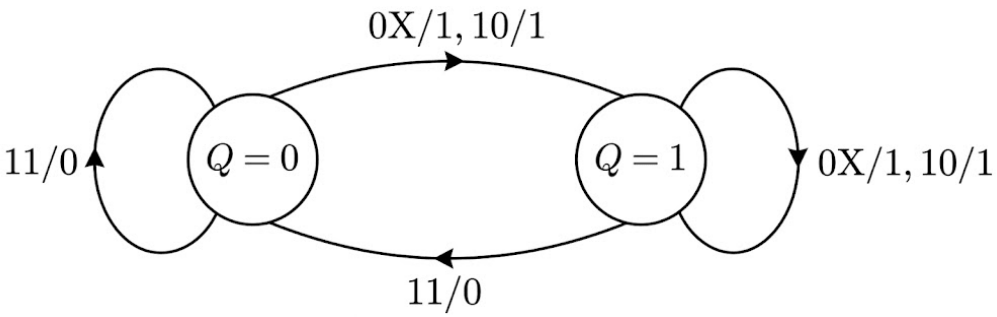


Fig. Solution Q5.14.1

If any one of the input is zero, output is logic '1' and if both inputs are '1', then output is logic '0'. Therefore, the state diagram represents the NAND gate.

Let A and B be the two inputs of the logic gate. From the state diagram,

$$\text{If } A = 1,\ B = 1 \quad \Rightarrow \quad Q = 0$$
$$\text{If } A = 0,\ B = X(0, 1) \quad \Rightarrow \quad Q = 1$$

i.e.

$$\left.\begin{matrix} A = 0,\ B = 0 \\ A = 0,\ B = 1 \end{matrix}\right\} \Rightarrow Q = 1$$
$$\text{If } A = 1,\ B = 0 \quad \Rightarrow \quad Q = 1$$

Thus, the truth table can be written as:

A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

Thus, the truth table represents NAND logic gate.

Key Points

- A logic circuit with output delay stores the previous input and outputs it one clock cycle later.
- Read the state transitions: the output pattern here matches a NAND gate.

Hence, the correct option is (D).

(D) NAND

Q5.14.2 MCQ 2014 1M Ans: C

Given:

(i) M_1 is a positive edge triggered monoshot multi-vibrator which generates a pulse of width T_1.

(ii) M_2 is a negative edge triggered monoshot multi-vibrator which generates a pulse of width T_2.

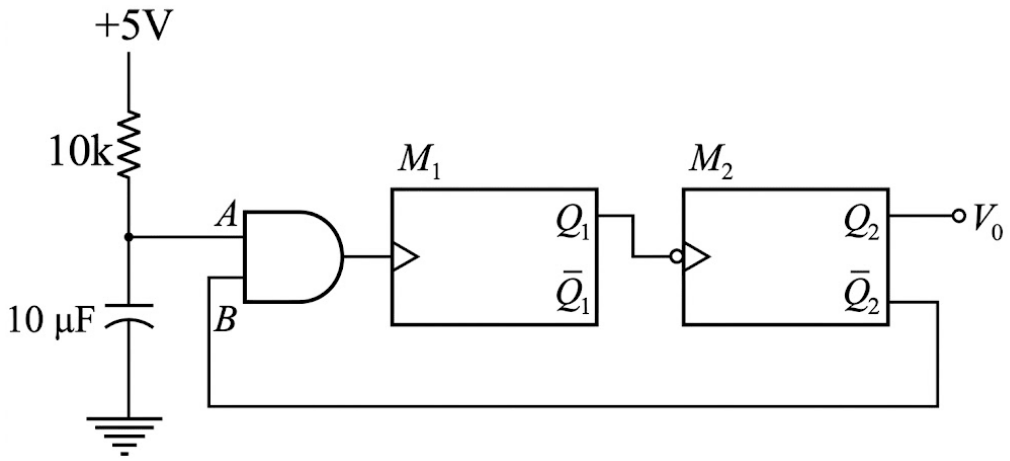


Fig. Solution Q5.14.2

(α) **Input A of AND gate** is always fixed at 1 because the capacitor is charged to +5 V, which represents a steady logic high level.

(α) Let initially $V_0 = 0 = Q_2$, which means the complementary output $\bar{Q}_2 = 1$. This value is fed directly back into **input B of the AND gate**. Since both inputs (A and B) of the AND gate are now 1, its output goes high. This transition activates the positive-edge-triggered monoshot M_1, causing it to generate a pulse on Q_1 of duration T_1.

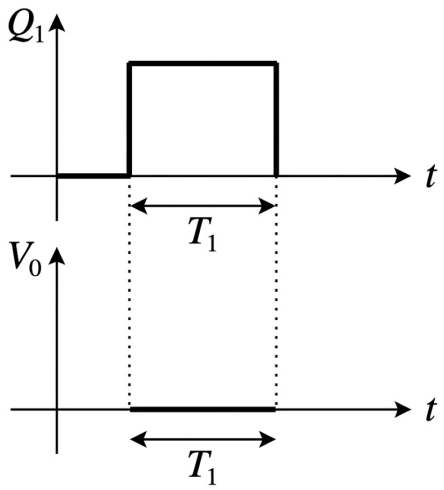


Fig. Solution Q5.14.2

For the duration of T_1, the output $V_0 = Q_2$ remains at 0 because M_2 responds exclusively to a negative-going edge.

(β) **When the output of Q_1 falls from 1 to 0** (i.e., at its negative-going edge), the monoshot M_2 is triggered. Upon activation, M_2 generates an output pulse on Q_2 (V_0) with a duration of T_2.

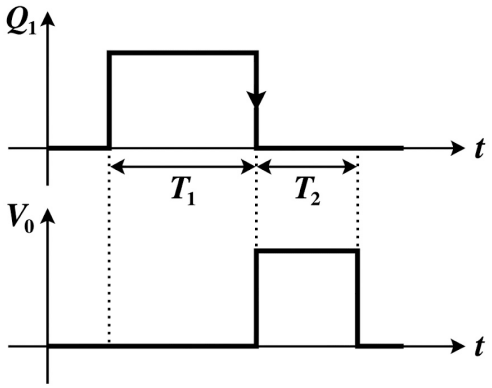


Fig. Solution Q5.14.2

(γ) **After the duration T_2 expires**, the output pulse drops, returning $V_0 = Q_2 = 0$. Consequently, $\bar{Q}_2$ transitions back to 1. This high signal is fed back into input B of the AND gate, restarting the entire cycle automatically.

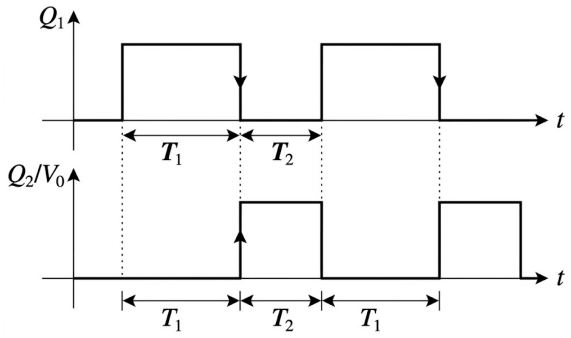


Fig. Solution Q5.14.2

The complete, repeating waveform sequence generated by this configuration matches perfectly with the characteristics described in **option (C)**.

Key Points

- Interlocking a positive-edge triggered monoshot with a negative-edge triggered monoshot in a closed feedback loop creates a self-sustaining, free-running astable multivibrator.
- The total periodic cycle time of the resulting clock output is equal to the sum of both individual monoshot pulse widths ($T_{total} = T_1 + T_2$).

Hence, the correct option is **(C)**.

(C) [Waveform with period $T_1 + T_2$, high for T_2 and low for T_1]

Q5.14.3 MCQ 2014 1M Ans: C ● ○ ○

Given: A cascade of three identical modulo 5 counters.
Overall modulus = $5 \times 5 \times 5 = 125$.

125

Key Point: For cascading of 'N' identical modulo of 'M' counters, then overall modulus = M^N.

Key Points

- Cascading n identical MOD-M counters gives overall MOD-M^n.
- This is because each stage divides by M, so the total division is $M \times M \times M = M^3$.

Hence, the correct option is (C).

(C) 125

Q5.14.4 MCQ 2014 2M Ans: B ● ○ ○

Key Point: To obtain J-K flip-flop from T flip-flop, following steps need to be taken:

1. Writing characteristic table of J-K flip-flop.
2. Obtaining excitation table for T flip-flop.

		Characteristic table of J-K flip-flop		Excitation table of T-flip-flop
J	K	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

K-map of function T in POS form is:

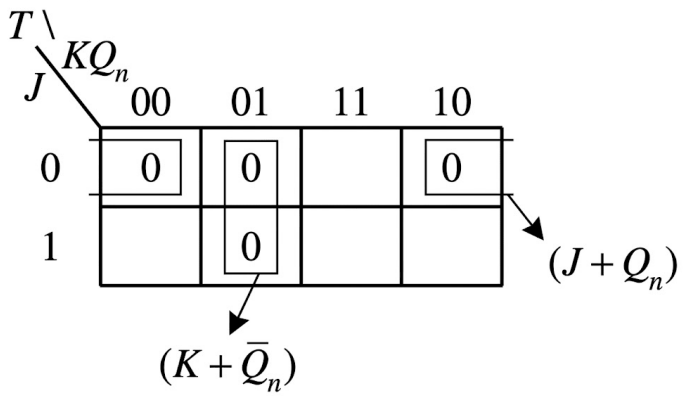


Fig. Solution Q5.14.4

Grouping: $(K + Q_n)$ from column, $(J + \bar{Q}_n)$ from rows.
Thus $T = (J + Q_n)(K + \bar{Q}_n)$.
Therefore from the above equation the J-K flip-flop can be implemented by T flip-flop as shown below.

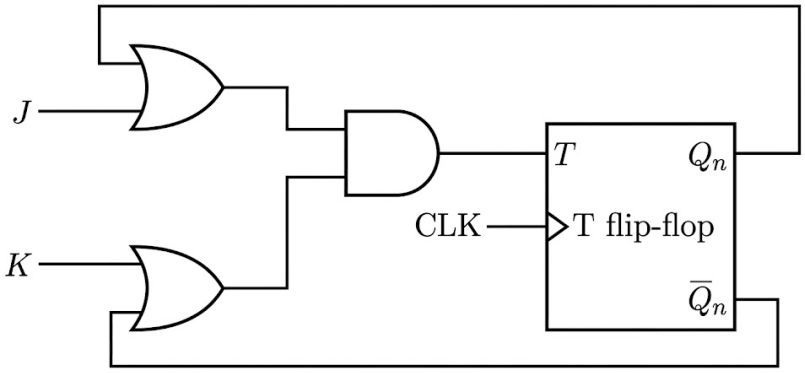


Fig. Solution Q5.14.4

Key Points

- For flip-flop conversion, equate the characteristic equations and solve for the excitation input of the target flip-flop.
- $T = J\bar{Q} + KQ$ is the standard conversion formula for JK-to-T.

Hence, the correct option is (B).

(B) [Circuit: $T = (J + Q_n)(K + \bar{Q}_n)$ implemented with T flip-flop]

Q5.13.1 MCQ | 2013 | 2M | Ans: B ● ● ○

Given: Clock frequency $f_{in} = 1$ kHz, and initial state $Q = 0$.

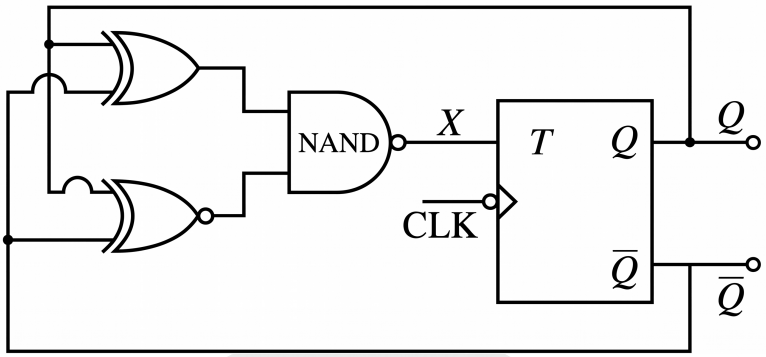


Fig. Solution Q5.13.1

1. Derivation of Combinational Logic Expression: To determine the input to the T flip-flop ($T = X$), we analyze the logic gates step-by-step using the complementary outputs Q and $\bar{Q}$:

- **XOR Gate Output:** The inputs are Q and $\bar{Q}$. Since one input is always the complement of the other, the output is:

$$Y_1 = Q \oplus \bar{Q} = 1$$

- **XNOR Gate Output:** The inputs are also Q and $\bar{Q}$. Since the inputs are always different, the output is:

$$Y_2 = Q \odot \bar{Q} = 0$$

- **NAND Gate Output (X):** The inputs to the NAND gate are the outputs of the XOR and XNOR gates (Y_1 and Y_2):

$$X = \overline{Y_1 \cdot Y_2} = \overline{1 \cdot 0} = \bar{0} = 1$$

Since $T = X$, the flip-flop input is permanently held at:

$$T = 1$$

This proves that the T flip-flop operates strictly in **toggle mode** for all clock events.

2. State Transition Table: Using $T = 1$, we trace the state transition sequence:

CLK	Present state Q	Flip-flop input $T = X$	Next state Q^+
1	0	1	1
2	1	1	0

The output sequence alternates indefinitely as:

$$0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow \ldots$$

Thus, the total number of distinct states in the repeating cycle is:

$$M = 2$$

This confirms that the circuit behaves as a **MOD-2 counter** (or a divide-by-2 frequency divider).

3. Output Frequency Calculation: The frequency of the output waveform Q is given by:

$$f_{out} = \frac{f_{in}}{M} = \frac{1\,\text{kHz}}{2} = 0.5\,\text{kHz}$$

Hence, the correct option is (B).

(B) 0.5

Key Points

- A T flip-flop with its input tied to a permanent high logic level ($T = 1$) remains constantly in toggle mode.
- Any sequential network forming a stable loop of 2 states acts as a MOD-2 counter, cutting the driving input clock frequency exactly in half ($f_{out} = f_{clk}/2$).

Q5.12.1 MCQ 2012 1M Ans: A

Given logic circuit is shown below,

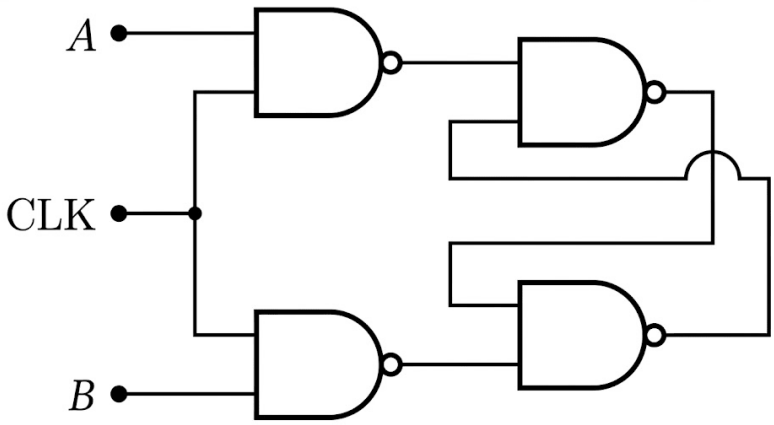


Fig. Solution Q5.12.1

Race around condition occurs in J-K flip-flop when both the inputs are "1" [i.e. $J = K = 1$] and flip-flops are level triggered.

Given circuit is a S-R flip-flop. Hence, race around condition never occurs.

Hence, the correct option is (A).

(A) does not occur.

Key Points

Race around condition :

- When the input to the J-K flip-flop is $J = 1$ and $K = 1$ (toggle mode), the race around condition occurs, i.e. it occurs when the time period of the clock pulse (T_{Clock}) is greater than the propagation delay (t_{pd}) of the flip flop.

  The state of the flip-flop will keep on changing from 0 to 1, 1 to 0, 0 to 1 and so on, and at the end of the clock pulse, its state will be uncertain to predict the output whether it is 1 or 0 at the end of clock pulse. This phenomenon is called the race around condition.
- This condition occurs only when flip-flop is level triggered and operated in toggle mode.
- Two avoid race around condition :

(i) Flip-flops are arranged in Master-Slave configuration.

(ii) Time period of clock (T_{Clock}) should be smaller than propagation delay (t_{pd}) of flip-flop. i.e.

$$T_{\text{Clock}} \leq t_{pd}$$

Q5.12.2 MCQ | 2012 | 2M | Ans: D

Given logic circuit is shown below,

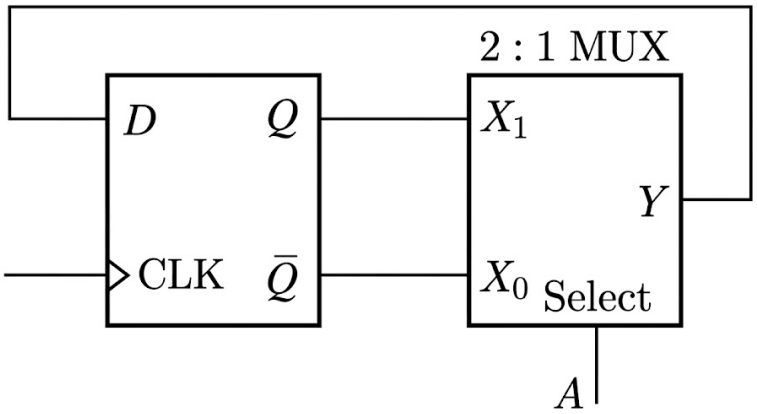


Fig. Solution Q5.12.2

The output equation of a 2 : 1 MUX with select line A, inputs X_0 and X_1 is given by:

$$Y = \bar{A}X_0 + AX_1$$

From the given circuit diagram: * $X_0 = \bar{Q}$ * $X_1 = Q$
Substituting these values into the MUX equation:

$$Y = \bar{A}\bar{Q} + AQ$$

Since the output of the multiplexer Y is connected directly to the input D of the flip-flop, the characteristic equation for the next state Q_{n+1} becomes:

$$Y = D = Q_{n+1} = \bar{A}\bar{Q}_n + AQ_n$$

The truth table for the sequential circuit is shown below:

Previous state Q_n	Input A	Next states $Y = D = Q_{n+1}$	States
0	0	1	Toggle
0	1	0	Hold
1	0	0	Toggle
1	1	1	Hold

The above truth table corresponds to the state diagram,

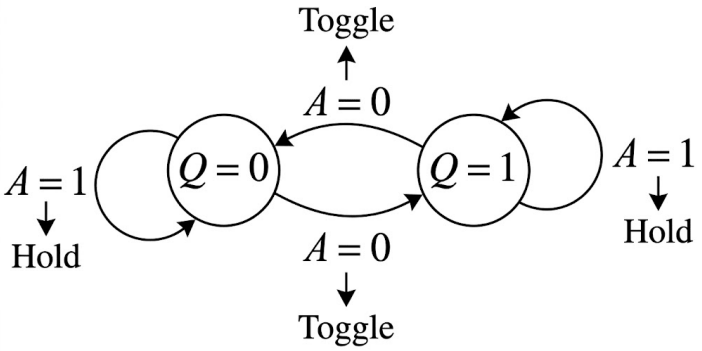


Fig. Solution Q5.12.2

Hence, the correct option is (D).

(D)

Q5.11.1 MCQ 2011 2M Ans: C

Given: 2-bit counter circuit. Initial state $Q_AQ_B = 10$. Find the state after 3 clock pulses.

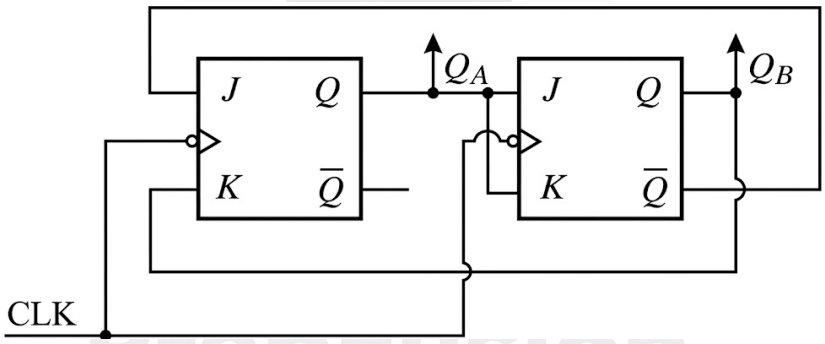


Fig. Solution Q5.11.1

From the circuit, the next-state equations are:

$$J_{B,n} = K_{B,n} = Q_{A,n}, \quad J_{A,n} = \bar{Q}_{B,n}, \quad K_{A,n} = Q_{B,n}$$

Transition table for the given circuit is shown below,

CLK Pulse	Present state		Flip-flop inputs				Next state	
	Q_A	Q_B	$J_A = \bar{Q}_B$	$K_A = Q_B$	$J_B = Q_A$	$K_B = Q_A$	Q_A^+	Q_B^+
1	1	0	1	0	1	1	1	1
2	1	1	0	1	1	1	0	0
3	0	0	1	0	0	0	1	0

From the above table it is clear that output after 3 clock pulses is $Q_AQ_B = 10$.
Hence, the correct option is (C).

(C) 10

Q5.08.1 MCQ | 2008 | 1M | Ans: A ●●○

Given truth table is shown below,

X	Y	Q	$\bar{Q}$
0	↑	⊓	⊓
↓	1	⊓	⊓

Fig. Solution Q5.08.1

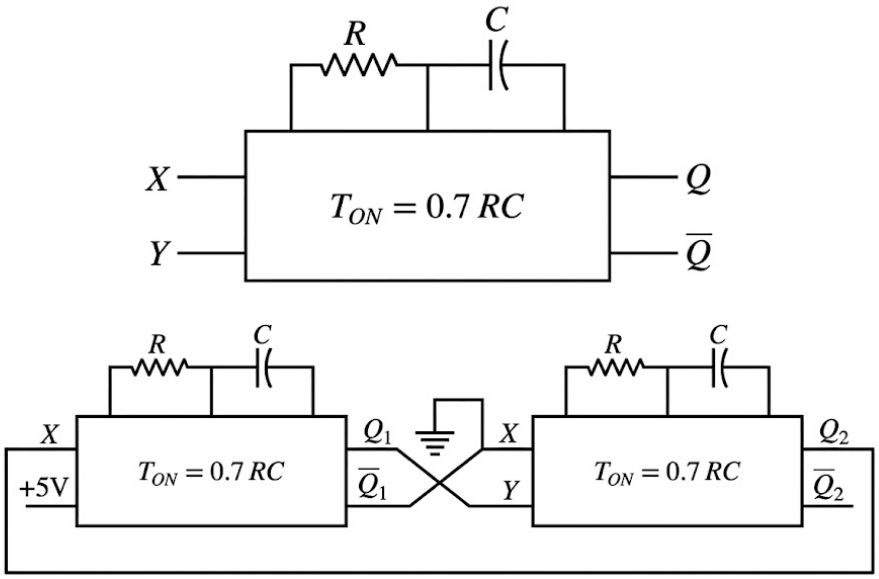


Fig. Solution Q5.08.1

1. Analyzing the Cross-Coupled Trigger Mechanism: From the circuit diagram, the connections between the two monoshots are:

$$Y = \bar{Q}_1 \quad \text{and} \quad X = Q_2$$

By evaluating the provided truth table, we can deduce the specific triggering edge conditions for each monoshot block:

- **Triggering Monoshot-2:** The table states that a response is initiated when input $X = 0$ and input Y transitions from $0 \rightarrow 1$. Since $Y = \bar{Q}_1$, this means Monoshot-2 triggers precisely on the **rising edge** ($\uparrow$) of $\bar{Q}_1$, which corresponds directly to the **falling edge** ($\downarrow$) of Q_1.
- **Triggering Monoshot-1:** The table states that a response is initiated when input $Y = 1$ and input X transitions from $1 \rightarrow 0$. Since $X = Q_2$, this means Monoshot-1 triggers precisely on the **falling edge** ($\downarrow$) of Q_2.

2. Self-Sustaining Loop Dynamics: These cross-coupled conditions form a continuous, alternating oscillation loop:

$$\text{Falling edge of } Q_1 \xrightarrow{\text{triggers}} \text{Monoshot-2 to fire } (Q_2 \text{ goes HIGH for } T_{ON_2})$$

$$\text{Falling edge of } Q_2 \xrightarrow{\text{triggers}} \text{Monoshot-1 to fire } (Q_1 \text{ goes HIGH for } T_{ON_1})$$

As each monoshot finishes its quasi-stable state time-out, its falling edge automatically triggers the next monoshot, generating a self-sustaining astable multivibrator waveform sequence.

3. Waveform and Mathematical Derivations: The resulting output waveforms for Q_1 and Q_2 are illustrated below:

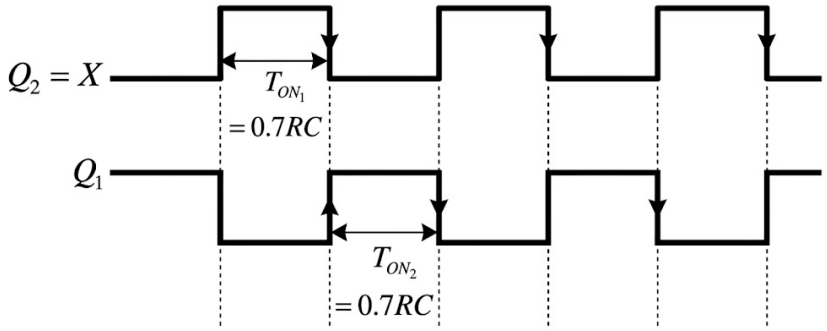


Fig. Solution Q5.08.1

From the timing diagram, one complete periodic cycle T consists exactly of the pulse duration of Monoshot-1 followed by the pulse duration of Monoshot-2:

$$T = T_{ON_1} + T_{ON_2}$$

Therefore, the operating frequency (f) of the waveform is:

$$f = \frac{1}{T} = \frac{1}{T_{ON_1} + T_{ON_2}}$$

The duty cycle (D) of the output waveform Q_1 is defined as the ratio of its active HIGH interval to the total cycle time. Since Q_1 remains HIGH only during the T_{ON_1} interval:

$$D = \frac{T_{ON}}{T} = \frac{T_{ON_1}}{T_{ON_1} + T_{ON_2}}$$

Hence, the correct option is (A).

(A) $\dfrac{1}{T_{ON_1} + T_{ON_2}}$ and $\dfrac{T_{ON_1}}{T_{ON_1} + T_{ON_2}}$

Key Points

- A rising transition ($0 \rightarrow 1$) on a complemented output signal ($\bar{Q}$) corresponds logically to a falling transition ($1 \rightarrow 0$) on the true output signal (Q).
- Cross-coupling two monostable multivibrators such that the trailing edge of one fires the input of the other sets up a continuous astable square-wave generator.

Q5.05.1 MCQ 2005 2M Ans: C ●●○

Given logic circuit is shown below,

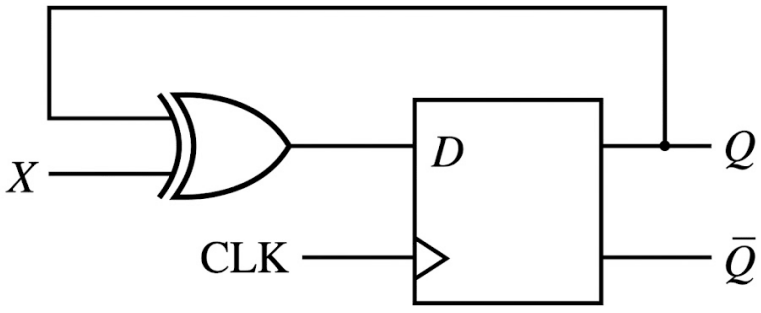


Fig. Solution Q5.05.1

From figure,

$$D = X \oplus Q_n \qquad \ldots (i)$$

For D flip-flop, the characteristic equation is given by,

$$Q_{n+1} = D \qquad \ldots (ii)$$

From equation (i) and (ii),

$$Q_{n+1} = X \oplus Q_n \qquad \ldots (iii)$$

The above equation (iii) represents the characteristic equation of T flip-flop.
Truth table of given flip-flop is shown below,

X	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Hence, the correct option is (C).

(C) T flip-flop

Q5.05.2 Descriptive 2005 1M Ans: *

Given: Waveforms for inputs X_1, X_2, and output Q.

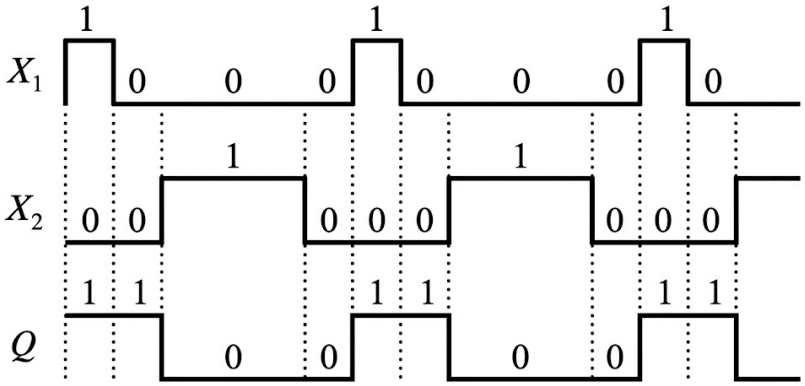


Fig. Solution Q5.05.2

1. Functional Analysis from Waveforms: By tracking the signal levels step-by-step from the given timing diagram, we observe the following behavior for output Q:

- When $(X_1, X_2) = (1, 0) \Rightarrow Q$ becomes 1 (**Set** operation).
- When $(X_1, X_2) = (0, 0) \Rightarrow Q$ maintains its value at 1 (**Hold** operation).
- When $(X_1, X_2) = (0, 1) \Rightarrow Q$ becomes 0 (**Reset** operation).
- When $(X_1, X_2) = (0, 0) \Rightarrow Q$ maintains its value at 0 (**Hold** operation).

Summarizing this operational sequence gives us the function table of an active-HIGH cross-coupled NOR latch where X_1 behaves as the **Set** (S) input and X_2 behaves as the **Reset** (R) input:

$$Q^+ = \overline{X_2 + \bar{Q}}, \quad \bar{Q}^+ = \overline{X_1 + Q}$$

2. Evaluation of Latch Options: Let us evaluate each option by directly testing the node explicitly labeled Q against the established mathematical latch equations:

- **Option (A):** This circuit shows a standard cross-coupled NOR latch configuration. Let the output of the bottom NOR gate be named W. The expressions for the two gates are:

  $$Q = \overline{X_1 + W} \quad \text{and} \quad W = \overline{X_2 + Q}$$

  Let us apply the first input condition from the waveform: $(X_1, X_2) = (1, 0)$. Substituting $X_1 = 1$ gives $Q = \overline{1 + W} = 0$. This directly contradicts the waveform sequence, which explicitly shows that Q must go HIGH (1) when $(X_1, X_2) = (1, 0)$. Thus, Option (A) is incorrect.

- **Option (B):** This circuit uses OR gates with cross-coupling instead of NOR gates. The labeled output node satisfies $Q = X_1 + W$. If $(X_1, X_2) = (0, 0)$, the circuit simplifies to $Q = 0 + W = W$, and $W = 0 + Q = Q$. This creates an un-complemented feedback loop that cannot hold a distinct stable $(Q, \bar{Q})$ pair. Thus, Option (B) is incorrect.

- **Option (C):** This configuration features an AND gate cross-coupled with a NAND gate. The node Q is driven by the AND gate: $Q = X_1 \cdot W$. If $(X_1, X_2) = (0, 1)$, then $X_1 = 0$ forces $Q = 0 \cdot W = 0$, which matches the waveform reset condition. However, when inputs transition to the hold state $(X_1, X_2) = (0, 0)$, $X_1 = 0$ continues to force $Q = 0$, completely failing to hold a previous state of 1. Thus, Option (C) is incorrect.

- **Option (D):** This circuit consists of cross-coupled NAND gates (an active-LOW latch). The node Q is given by $Q = \overline{X_1 \cdot W}$. Applying $(X_1, X_2) = (1, 0)$ gives $W = \overline{0 \cdot Q} = 1$, which leads to $Q = \overline{1 \cdot 1} = 0$. This produces a low output instead of the required high state. Thus, Option (D) is incorrect.

Consequently, none of the provided circuit options produce the precise functional behavior required at the labeled terminal Q, rendering all options incorrect.

None of the options are correct.

Key Points

- Derive the functional truth table behavior directly from the waveform transitions: $(X_1, X_2) = (1, 0) \rightarrow$ Set $(Q = 1)$, and $(0, 1) \rightarrow$ Reset $(Q = 0)$.
- To verify a circuit configuration, evaluate the boolean logic equation specifically at the node labeled Q to ensure it satisfies the functional criteria.
- The orientation or placement of terminal nodes can vary arbitrarily; the input steering connections relative to the labeled terminal dictate the final stable state.

Hence, none of the options are correct.

None of the options are correct

Q5.05.3 MCQ 2005 2M Ans: D ● ○ ○

Given: Circuit with $X_1 = X_2 = 1$ maintained. Find the behavior of output Q.

For $X_1 = X_2 = 1$, the given figure will be equivalent to the circuit shown below,

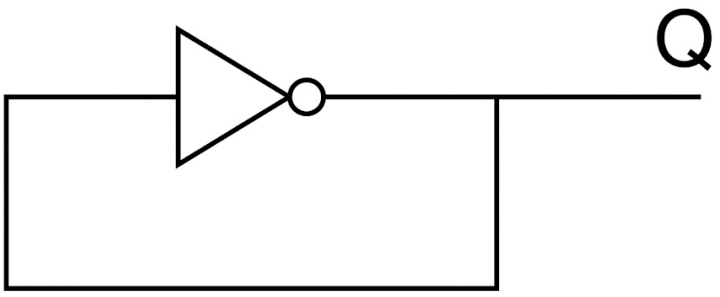


Fig. Solution Q5.05.3

It always switches its state from '1' to '0' and from '0' to '1'.

It always switches its state from '1' to '0' and from '0' to '1'.

Key Points

- When $X_1 = X_2 = 1$, the logic structure effectively simplifies into a single inverter loop back configuration.
- This configuration acts as an astable multivibrator or ring oscillator behavior, continuously toggling between high and low logic states.

Hence, the correct option is (D).

(D) Unstable.

Q5.04.1 MCQ 2004 1M Ans: A ● ○ ○

Given: A logic circuit consisting of two inverters connected in a feedback loop.

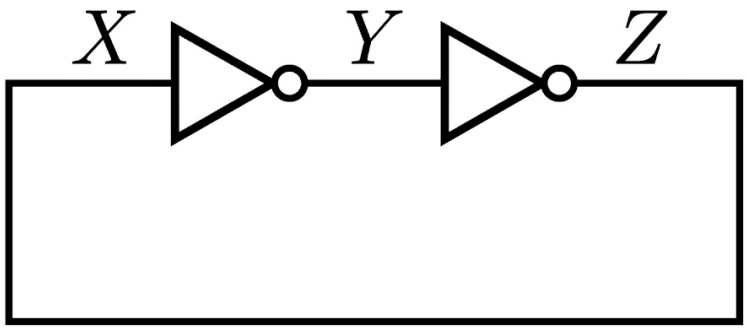


Fig. Solution Q5.04.1

Suppose we put input $X = 0$, then $Y = 1$ and output $Z = 0$. It shows that input X and output Z both are at logic 0 states, so we can say our circuit remains at a logic 0 stable state.

In the same way, if we put input $X = 1$, then $Y = 0$ and output $Z = 1$. It shows that input X and output Z both are at logic 1 states, so we can say our circuit remains at a logic 1 stable state.

Hence, the truth table for the given circuit is shown below:

X	Y	Z = X
0	1	0
1	0	1

Hence, the number of stable states is two, i.e., 0 and 1.
Thus, it is a bistable multivibrator.

Key Points

- **Even** number of inverters in feedback → **bistable** multivibrator.
- **Odd** number of inverters in feedback → **astable** multivibrator (ring oscillator).
- Bistable multivibrators serve as the core storage element in latches and flip-flops.

Hence, the correct option is (A).

(A) a bistable multivibrator

Q5.04.2 MCQ 2004 5M Ans: B

Given J-K flip-flop is shown below,

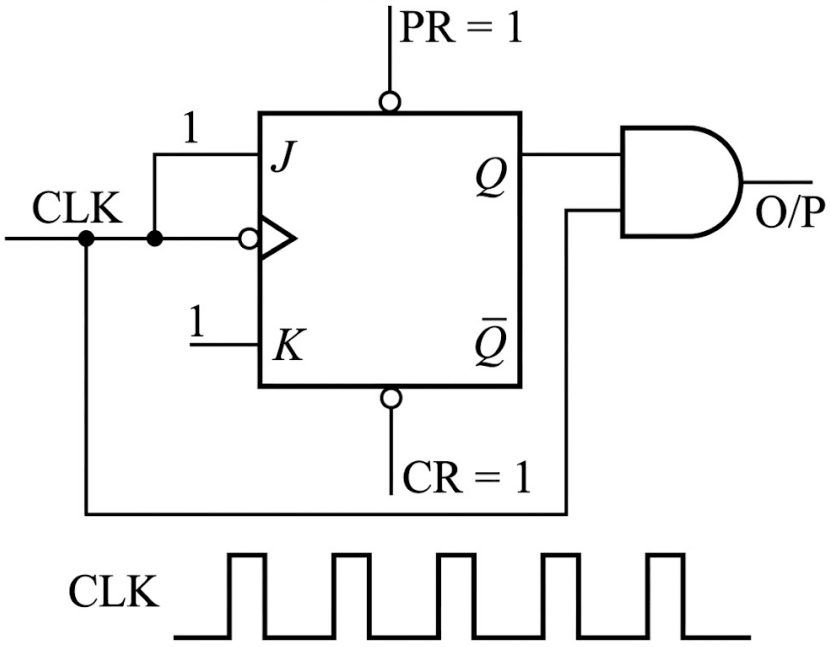


Fig. Solution Q5.04.2

1. Functional Logic Derivation: We analyze the behavior of the sequential circuit step-by-step:

- **Steering Inputs:** The inputs are tied to a permanent high logic level ($J = 1$ and $K = 1$). This configures the J-K flip-flop to operate permanently in **toggle mode**.
- **Clock Triggering:** The clock terminal features an inversion bubble, meaning the flip-flop is **negative-edge triggered**. Consequently, the output Q will change state only at the falling edge (↓) of the CLK signal.
- **Output Expression:** The combinational logic stage at the output features a 2-input AND gate whose inputs are driven directly by the true output Q of the flip-flop and the external clock signal (CLK). Thus, the explicit output expression is:

$$\text{O/P} = Q \cdot \text{CLK}$$

From this expression, it is clear that the final network output (O/P) can be high (logic 1) if and only if both Q and CLK are simultaneously high.

2. Initial State and Waveform Analysis: Because no initial state information is specified for the flip-flop output Q, we can arbitrarily choose an initial condition to trace the timing sequence. Let us assume the initial state is reset ($Q = 0$). Note that choosing the alternative state ($Q = 1$) merely shifts the starting phase, but the ultimate periodic, steady-state characteristics of the output waveform remain identical.

With $Q = 0$ initially:

- During the first clock cycle, $Q = 0$. Since O/P = $0 \cdot \text{CLK} = 0$, the overall circuit output stays low.
- At the first negative edge (↓) of CLK, the flip-flop toggles, making $Q = 1$.
- During this second clock cycle, Q remains high. Therefore, when CLK goes high, the output follows: O/P = $1 \cdot 1 = 1$. When CLK drops low, O/P = $1 \cdot 0 = 0$.
- This process alternates continuously, resulting in the output pulse being generated only during the intervals where the high phases of CLK and Q overlap.

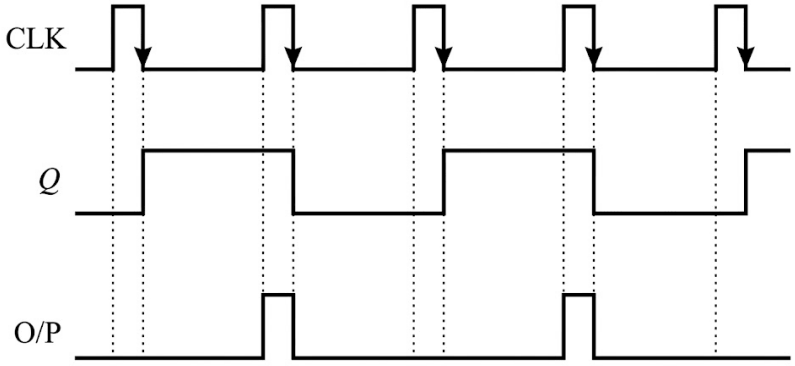


Fig. Solution Q5.04.2

Hence, the correct option is (B).

(B) [Waveform: AND output high only during overlap of CLK high and Q high]

Key Points

- When $J = K = 1$, the flip-flop halves the frequency of the input clock at its Q output terminal.
- Gating the output Q with the original CLK through an AND gate clips the pulse duration, matching the exact overlapping interval when both signals are high.

Q5.03.1 MCQ 2003 2M Ans: B ●●○

Given circuit diagram is shown below,

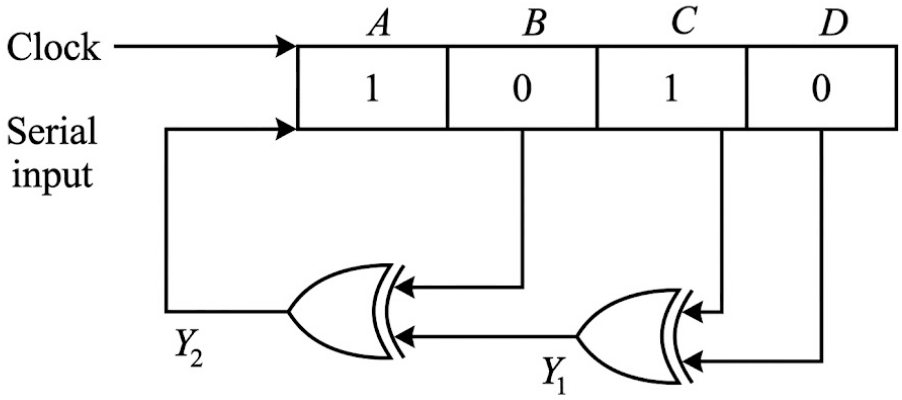


Fig. Solution Q5.03.1

For the above figure,

$$Y_1 = C \oplus D$$

$$Y_2 = B \oplus Y_1$$

$$Y_2 = B \oplus C \oplus D$$

Thus, Serial input = $Y_2 = B \oplus C \oplus D$

CLK pulse	Serial input $= Y_2 = B \oplus C \oplus D$	A	B	C	D
0	1	1	0	1	0
1	0	1	1	0	1
2	0	0	1	1	0
3	0	0	0	1	1
4	1	0	0	0	1
5	0	1	0	0	0
6	1	0	1	0	0
7	1	**1**	**0**	**1**	**0**

Hence, the pattern 1010 is obtained after 7th clock pulse.
Hence, the correct option is (B).

(B) 7

Q5.03.2 MCQ 2003 1M Ans: D ● ○ ○

Given: The truth table for an XY flip-flop. We need to find the equivalent inputs in terms of a J-K flip-flop.
Truth table for XY flip-flop is shown below,

X	Y	Q_n	Q_{n+1}
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

K-map of function Q_{n+1} in SOP form is,

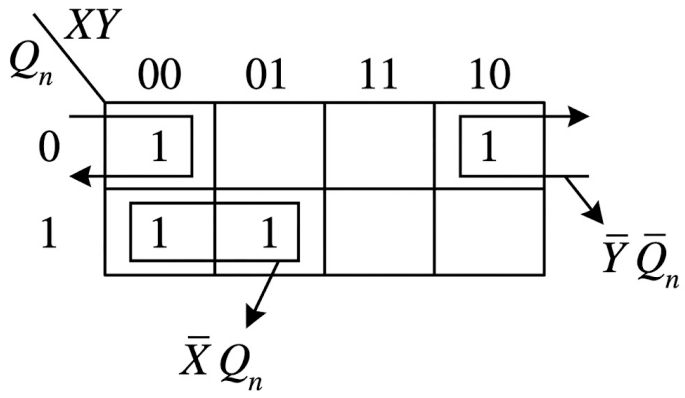


Fig. Solution Q5.03.2

$$Q_{n+1} = \bar{X}Q_n + \bar{Y}\bar{Q}_n \qquad \ldots (i)$$

For J-K flip-flop, the characteristic equation is given by,

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n \qquad \ldots (ii)$$

Comparing equation (i) and (ii),

$$J = \bar{Y} \text{ and } \bar{K} = \bar{X}$$

$$J = \bar{Y},\ K = X$$

Hence, the correct option is (D).

$$\boxed{\text{(D) } J = \bar{Y},\ K = X}$$

Q5.02.1 MCQ 2002 1M Ans: D ●●○

Given: Rising edge triggered D flip-flop. Clock frequency = 10 kHz. $\bar{Q}$ is fed back to D.

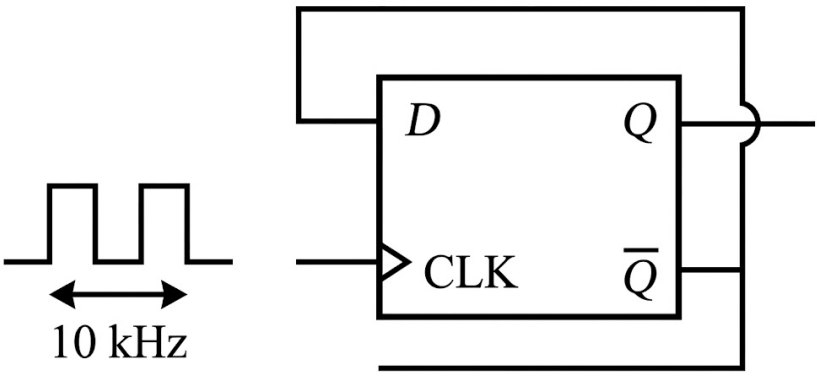


Fig. Solution Q5.02.1

$$D = \bar{Q}_n \implies Q_{n+1} = \bar{Q}_n$$

Q toggles on every rising clock edge (MOD-2 counter):

$$f_Q = \frac{f_{clk}}{2} = \frac{10 \text{ kHz}}{2} = \boxed{5 \text{ kHz}}$$

Key Points

- A D flip-flop with $D = \bar{Q}$ is a **toggle flip-flop** (T FF with $T = 1$).
- Output frequency = $f_{clk}/2$ (divides by 2).

Hence, the correct option is (D).

(D) 5 kHz

Q5.02.2 MCQ 2002 5M Ans: B

The given ripple counter is shown below,

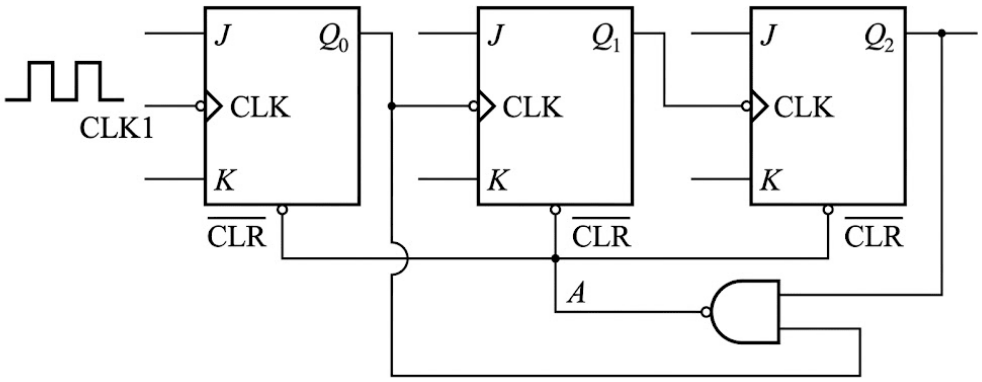


Fig. Solution Q5.02.2

The above circuit is a negative edge-triggered asynchronous counter.
Here, the output Q of one flip-flop is connected to the clock input of the next flip-flop.
Hence, it acts as an up counter.
A 3-bit up counter counts from

$$000 \rightarrow 111$$

i.e.,

$$0 \rightarrow 7$$

If $A = 0$, then all the flip-flops will be cleared.

$$A = 0$$

This is possible only when

$$Q_2 = 1, \quad Q_1 = 0, \quad Q_0 = 1$$

Therefore, the counter is cleared when

$$Q_2Q_1Q_0 = 101$$

i.e., at count 5.
Thus, at the 5th clock pulse, the counter resets to

$$Q_2Q_1Q_0 = 000$$

Hence, it is a MOD-5 up counter.
Therefore, the correct option is (B).

(B) 5

Q5.01.1 MCQ | 2001 | 5M | Ans: A | ● ● ○

Given: MOD-N counter circuit with logic gates.

From figure,

$$A = \overline{Q_3 + Q_2 + Q_1}$$

$$D_3 = A = \overline{Q_3 + Q_2 + Q_1}$$

$$D_2 = Q_3$$

$$D_1 = Q_2 \quad \text{and} \quad D_0 = Q_1$$

The truth table for the given circuit is shown below,

CLK	Present state				Flip-flop inputs				Next state			
	Q_3	Q_2	Q_1	Q_0	D_3	D_2	D_1	D_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+
1	1	1	1	0	0	1	1	1	0	1	1	1
2	0	1	1	1	0	0	1	1	0	0	1	1
3	0	0	1	1	0	0	0	1	0	0	0	1
4	0	0	0	1	1	0	0	0	1	0	0	0
5	1	0	0	0	0	1	0	0	0	1	0	0
6	0	1	0	0	0	0	1	0	0	0	1	0
7	0	0	1	0	0	0	0	1	0	0	0	1
8	0	0	0	1	1	0	0	0	1	0	0	0

4 states

Fig. Solution Q5.01.1

State Trajectory and Modulus Analysis: Tracing the sequence from the table starting with the initial configuration $Q_3Q_2Q_1Q_0 = 1110$:

- The circuit transitions through a series of initial transient states: $1110 \rightarrow 0111 \rightarrow 0011 \rightarrow 0001$.
- Once the circuit reaches the state 0001, it enters a closed, self-sustaining loop.

The explicitly tracked repeating cycle obtained from the state table is:

$$0001 \rightarrow 1000 \rightarrow 0100 \rightarrow 0010 \rightarrow 0001 \rightarrow \ldots$$

Since this repeating cycle contains exactly four distinct states, the stable count sequence length is 4. The initial states 1110, 0111, and 0011 are transient states that occur only once during power-up and do not factor into the periodic counting behavior.

Therefore, it is a MOD-4 counter.

Hence, the correct option is (A).

(A) 4

Key Points

- The modulus (MOD value) of a sequential counter is determined exclusively by the number of unique states contained within its recurring/periodic loop.
- Transient states are states that lead into the main cycle but are never revisited once the stable loop is entered.

Q5.99.1 MCQ | 1999 | 1M | Ans: A | ● ○ ○

Given: Flip-flop circuit consisting of cross-coupled NAND gates.

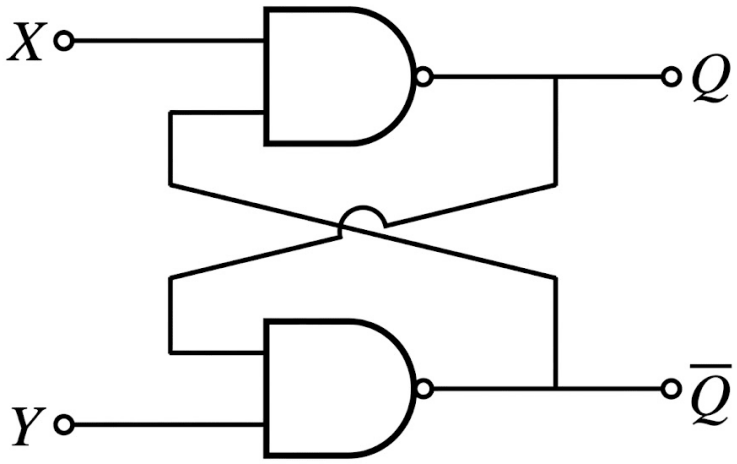


Fig. Solution Q5.99.1

The above figure represents an S-R NAND latch where input variable X acts as the active-LOW Set ($\bar{S}$) line and input variable Y acts as the active-LOW Reset ($\bar{R}$) line.

Truth table for this S-R NAND latch configuration is shown below:

X	Y	Q_n	Q_{n+1}	State
0	0	0	Invalid state	Invalid/Unstable
0	0	1	Invalid state	Invalid/Unstable
0	1	0	1	Set
0	1	1	1	Set
1	0	0	0	Reset
1	0	1	0	Reset
1	1	0	0	Hold
1	1	1	1	Hold

When both inputs are driven low simultaneously ($X = 0, Y = 0$), both cross-coupled NAND outputs are forced to a logic high level ($Q = 1$ and $\bar{Q} = 1$). This violates the complementary property ($Q = \bar{Q}$) required for standard latch stability, creating an invalid state configuration.

Hence, the correct option is (A).

Key Points

- Latches are asynchronous, level-triggered sequential circuits, whereas flip-flops are synchronous, edge-triggered circuits.
- A latch serves as a fundamental 1-bit memory storage element built using cross-coupled feedback loops.
- An S-R NAND latch reaches an invalid/unstable condition when both external control inputs are held at $(0, 0)$.
- An S-R NOR latch reaches an invalid/unstable condition when both external control inputs are held at $(1, 1)$.

(A) $X = 0,\ Y = 0$

Q5.95.1 NAT | 1995 | 1M | Ans: 0.5 | ● ● ○

Given : Clock frequency, $f_{CLK} = 1$ MHz

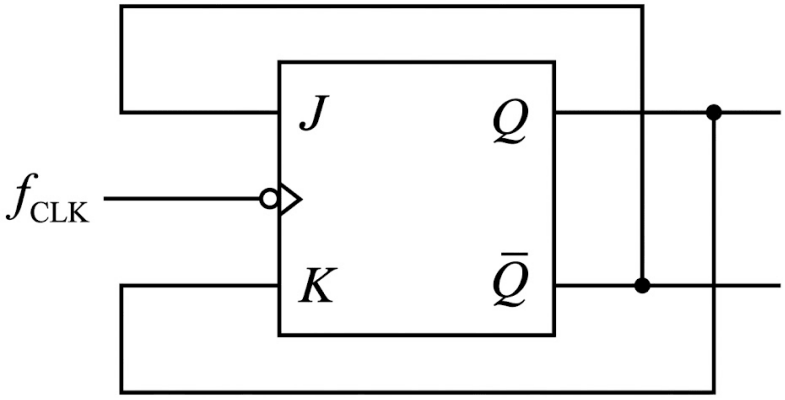


Fig. Solution Q5.95.1

From figure, $J_n = \bar{Q}_n$, $K_n = Q_n$
For a J-K FF, the characteristic equation is given by,

$$Q_{n+1} = J_n\bar{Q}_n + \bar{K}_nQ_n$$

$$Q_{n+1} = \bar{Q}_n\bar{Q}_n + \bar{Q}_nQ_n = \bar{Q}_n + 0 = \bar{Q}_n$$

The above given flip-flop counts only two states i.e. 0 and 1, hence it is MOD-2 counter.
For any MOD-M counter, output frequency is given by,

$$f_0 = \frac{f_{CLK}}{M} = \frac{10^6}{2}\text{ Hz} = 0.5\text{ MHz}$$

Hence, the output frequency of Q will be **0.5 MHz**.

Key Points

- With $J = \bar{Q}$ and $K = Q$: $Q_{n+1} = \bar{Q}_n$ always (toggles on every clock pulse).
- This configuration acts as a **divide-by-2** circuit (MOD-2 counter), which halves the input clock frequency.

Q5.91.1 MCQ | 1991 | 1M | Ans: C | ● ● ○

Given: 3-stage Johnson ring counter, clock frequency f_C, starting state $Q_0\,Q_1\,Q_2 = 100$.

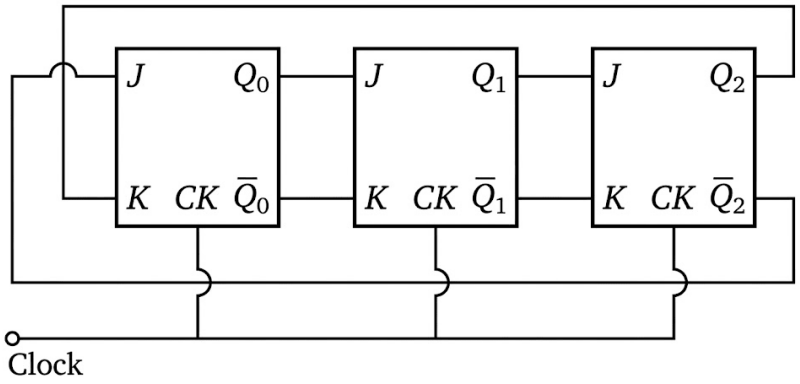


Fig. Solution Q5.91.1

For a 3-bit Johnson counter ($n = 3$), the number of states = $2n = 6$.

CLK	Q_0	Q_1	Q_2
0	1	0	0
1	1	1	0
2	1	1	1
3	0	1	1
4	0	0	1
5	0	0	0
6	1	0	0

Each output completes one full cycle in every 6 clock periods:

$$f_{Q_0} = f_{Q_1} = f_{Q_2} = \frac{f_C}{6}$$

$$\boxed{f_{Q_0} = f_{Q_1} = f_{Q_2} = \frac{f_C}{6}}$$

Key Points

- An n-stage Johnson counter cycles through $2n$ states; all outputs have frequency $f_C/(2n)$.
- The outputs are phase-shifted versions of the same square wave.

Hence, the correct option is (C).

$$\boxed{\text{(C) } \frac{f_C}{6}, \frac{f_C}{6}, \frac{f_C}{6}}$$

PrepFusion

SOLUTIONS

VI

ADC & DAC

Topics

1. Introduction to DAC — Weighted Resistor and R-2R Ladder Type
2. Introduction to ADC — Counter Type
3. SAR Type ADC
4. Flash Type ADC
5. Single Slope and Dual Slope ADC
6. Sample and Hold Operation in ADC

Unit Snapshot*

Stream: EE	**Questions:** 11	**MCQ:** 9	**MSQ:** 0	**NAT:** 2
1M: 7	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2016)–1M(2016)

**See preface for how Avg Weightage and Range are calculated.*

Q6.16.1 MCQ | 2016 | 2M | Ans: A | ● ○ ○

Given: A 2-bit flash analog to digital converter with an input voltage range of $0 \le V_{in} \le 3$ V.
The reference voltages for the comparators are derived from the voltage divider network:

$$I = \frac{3}{100 + 200 + 200 + 100} = 5 \text{ mA}$$

$$V_0 = 100 \times 5 \times 10^{-3} = 0.5 \text{ V}$$

$$V_1 = 300 \times 5 \times 10^{-3} = 1.5 \text{ V}$$

$$V_2 = 500 \times 5 \times 10^{-3} = 2.5 \text{ V}$$

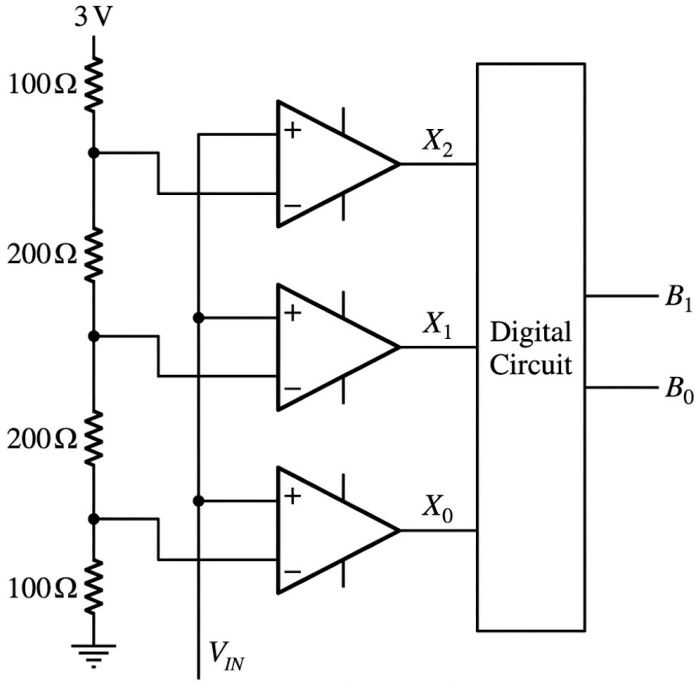


Fig. Solution Q6.16.1

Based on the reference levels, the output of the comparators (X_2, X_1, X_0) and the standard digital circuit outputs (B_1, B_0) are given as:

V_{in} (Volt)	X_2	X_1	X_0	B_1	B_0
$0 \le V_{in} \le 0.5$	0	0	0	0	0
$0.5 < V_{in} \le 1.5$	0	0	1	0	1
$1.5 < V_{in} \le 2.5$	0	1	1	1	0
$2.5 < V_{in} \le 3.0$	1	1	1	1	1

The above solution is not in the minimized form; the solution in the question is solved according to the specific options provided. From the options, we can represent B_0 in SOP form:

$$B_0 = \overline{X_2} \cdot \overline{X_1} \cdot X_0 + X_2 \cdot X_1 \cdot X_0$$

$$= (\overline{X_2} \cdot \overline{X_1} + X_2 \cdot X_1) X_0$$

$$= (\overline{X_2 \oplus X_1}) X_0$$

$$B_0 = X_0 \, [\, \overline{X_2 \oplus X_1} \,]$$

(A) $X_0 \, [\, \overline{X_2 \oplus X_1} \,]$

Key Points

- An n-bit flash ADC uses $2^n - 1$ comparators. For a 2-bit flash ADC, $2^2 - 1 = 3$ comparators are required.
- The valid input combinations for a 2-bit ADC are thermometer-coded: only 000, 001, 011, and 111 are valid states. All others are don't-care conditions.
- The XOR function $(\overline{X_2} \oplus X_1)$ detects the boundary condition where B_0 transitions between 0 and 1.

Mistakes to be avoided

- Do not forget that input combinations like 010, 100, 101, and 110 are **invalid (don't-care)** in a thermometer-coded flash ADC. Treating them as 0s will lead to an incorrect Boolean expression.
- Ensure the K-map is filled with don't-cares (X) for these combinations before simplifying, as they can be used to obtain a minimal expression.

Q6.15.1 MCQ 2015 2M Ans: A ●○○

Given:

- Number of bits: $n = 8$
- Input voltage: $V_a = 3.5\,\text{V}$
- Reference voltage: $V_{ref} = +5\,\text{V}$

Method 1

Step-by-step SAR algorithm:

Resolution:

$$R = \frac{V_{FS}}{2^n - 1} = \frac{5}{2^8 - 1} = \frac{5}{255} \approx 0.0196\,\text{V}$$

1st clock pulse: MSB is set.

$$\text{Output} = (10000000)_2 = (128)_{10}$$

$$V_C = 128 \times \frac{5}{255} = 2.5098\,\text{V}$$

Since $V_0 < V_a$ (i.e., 2.5098 < 3.5), MSB **remains set** (1).

2nd clock pulse: Next MSB is set.

$$\text{Output} = (11000000)_2 = (192)_{10}$$

$$V_C = 192 \times \frac{5}{255} = 3.7647\,\text{V}$$

Since $V_0 > V_a$ (i.e., 3.7647 > 3.5), 2nd MSB is **reset** (0).

3rd clock pulse: Next bit is set.

$$\text{Output} = (10100000)_2 = (160)_{10}$$

$$V_C = 160 \times \frac{5}{255} = 3.137\,\text{V}$$

Since $V_0 < V_a$ (i.e., 3.137 < 3.5), 3rd MSB **remains set** (1).

Therefore, the output after the 3rd clock pulse is:

$$\boxed{10100000}$$

Method 2

Using the SAR algorithm flowchart:

This method interprets the same SAR conversion process using the binary-search concept. The comparison voltages (2.5098 V, 3.7647 V, 3.137 V) are the DAC outputs corresponding to the trial codes generated by the SAR register in Method 1.

The SAR ADC uses a binary search algorithm. Starting from $V_{ref}/2$ each time:

- Clock 1: Compare 3.5 V vs 2.51 V $\Rightarrow V_a > V_C \Rightarrow$ bit 7 = **1** $\Rightarrow$ 10000000
- Clock 2: Compare 3.5 V vs 3.76 V $\Rightarrow V_a < V_C \Rightarrow$ bit 6 = **0** $\Rightarrow$ 10000000
- Clock 3: Compare 3.5 V vs 3.14 V $\Rightarrow V_a > V_C \Rightarrow$ bit 5 = **1** $\Rightarrow$ 10100000

(A) 1010 0000

Key Points

- **SAR Rule:** If $V_a > V_C$: current bit remains **1**, next bit is tested. If $V_a < V_C$: current bit is **reset to 0**, next bit is tested.
- An n-bit SAR ADC requires exactly n clock pulses to complete one conversion.
- Maximum conversion time: $T_{max} = n \cdot T_{CLK}$

Mistakes to be avoided

- Do not use 2^n in the denominator for resolution. The correct formula uses $2^n - 1$ for unipolar ADCs.
- The algorithm continues even after resetting a bit — the next lower bit is always tested in the subsequent clock cycle.

Q6.06.1 MCQ 2006 2M Ans: C

Given: A 3-bit binary down counter is connected to an R-2R ladder DAC as shown below.

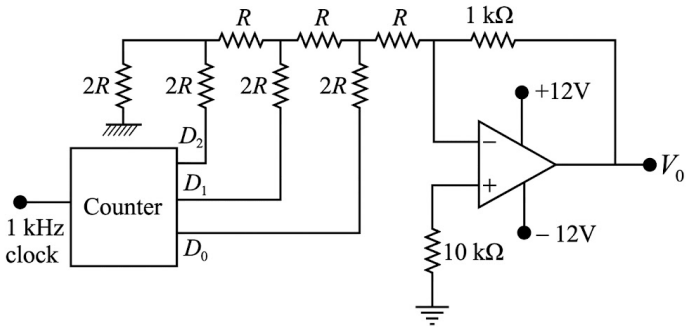


Fig. Solution Q6.06.1

The observed staircase waveform is shown below.

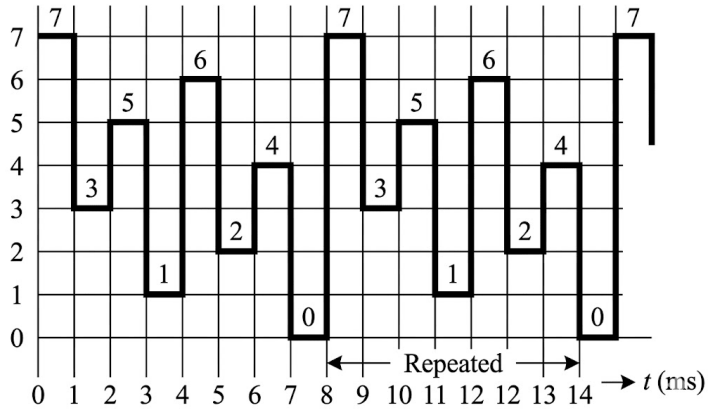


Fig. Solution Q6.06.1

Since it is a down counter, under ideal conditions it should count from 7 down to 0:

$$(111)_2 \rightarrow (110)_2 \rightarrow (101)_2 \rightarrow (100)_2 \rightarrow (011)_2 \rightarrow (010)_2 \rightarrow (001)_2 \rightarrow (000)_2$$

Hence, the expected DAC output sequence is:

$$7,\ 6,\ 5,\ 4,\ 3,\ 2,\ 1,\ 0$$

Initially, the counter output is $(111)_2 = (7)_{10}$ and the DAC output is also 7, which is correct.

For the next count, the counter output becomes $(110)_2 = (6)_{10}$. Thus, the DAC output should be 6 V. However, the observed output is 3 V. Since

$$(110)_2 \xrightarrow{\text{MSB-LSB swap}} (011)_2 = (3)_{10}$$

the DAC is effectively receiving $(011)_2$ instead of $(110)_2$.

For the next count, the counter output is $(101)_2 = (5)_{10}$. Interchanging the MSB and LSB leaves the code unchanged:

$$(101)_2 \xrightarrow{\text{MSB-LSB swap}} (101)_2 = (5)_{10}$$

Hence, the observed output remains 5 V.

For the next count, the counter output is $(100)_2 = (4)_{10}$. Interchanging the MSB and LSB gives

$$(100)_2 \xrightarrow{\text{MSB-LSB swap}} (001)_2 = (1)_{10}$$

which matches the observed output. Tracing all states systematically yields the complete mapped sequence:

Counter Output	DAC Output	Correct Sequence
111	7	7
011	3	6
101	5	5
001	1	4
110	6	3
010	2	2
100	4	1
000	0	0

Analysis and Elimination of Alternative Causes: To definitively confirm the root error, let us evaluate why the remaining options are inconsistent with the observed behavior:

- **Option A (Incorrect resistance values):** Mismatched or out-of-tolerance resistors alter the analog voltage step heights (amplitudes), leading to uneven or non-monotonic steps. However, changing resistance values cannot systematically reorder the digital state sequences to flip distinct voltage levels (like mapping a 6 to a 3).
- **Option B (Counter is not working properly):** A faulty counter (e.g., stuck bits or skipping states) would fail to loop through all 8 unique states smoothly or would generate irregular, erratic jumps. The waveform instead shows a highly structured, repeating 8-level permutation pattern, confirming the counter itself is counting correctly.
- **Option D (R and 2R are interchanged):** Swapping the position of R and $2R$ components changes the fundamental input weighting criteria of the ladder network, drastically shifting the individual step scaling factors, but it cannot structurally generate the exact bit-swapped code sequence observed.

This elimination explicitly confirms that the issue is restricted to the routing interface between the two systems. Specifically, the interconnect lines corresponding to bits D_2 and D_0 are interchanged.

(C) The connection from the counter to DAC is not proper.

Key Points

- In an R-2R ladder DAC, each input bit has a fixed binary weight. Any interchange of bit lines changes the analog output sequence.
- A 3-bit down counter normally generates a monotonic downward counting profile.
- Swapping the MSB and LSB transforms the count sequence seen by the DAC into a permutation that exactly tracks the observed staircase waveform.
- Incorrect resistor values or interchanged R-2R positions skew the voltage amplitudes/linearity but do not re-index the sequence of states.

Mistakes to be avoided

- Do not assume the counter is malfunctioning; first verify whether the waveform can be explained by a wiring error.
- Do not confuse DAC bit significance with counter bit positions. Always trace the actual interconnections.
- Incorrect resistor values affect output amplitudes, whereas interchanged bit connections affect the order of the staircase levels.

Q6.05.1 NAT 2005 2M Ans: 8 ● ○ ○

Given:

- Full-scale output voltage of DAC: $V_{FS} = 3.5\,\text{V}$
- Resolution: $R \approx 14\,\text{mV} = 14 \times 10^{-3}\,\text{V}$

Resolution of a DAC is given by:

$$R = \frac{V_{fullscale}}{2^n - 1}$$

Substituting the given values:

$$2^n - 1 = \frac{3.5}{14 \times 10^{-3}} = \frac{3.5}{0.014} = 250$$

$$2^n = 251$$

Taking the natural logarithm on both sides:

$$n \ln(2) = \ln(251)$$

$$n = \frac{\ln(251)}{\ln(2)} = \frac{5.5255}{0.6931} \approx 7.97 \approx 8$$

Since n must always be an integer:

Its bit size is **8**.

Key Points

- **DAC Resolution:** $R = \dfrac{V_{FS}}{2^n - 1}$, where n is the number of bits and V_{FS} is the full-scale output voltage.
- Resolution can also be expressed as a percentage: $R(\%) = \dfrac{1}{2^n - 1} \times 100\%$.
- For an n-bit DAC, there are 2^n possible output levels and $2^n - 1$ steps.

Q6.04.1 MCQ 2004 1M Ans: A ● ○ ○

A voltage comparator has two inputs V_1 and V_2 and one output V_0:

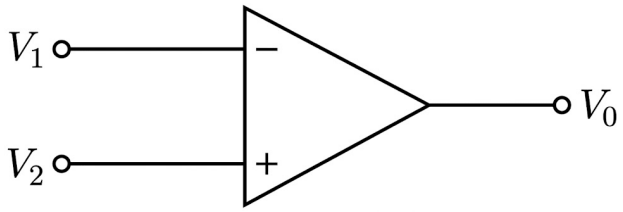


Fig. Solution Q6.04.1

Case I: When $V_1 < V_2$, the output voltage V_0 is $+V_{set}$, i.e., High (logic-1).
Case II: When $V_1 > V_2$, the output voltage V_0 is $-V_{set}$, i.e., Low (logic-0).
V_0 can assume only one of two possible values: High or Low, corresponding to logic-1 and logic-0. This represents a **1-bit** digital output.

Therefore, a voltage comparator can be used as a **1-bit quantizer** in analog-to-digital conversion.

(A) A 1-bit quantizer

Key Points

- A single comparator produces a 1-bit output (two levels: high or low).
- A **flash ADC** uses $2^n - 1$ comparators to produce an n-bit digital output simultaneously.
- For an 8-bit flash ADC, $2^8 - 1 = 255$ comparators are required.

Q6.01.1 MCQ 2001 1M Ans: C ● ○ ○

The speed of various ADC types in **increasing order** is:

$$\underbrace{\text{Dual slope (Integrating)}}_{\text{Slowest}} < \text{Counting (Counter) type} < \text{Successive approximation} < \underbrace{\text{Parallel comparator (Flash)}}_{\text{Fastest}}$$

The **integrating type** (dual slope) ADC is the slowest because it requires charging and discharging of a capacitor over a fixed time period. It needs 2^{n+1} clock pulses for an n-bit conversion.

(C) Integrating type

Key Points

- **Conversion time comparison:**
  - Flash ADC: $1 \cdot T_{CLK}$ (fastest)
  - SAR ADC: $n \cdot T_{CLK}$
  - Counter type: $(2^n - 1) \cdot T_{CLK}$ (average)
  - Dual slope (Integrating): $2^{n+1} \cdot T_{CLK}$ (slowest)
- Despite being the slowest, the dual slope ADC provides the **highest accuracy** and **best noise immunity** among all ADC types.

Q6.00.1 MCQ | 2000 | 1M | Ans: B | ● ○ ○

Given: An n-bit dual-slope ADC.

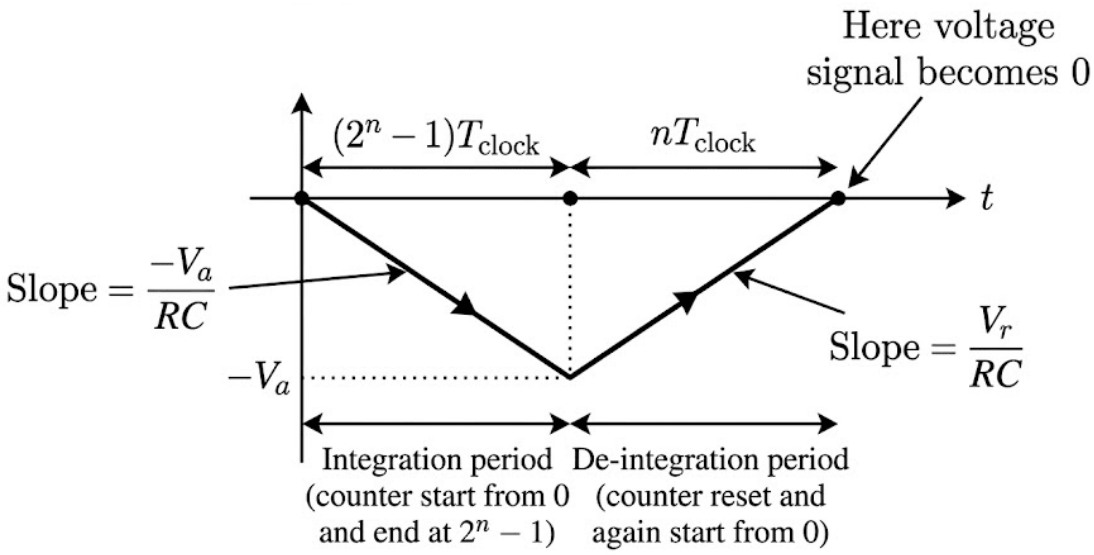


Fig. Solution Q6.00.1

In a dual-slope ADC, the conversion takes place in two phases:

1. **Integration phase:** The unknown input voltage V_a is applied to the integrator for a fixed duration.
2. **De-integration phase:** A known reference voltage $-V_r$ is applied and the time required for the integrator output to return to zero is measured.

As shown in the figure, the integration interval is controlled by an n-bit counter.
An n-bit counter can count from

$$0 \text{ to } 2^n - 1.$$

Therefore, during the integration phase, the counter is allowed to count up to

$$\boxed{2^n - 1}$$

before the ADC switches to the de-integration phase.
Hence, the correct option is **(B)**

$$\boxed{\text{(B) } 2^n - 1}$$

Key Points

(i) A dual-slope ADC uses integration followed by de-integration.

(ii) The integration time is fixed and is determined by the counter.

(iii) An n-bit counter has a maximum count value of $2^n - 1$.

(iv) The de-integration time is proportional to the input voltage.

(v) Dual-slope ADCs provide high accuracy and excellent noise immunity.

(vi) Dual-slope ADCs are comparatively slow.

Mistakes to be avoided

- Do not confuse the maximum counter value $(2^n - 1)$ with the total number of states (2^n).
- The question asks for the maximum count value, not the integration time.
- The quantity proportional to V_a is the de-integration time.

Q6.94.1 MCQ 1994 1M Ans: B ●○○

Given: A 4-bit flash type A/D converter.

For an n-bit flash ADC, the number of comparators required is:

$$C = 2^n - 1$$

For $n = 4$:

$$C = 2^4 - 1 = 16 - 1 = \boxed{15}$$

(B) 15

Key Points

- An n-bit flash ADC requires: $2^n - 1$ comparators, 2^n resistors (for the voltage divider), and a $2^n \times n$ priority encoder.
- The flash ADC is the **fastest** ADC type since all comparisons are made simultaneously in a single clock cycle.
- The disadvantage is hardware complexity: for large n, the number of comparators grows exponentially.

Q6.93.1 MCQ 1993 1M Ans: D ●○○

Given:

- Number of bits: $n = 10$
- Input voltage range: $V_{min} = 0\,\text{V}$ to $V_{max} = 5\,\text{V}$

1. Resolution and LSB Calculation: The resolution of the ADC represents the smallest analog voltage change corresponding to a change of 1 Least Significant Bit (LSB) in the digital output code:

$$\text{LSB} = \frac{V_{max} - V_{min}}{2^n} = \frac{5 - 0}{2^{10}} = \frac{5}{1024} \approx 0.00488\,\text{V} \approx 4.88\,\text{mV}$$

Note: If the peak-to-peak power supply ripple voltage exceeds approximately one LSB ($\approx 5\,\text{mV}$), the internal reference references or comparison levels will fluctuate significantly. This variation can cause the digital output code to toggle or drift by one or more counts even when the actual analog input signal remains perfectly constant.

Therefore, to prevent spurious fluctuations and maintain a stable conversion, the maximum allowable peak-to-peak ripple voltage must be restricted to approximately equal to one LSB, which is about 5 mV.

(D) Nearly 5 mV

Key Points

- The resolution of an ADC represents the smallest analog change that produces a change in digital output.
- Any supply ripple exceeding the resolution will cause the ADC output to fluctuate by at least 1 LSB, introducing errors.
- For high-precision design purposes, supply ripple should be kept well below 1 LSB (ideally ≤ 0.5 LSB) to preserve accuracy.

Q6.92.1 MCQ 1992 1M Ans: C ● ○ ○

Given: A parallel conversion (flash) type 8-bit A/D converter.

For an n-bit parallel (flash) ADC, the number of comparators required is:

$$C = 2^n - 1$$

For $n = 8$:

$$C = 2^8 - 1 = 256 - 1 = \boxed{255}$$

$$\boxed{\text{(C) } 255}$$

Key Points

- An n-bit flash ADC also requires 2^n resistors and a $2^n \times n$ priority encoder in addition to $2^n - 1$ comparators.
- The flash ADC achieves 1-clock-cycle conversion, making it the fastest, but the hardware grows exponentially with bit-count.

Q6.91.1 NAT 1991 1M Ans: 0.392 ● ○ ○

Given: An 8-bit A/D converter ($n = 8$).

The percentage resolution of an n-bit ADC is defined as:

$$\text{Resolution}(\%) = \frac{1}{2^n - 1} \times 100\%$$

For $n = 8$:

$$\text{Resolution}(\%) = \frac{1}{2^8 - 1} \times 100\% = \frac{1}{255} \times 100\%$$

$$\text{Resolution}(\%) = 0.392\%$$

The resolution of an 8-bit A/D converter is **0.392**%.

Key Points

- **Percentage Resolution** = $\frac{1}{2^n - 1} \times 100\%$ (uses $2^n - 1$ steps).
- **Absolute Resolution** = $\frac{V_{FS}}{2^n - 1}$ in volts.
- Higher the number of bits, finer the resolution (smaller percentage), and greater the accuracy of the ADC.

SOLUTIONS

VII

Logic Families and Semiconductor Memories

Topics

1. Logic Gate Implementation Using CMOS
2. Characteristics of Logic Families
3. Hazards in Combinational Circuits
4. Programmable Logic Devices (PLD, PLA, PAL)
5. ROM, SRAM and DRAM

Unit Snapshot*

Stream: EE	**Questions:** 8	**MCQ:** 8	**MSQ:** 0	**NAT:** 0
1M: 5	**2M:** 3	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2017)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q7.26.1 MCQ | 2026 | 2M | Ans: D ● ● ○

Given MOSFET switch circuit is shown below,

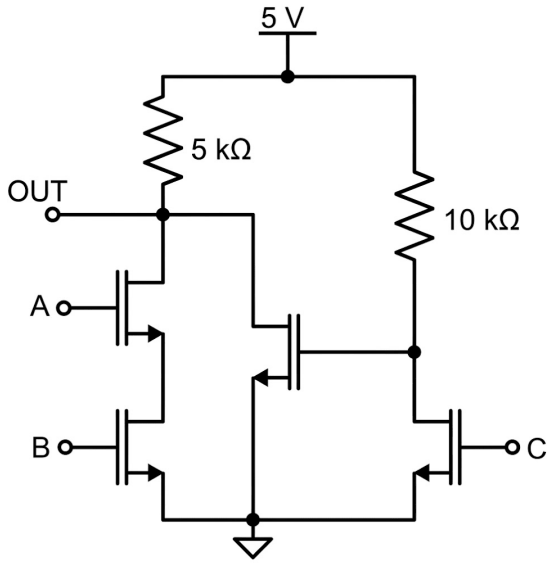


Fig. Solution Q7.26.1

Let the intermediate node connected to the 10 kΩ resistor be X.
When $C = 1$, the NMOS transistor turns ON and pulls node X to ground.
When $C = 0$, the NMOS transistor turns OFF and the pull-up resistor pulls node X to 5 V.
Therefore,

$$X = \overline{C}$$

Now consider the pull-down network connected to the output node.

- The series combination of NMOS transistors controlled by A and B conducts when

$$AB = 1$$

- The middle NMOS transistor conducts when

$$X = \overline{C} = 1$$

Hence, the pull-down condition is

$$AB + \overline{C}$$

Since the output node has a resistive pull-up, the output is the complement of the pull-down function:

$$\text{OUT} = \overline{AB + \overline{C}}$$

Applying De Morgan's theorem,

$$\text{OUT} = \overline{AB} \cdot C$$

which corresponds to the expression given in option (D).

Maximum Power Consumption

Static power is consumed whenever a conducting path exists from the 5 V supply to ground.

For the 5 kΩ branch,

$$P_1 = \frac{V^2}{R} = \frac{5^2}{5 \times 10^3} = 5\,\text{mW}$$

For the 10 kΩ branch,

$$P_2 = \frac{V^2}{R} = \frac{5^2}{10 \times 10^3} = 2.5\,\text{mW}$$

Both branches conduct simultaneously for

$$A = 1, \quad B = 1, \quad C = 1$$

Therefore,

$$P_{\max} = P_1 + P_2 = 5 + 2.5 = 7.5\,\text{mW}$$

$$\boxed{P_{\max} = 7.5\,\text{mW}}$$

$$\boxed{\text{(D) } \overline{AB}\,C, \quad P = 7.5\,\text{mW}}$$

Key Points

- An NMOS with a pull-up resistor acts as an inverter.
- Series NMOS transistors implement an AND pull-down function.
- Parallel pull-down paths combine using OR operation.
- The output of resistive-load NMOS logic is the complement of the pull-down function.
- Maximum static power occurs when all possible current paths from V_{DD} to ground conduct simultaneously.

Mistakes to be avoided

- Do not forget to complement the pull-down network while obtaining the output logic.
- Do not calculate power using only one conducting branch; check whether multiple branches can conduct simultaneously.
- Apply De Morgan's theorem carefully while simplifying the Boolean expression.

Q7.17.1 MCQ | 2017 | 1M | Ans: B | ● ○ ○

Given logic circuit is shown below,

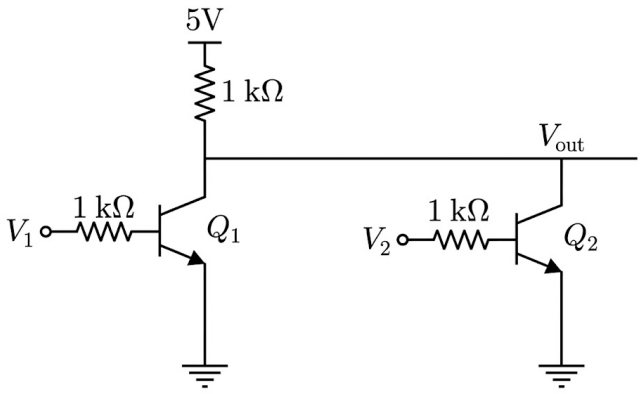


Fig. Solution Q7.17.1

Condition 1 : When $V_1 = 0$ and $V_2 = 0$. Transistor (Q_1) and transistor (Q_2) both are operated under cutoff mode it means both transistor behave as an open circuit (OFF) as shown below,

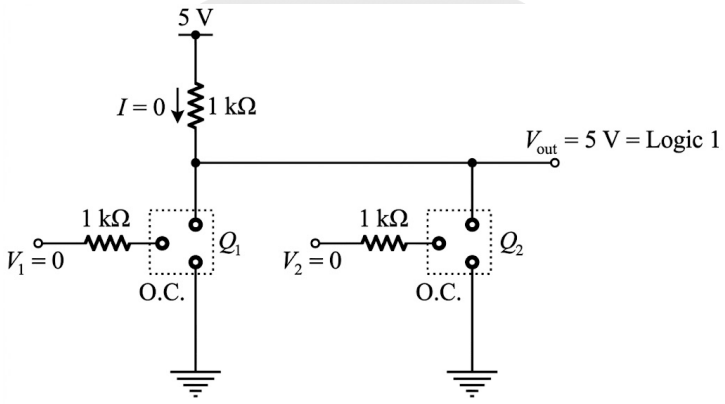


Fig. Solution Q7.17.1

Hence, $V_1 = 0$ and $V_2 = 0 \rightarrow V_{out} = 5$ V = logic 1

Condition 2 : When $V_1 = 0$ and $V_2 = 5$ V (logic 1). Transistor (Q_1) is in cutoff mode it means open circuit (OFF) and transistor (Q_2) is in saturation mode it means short circuit (ON) as shown below,

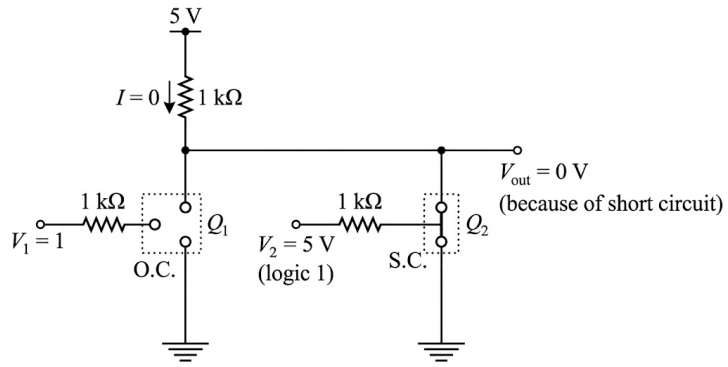


Fig. Solution Q7.17.1

Hence, $V_1 = 0$ and $V_2 = 5$ V (logic 1) $\rightarrow V_{out} = 0$

Condition 3 : When $V_1 = 5$ V (logic 1) and $V_2 = 0$. Transistor (Q_1) is in saturation mode it means short circuit (ON) and transistor (Q_2) is in cutoff mode it means open circuit (OFF) as shown below,

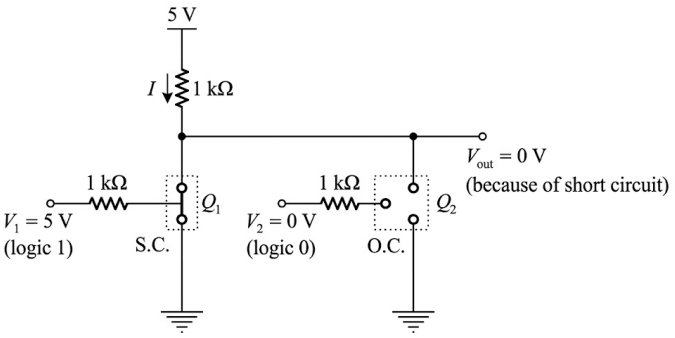


Fig. Solution Q7.17.1

Hence, $V_1 = 5$ V (logic 1) and $V_2 = 0 \rightarrow V_{out} = 0$ V.

Condition 4 : When $V_1 = 5$ V (logic 1) and $V_2 = 5$ V (logic 1) Transistor (Q_1) and transistor (Q_2) both are operated in saturation mode it means both are short circuited (ON) as shown below,

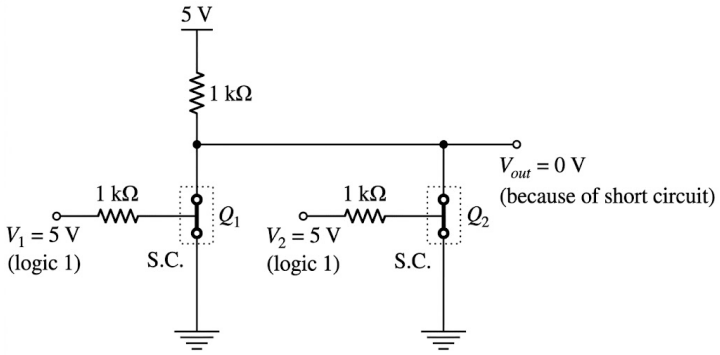


Fig. Solution Q7.17.1

Hence, $V_1 = 5$ V (logic 1) and $V_2 = 5$ V (logic 1) $\rightarrow V_{out} = 0$ V.

Based on above four condition we can construct truth table as,

Input		Transistor		Output
V_1	V_2	Q_1	Q_2	V_{out}
0	0	OFF	OFF	1
0	1	OFF	ON	0
1	0	ON	OFF	0
1	1	ON	ON	0

Thus, we conclude that, given circuit follow the truth table of NOR gate.

(B) NOR

Key Points

(i) Common interconnecting node with pull-up resistor formed Wired-AND-Logic.

(ii) Common interconnecting node with pull-down resistor formed Wired-OR-Logic.

Q7.13.1 MCQ | 2013 | 1M | Ans: B ●●○

Given circuit is shown below,

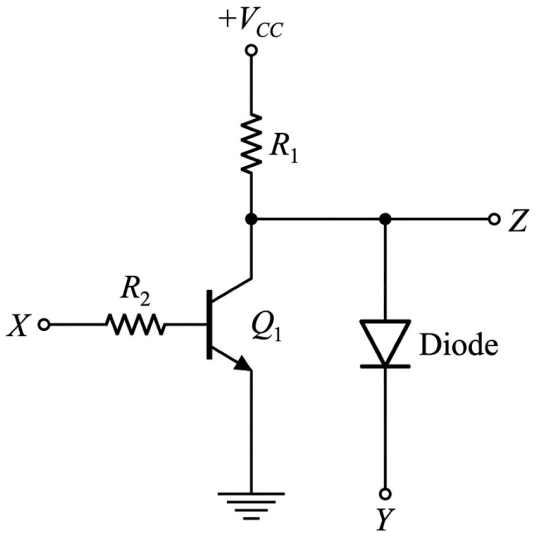


Fig. Solution Q7.13.1

Here, transistor (Q_1) and diode (D) both can work as a switch and transistor (Q_1) is controlled by input (X) and diode (D) is controlled by input (Y).

Condition 1 : When $X = 0$ and $Y = 0$. Transistor Q_1 is operated under cutoff mode, it means open circuited (OFF) and diode D is in forward bias mode, it means short circuited (ON).

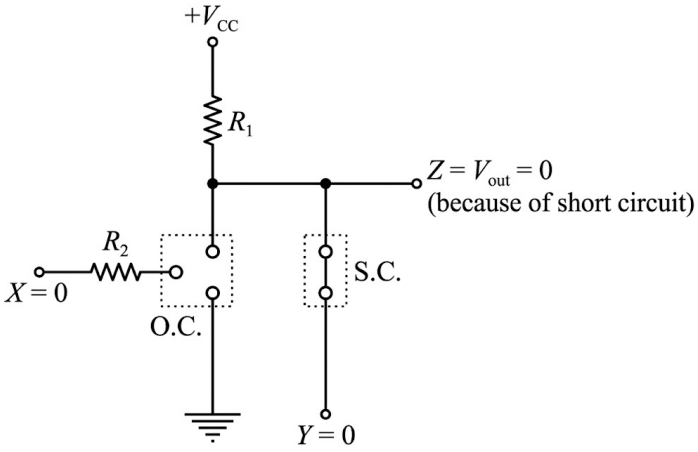


Fig. Solution Q7.13.1

Hence, $X = 0$ and $Y = 0 \rightarrow Z = V_{out} = 0$ (because of short circuit)

Condition 2 : When $X = 0$ and $Y = 1$. Transistor Q_1 is operated under cutoff mode, it means open circuited (OFF) and diode D is in reverse bias mode, it means open circuited (OFF).

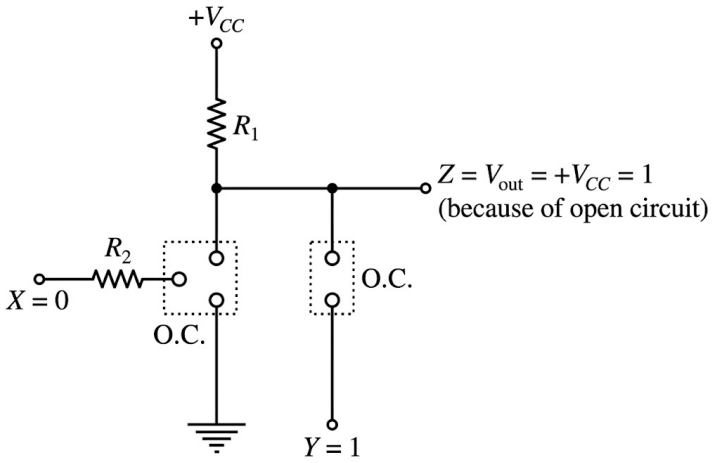


Fig. Solution Q7.13.1

Hence, $X = 0$ and $Y = 1 \rightarrow Z = V_{out} = +V_{cc} = 1$ (because of open circuit)

Condition 3 : When $X = 1$ and $Y = 0$. Transistor Q_1 is operated under saturation mode, it means short circuited (ON) and diode D is in forward bias mode, it means short circuited (ON).

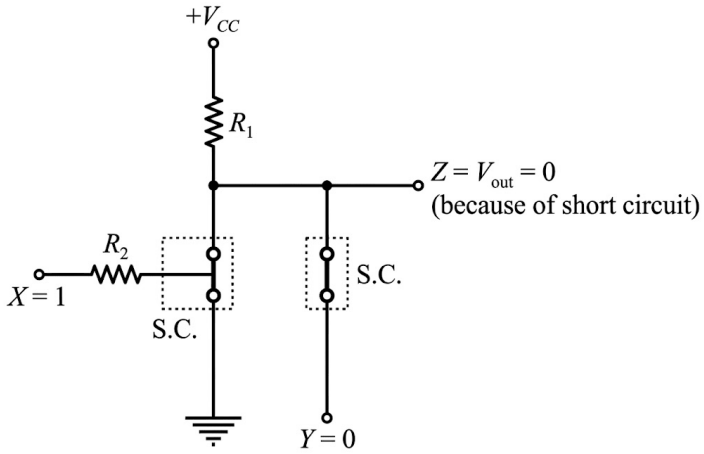


Fig. Solution Q7.13.1

Hence, $X = 1$ and $Y = 0 \rightarrow Z = V_{out} = 0$ (because of short circuit)

Condition 4 : When $X = 1$ and $Y = 1$. Transistor Q_1 is operated under saturation mode, it means short circuited (ON) and diode D is in reverse bias mode, it means open circuited (OFF).

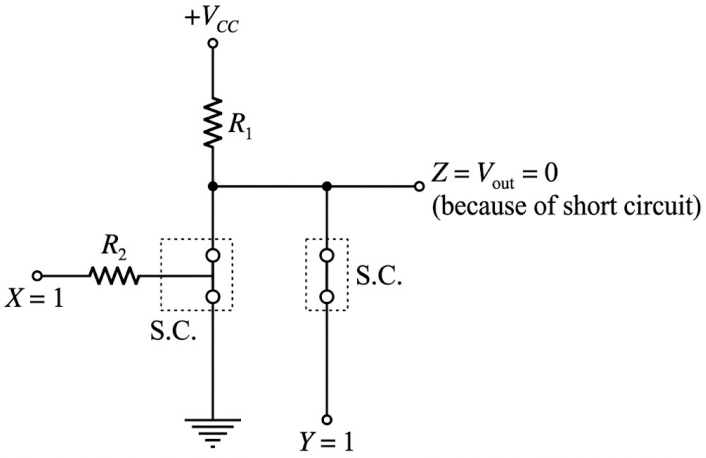


Fig. Solution Q7.13.1

Hence, $X = 1$ and $Y = 1 \rightarrow Z = V_{out} = 0$ (because of short circuit)
Based on above four condition we can form truth table as,

Input		Transistor	Diode	Output
X	Y	Q	D	Z
0	0	OFF	ON	0
0	1	OFF	OFF	1
1	0	ON	ON	0
1	1	ON	OFF	0

K-map of function Z in SOP form is,

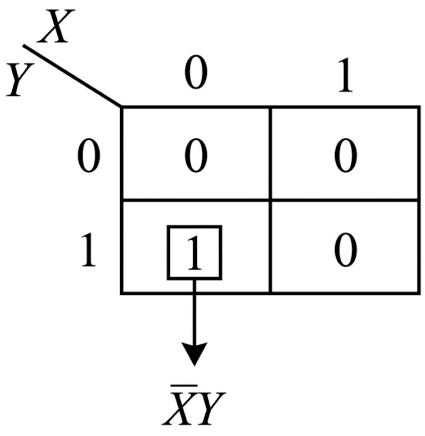


Fig. Solution Q7.13.1

Thus, $Z = \overline{X}Y$

(B) $\bar{X}Y$

Key Points

- A transistor under cutoff is an open circuit (OFF); under saturation it is a short circuit (ON).
- A diode in forward bias is a short circuit; in reverse bias it is an open circuit.
- **Wired-AND** at the common interconnecting node: $Z = \bar{X} \cdot Y$.

Mistakes to be avoided

- The diode is controlled by Y directly (not $\bar{Y}$): when $Y = 1$, diode is reverse biased (OFF); when $Y = 0$, diode is forward biased (ON). Do not confuse the bias direction.

Q7.10.1 MCQ 2010 2M Ans: A ●●○

Given logic circuit is shown below,

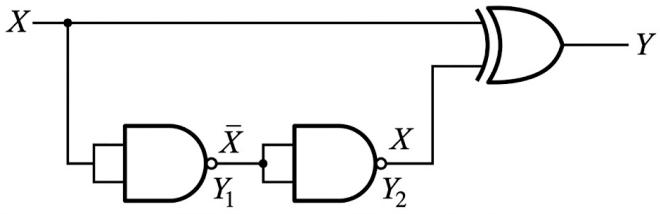


Fig. Solution Q7.10.1

From figure,

$$Y_1 = \overline{X}$$
$$Y_2 = X$$
$$Y = X \oplus Y_2$$
$$Y = X \oplus X = 0$$

The waveforms for output of different gates are shown below,

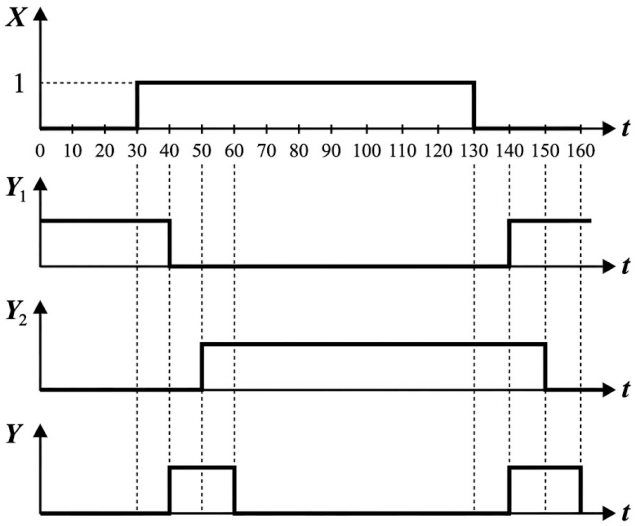


Fig. Solution Q7.10.1

(A) [Waveform A — output stays LOW with narrow glitch pulses at transitions]

Key Points

- **Hazard / Glitch:** When a signal and its complement are combined through gates with propagation delays, narrow spurious pulses (glitches) appear at the output during transitions.
- **Static hazard:** Output momentarily changes when it should remain constant.

- The glitch width equals the propagation delay of the inverter in the path.

Q7.06.1 MCQ | 2006 | 2M | Ans: C ●●○

Given logic circuit is shown below,

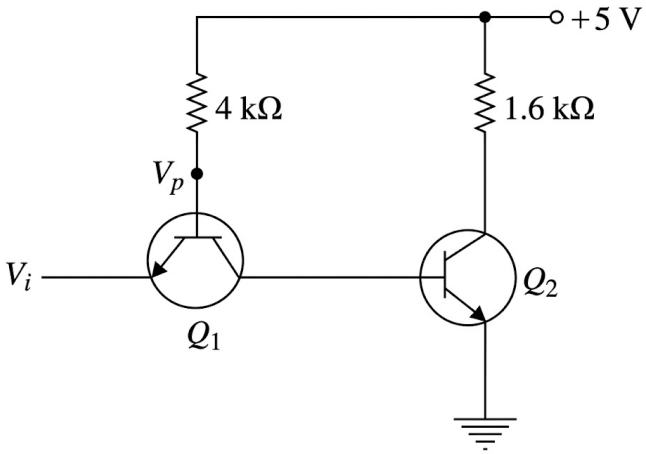


Fig. Solution Q7.06.1

Given : $V_i = 3$ V, $V_{BE} = 0.7$ V

For a standard TTL gate, 0 to 0.8 V is treated as logic 0 and 2 V to 5 V is treated as logic 1. Therefore, the applied input $V_i = 3$ V functions as a logic 1 (HIGH).

Detailed Biasing & Current Steering Analysis:

1. **Base Potential Calculation:** Current flows from the +5 V rail through the 4 kΩ base resistor into the base-collector junction of Q_1, which then drives the base-emitter junction of Q_2. Since Q_2's emitter is grounded, its base-emitter junction is forward biased ($V_{BE2} = 0.7$ V). The base voltage of Q_1 is pinned at:

$$V_{B1} = V_{BC1} + V_{BE2} = 0.7 \text{ V} + 0.7 \text{ V} = 1.4 \text{ V}$$

2. **Base-Emitter Junction of Q_1:** The voltage across the base-emitter junction of Q_1 is:

$$V_{BE1} = V_{B1} - V_i = 1.4 \text{ V} - 3.0 \text{ V} = -1.6 \text{ V}$$

Since $V_{BE1} < 0$, the base-emitter junction is strongly **reverse biased**. Normal emitter electron injection into the base does not occur.

3. **Base-Collector Junction of Q_1:** Because the base terminal sits at 1.4 V and the collector terminal is tied to the base of the active transistor Q_2 ($V_{C1} = V_{B2} = 0.7$ V), the base-collector voltage is:

$$V_{BC1} = V_{B1} - V_{C1} = 1.4 \text{ V} - 0.7 \text{ V} = +0.7 \text{ V}$$

This confirms that the collector-base junction is successfully **forward biased**.

With its base-emitter junction reverse biased and its base-collector junction forward biased, Q_1 operates in the reverse-active region (also termed "reverse ON"), steering the base current via its collector terminal straight into Q_2 to turn Q_2 fully ON.

(C) Q_1 reverse ON and Q_2 ON

Key Points

- In a TTL gate with HIGH input ($V_i > 2$ V): Q_1 operates in **reverse active mode** (B-E reverse biased, B-C forward biased), and Q_2 turns ON (saturation).
- In a TTL gate with LOW input ($V_i < 0.8$ V): Q_1 operates in **normal active/saturation mode**, and Q_2 is cut off.
- The multi-emitter transistor Q_1 in a TTL gate acts as a current steering element.

Mistakes to be avoided

- Do not confuse "reverse ON" with "OFF". In reverse active mode, the transistor is conducting but with emitter and collector roles swapped - the collector-base junction is forward biased, not the emitter-base junction.

Q7.05.1 MCQ | 2005 | 1M | Ans: D

Given logic circuit is shown below,

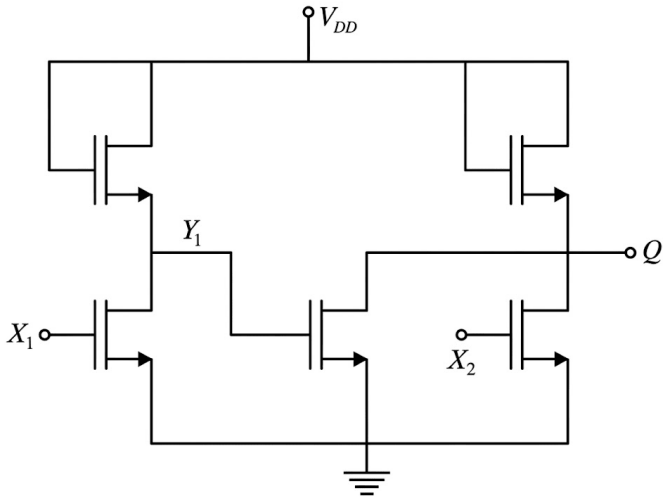


Fig. Solution Q7.05.1

This is a digital logic circuit implemented using NMOS transistors with enhancement-load NMOS pull-up devices.
Consider the left part of the circuit.
When $X_1 = 1$, transistor Q_1 turns ON and pulls node Y_1 to ground.
Hence,

$$Y_1 = 0$$

When $X_1 = 0$, transistor Q_1 turns OFF and the load transistor pulls Y_1 HIGH.
Hence,

$$Y_1 = 1$$

Therefore, the left stage acts as an inverter:

$$Y_1 = \overline{X_1}$$

Now consider the right part of the circuit.
Transistors Q_2 and Q_3 are connected in parallel between the output node and ground.
If either Y_1 or X_2 is HIGH, the corresponding NMOS transistor turns ON and pulls the output LOW.
Thus, the right stage implements a NOR gate.

Hence,

$$Q = \overline{Y_1 + X_2}$$

Substituting $Y_1 = \overline{X_1}$,

$$Q = \overline{\overline{X_1} + X_2}$$

Using De Morgan's theorem,

$$Q = X_1 \cdot \overline{X_2}$$

Therefore,

$$\boxed{\text{(D) } X_1 \cdot \overline{X_2}}$$

Key Points

- An NMOS transistor acts as a pull-down device. When its gate input is HIGH, it connects the output node to ground.
- An enhancement-load NMOS inverter realizes the logic function

$$Y = \overline{X}.$$

- Parallel NMOS pull-down transistors implement a NOR function because any HIGH input can pull the output LOW.
- The overall circuit consists of an inverter followed by a NOR gate.

Mistakes to be avoided

- Do not analyze the right stage directly using X_1 and X_2; first determine the intermediate node Y_1.
- Remember that parallel NMOS pull-down paths realize a NOR gate, whereas series NMOS pull-down paths realize a NAND gate.
- Apply De Morgan's theorem carefully:

$$\overline{A + B} = \overline{A}\,\overline{B}.$$

PrepFusion

Q7.98.1 MCQ 1998 1M Ans: A

Method 1

A pull-up resistor is used to stop the random floating of input of logic gate, when there is no input present at logic gate, so we can say that pull-up resistor are basically used to give correct biasing for input of logic gates.

Basically fixed value of resistor is used in pull-up resistors and pull-up resistors are connected between voltage supply and common interconnecting node.

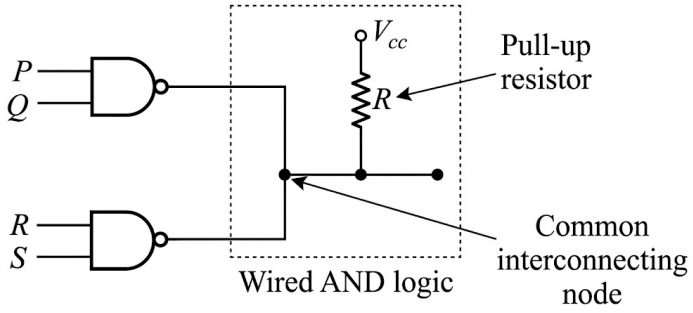


Fig. Solution Q7.98.1

If output of logic gates are connected together at common interconnecting node with pull-up resistor than additional logic is performed without additional hardware. This type of connection is referred as Wired-AND-Logic/Implied-AND-Logic.

The wired AND logic acts as AND gate. Therefore, the equivalent circuit is shown below,

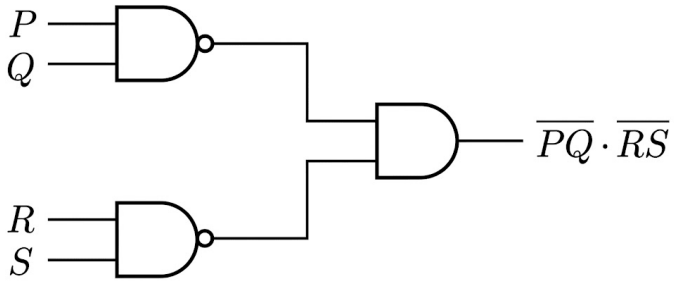


Fig. Solution Q7.98.1

The output is equal to $\overline{PQ} \cdot \overline{RS}$.

(A) $\overline{PQ} \cdot \overline{RS}$

Q7.98.2 MCQ 1998 1M Ans: B

In standard TTL gates, the **totem pole output stage** serves two primary purposes:

1. **Decrease output switching delay:** During a low-to-high output transition, the upper transistor of the totem pole actively charges the load capacitance, providing a low-impedance path to V_{CC}. Without the totem pole, the output would rise slowly through the passive pull-up resistor alone. The active pull-up significantly reduces the rise time and hence the switching delay.
2. **Provide sufficient power for high fanout:** The totem pole output can source and sink large currents, supporting a high fanout (typically fanout = 10 for standard TTL).

(B) Decrease the output switching delay

Key Points

- **Totem pole output:** Upper transistor (active pull-up) + lower transistor (active pull-down), separated by a diode to prevent both from being ON simultaneously.
- **Active pull-up** charges load capacitance faster than a passive resistor, reducing rise time and propagation delay.
- **Limitation:** Totem pole outputs **cannot** be wired together (no Wired-AND possible) — this requires open-collector outputs instead.

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SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

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